Black holes

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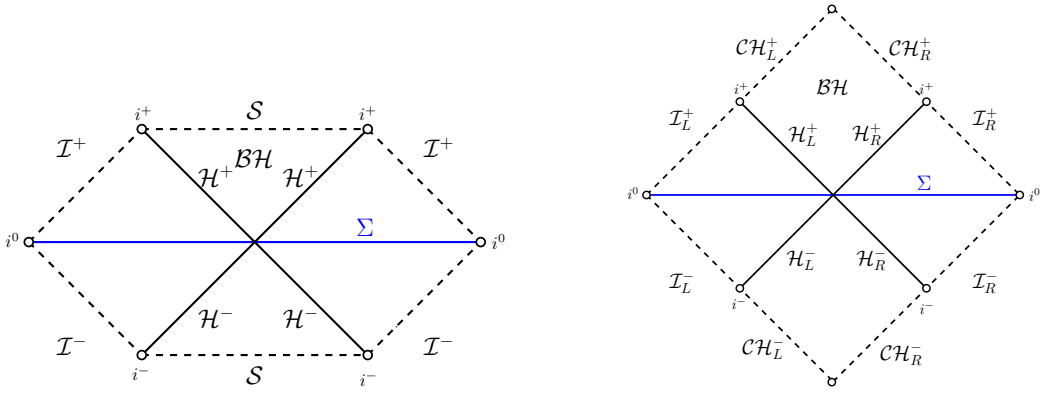
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1 Introduction

Black holes stand as objects *par excellence*, showcasing the radical divergence between Einstein's and Newton's theories of gravity. They are spacetime solutions to Einstein's equations of general relativity that are fascinating from a geometric, analytic and astrophysical perspective and form the stage for an interplay between mathematics, theoretical physics and astrophysics.

In these lectures, we will cover the following topics on black holes:

- The geometric properties of **Schwarzschild black hole spacetimes** and the dynamics of geodesics in these spacetimes. We will see how to make sense of the following pictures, representing the maximally-extended Schwarzschild spacetime and the (sub-extremal) electromagnetically charged **Reissner–Nordström spacetimes**, respectively:



(a) Maximally-extended Schwarzschild with asymptotically flat two-ended initial data hypersurface Σ

(b) Sub-extremal Reissner–Nordström with asymptotically flat two-ended initial data hypersurface Σ

Figure 1: Penrose diagrams of Schwarzschild and Reissner–Nordström spacetimes

- The **Einstein equations coupled to matter in spherical symmetry**. We will prove that in the absence of matter, all spherically symmetric solutions to the Einstein equations are locally isometric to a Schwarzschild or the Minkowski spacetime (*Birkhoff's theorem*). We will discuss how to represent general, dynamical spherically symmetric black hole spacetimes pictorially and understand the following picture representing the formation of a dynamical black hole:

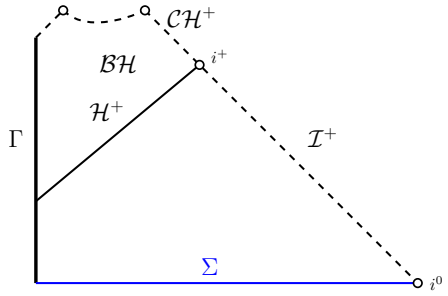


Figure 2: A possible Penrose diagram of a spacetime describing gravitational collapse to a black hole in spherical symmetry arising from initial data on the spacelike hypersurface Σ.

In this part of the course, we will encounter "baby versions" of some of the big theorems in

black hole dynamics.

- The geometric properties of rotating **Kerr black hole spacetimes** and the dynamics of geodesics in these spacetimes;
- The **initial value** or **Cauchy problem**, which provides a systematic way to study dynamics of spacetimes;
- **Asymptotically flat initial data** and notion of *mass/energy*, *momentum* and *angular momentum* in general relativity;
- Trapped surfaces, causal geodesic incompleteness of spacetimes and the stability of the property of "black hole-ness" via **Penrose's incompleteness theorem**.
- Formulations of the **weak and strong cosmic censorship conjectures**.

We will discuss the above topics in a mathematically precise style. This means that we will organize the material with definitions, propositions, theorems and lemmas and that we will try to keep precise track of the various domains and codomains of the functions and maps that we encounter. This will serve as a way to eliminate any ambiguities and confusions that have historically appeared in the study of black holes.

Throughout the lecture notes, there will be EXERCISES that are meant to encourage an active reading of the text.

These lecture notes are still a work in progress! Further chapters may be added as the course progresses. For this reason, there will inevitably be typos throughout the text. If you spot any typos or other errors, please do let me know.

1.1 Some useful literature

These lectures will not follow any existing textbooks too closely. Nevertheless, the list below *complements* the lectures and also provides different styles of presentation on some of the topics that we will cover.

- Robert M. Wald, *General relativity*. University of Chicago press, 1984.
  A classic text on general relativity. Note the use of "abstract index notation" for tensor fields that we will not follow in this course.
- Stephen W. Hawking and George F.R. Ellis, *The large scale structure of space-time*. Cambridge University Press, 1973.
  A complete discussion on many topics in general relativity. Includes Penrose diagrams of several solutions to the Einstein equations that we will also discuss. Contains a discussion on causality and Lorentzian geometry that goes beyond this course.
- Harvey S. Reall, *Mathematical Tripos Part III: General relativity*. https://www.damtp.cam.ac.uk/user/hsr1000/part3_gr_lectures.pdf, 2022
  Lecture notes for a course on general relativity taught at the University of Cambridge.
- Harvey S. Reall, *Mathematical Tripos Part III: Black holes*. http://www.damtp.cam.ac.uk/user/hsr1000/black_holes_lectures_2020.pdf, 2020
  Lecture notes for a course on black holes taught at the University of Cambridge. Contains a discussion on quantum field theory on curved spacetimes that will not be covered in this course.

- John M. Lee, *Introduction to Riemannian manifolds.* Springer, 1997.
  In contrast with the above references, this is a mathematics textbook. Introduces concepts of Lorentzian Riemannian geometry in a mathematically precise manner.

- Stefanos Aretakis, *General Relativity.* https://www.math.toronto.edu/aretakis/General%20Relativity-Aretakis.pdf, 2018
  Lecture notes on general relativity with a more mathematical focus and an accessible discussion of Penrose's incompleteness theorem.

- Demetrios Christodoulou, *Mathematical problems of general relativity I.* European Mathematical Society, 2008.
  An introduction to general relativity from a mathematical and PDE (partial differential equations) angle. Contains a discussion of conserved quantities on asymptotically flat spacetimes.

- Piotr T. Chruściel, *Geometry of black holes.* Oxford University Press, 2020
  Extensive, mathematically rigorous discussion on geometric properties of several families of black hole spacetimes.

- Barrett O'Neill, *Semi-Riemannian geometry with applications to relativity.* Academic press, 1983.
  Exhaustive, mathematically rigorous treatment covering various aspects of Lorentzian geometry in detail.

2 Preliminaries: basics of spacetime geometry

Before we start discussing black hole spacetimes, we will review the main mathematical machinery that is required to introduce black hole spacetimes. The material in this section would typically be covered in a general relativity or differential geometry course. This section is by no means a complete discussion and is meant to serve as a review as well as an opportunity to set the geometric notation used throughout the course.

The main mathematical objects in the field of general relativity are so-called "spacetimes", which are pairs consisting of a 4-dimensional *smooth manifold* $\mathcal{M}$ and a Lorentzian metric g:

$$(\mathcal{M}, g),$$

together with a "time-orientation", an unambiguous notion of past and future. We will define each of these objects.

2.1 Manifolds

An n-dimensional smooth manifold is a mathematical object that locally "looks like" $\mathbb{R}^n$. This roughly means the following: $\mathcal{M}$ is a topological space[1] that can be covered by open sets $U_\alpha \subset \mathcal{M}$ which come equipped with coordinate charts $\phi_\alpha = (x^1, \dots, x^n) : U_\alpha \to \mathbb{R}^n$. This means that $\phi_\alpha : U_\alpha \to \phi(U)$ are homeomorphisms (they are continuous, bijective, with continuous inverse) and for any other coordinate chart $\phi_\beta : U_\beta \to \mathbb{R}^n$, the coordinate transformation map $\phi_\beta \circ \phi_\alpha^{-1} : \phi_\beta(U_\alpha \cap U_\beta) \to \phi(U_\alpha \cap U_\beta)$ is smooth on an appropriate subset of $\mathbb{R}^n$.[2]

Rather than giving a detailed exposition of the above construction, let us note that in the study of black holes in 4 spacetime dimensions, we do not require an understanding of exotic manifolds. The following two special cases are sufficient to capture all the intricate properties of single black holes and their dynamics in four dimensions:[3]

$$\mathcal{M} \cong \mathbb{R} \times \mathbb{R}^3,$$
$$\mathcal{M} \cong \mathbb{R}^2 \times \mathbb{S}^2.$$

Here $\mathbb{S}^n$ denotes the unit round n-sphere: $\mathbb{S}^n = \{x \in \mathbb{R}^{n+1}, \sum_{i=1}^n x_i^2 = 1\}$, which is an example of an n-dimensional manifold. The symbol $\cong$, in the context of manifolds, indicates equality *up to smooth diffeomorphism*. For example, $\mathcal{M} \cong \mathbb{R} \times \mathbb{R}^3$ means that there exists a map $\psi : \mathcal{M} \to \mathbb{R} \times \mathbb{R}^3$ that is smooth, bijective and has a smooth inverse ψ^{-1}. A C^k-diffeomorphism is k times continuously differentiable and a smooth or C^∞-diffeomorphism is arbitrarily many times continuously differentiable.

We will also use the notation $\mathcal{M} = \mathbb{R} \times \Sigma$, with $\Sigma = \mathbb{R}^3$ or $\Sigma = \mathbb{R} \times \mathbb{S}^2$, to treat these two cases simultaneously. Later in the course, we will see that $(n+1)$-dimensional *globally hyperbolic spacetimes*, the class of spacetimes relevant for studying dynamical aspects of general relativity, always take the form $\mathcal{M} = \mathbb{R} \times \Sigma$, where Σ is an n-dimensional manifold.

In the case $\Sigma \cong \mathbb{R} \times \mathbb{S}^2$, we already notice an important property: *there exists no global coordinate chart on* Σ because there exists no global coordinate chart on the sphere $\mathbb{S}^2$. Instead

[1] It comes with a notion of "open subset" and, related to this, a notion of continuity of maps on $\mathcal{M}$.

[2] More precisely, a smooth manifold is a topological space that is *Hausdorff*, *second-countable*, *locally Euclidean* and satisfies the above properties.

[3] In the case of spacetimes describing gravitational collapse to a black hole, $\mathcal{M} \cong \mathbb{R}^4$. It can also be shown that under very general assumptions (Hawking 1972), the exterior region of a stationary (time-independent) 4-dimensional black hole spacetime with reasonable matter is diffeomorphic to $\mathbb{R}^2 \times \mathbb{S}^2$. The event horizon (the boundary of the black hole region) at any fixed time, is diffeomorphic to $\mathbb{S}^2$. For dynamical spacetimes containing black hole regions, or in the case of higher spacetimes dimensions, the topology of the horizon at an instant in time may be more complicated.

we will often consider *standard spherical coordinates* (θ, φ) on $\mathbb{S}^2$, which are defined as follows:

$$\begin{aligned} x &= \cos\varphi \sin\theta, \\ y &= \sin\varphi \sin\theta, \\ z &= \cos\theta. \end{aligned}$$

Since we cannot always work in a single coordinate chart it will be convenient to introduce a few geometric objects that make sense even if we do not specify exactly what coordinate chart we happen to be working in.

2.2 Tensors and tensor fields

In the physics literature, tensor fields are often treated as objects that satisfy the following property:

A tensor field is a list of functions that transforms like a tensor field.

We will, however, define tensors and tensor fields in a different, less utilitarian manner. This has the notational advantage of being able to write down statements without prescribing a coordinate chart.

Before discussing tensor *fields*, we first need to introduce *tensors*. Before we introduce tensors, we recall the definition of *vectors* and *covectors*.

Definition 2.1. *Let $(V, +)$ be a finite-dimensional vector space over $\mathbb{R}$. Then elements $v \in V$ are referred to as* vectors. *The* dual space *V^* is defined as follows:*

$$V^* := \{f : V \to \mathbb{R}\,,\ f \text{ is linear}\}.$$

Elements $f \in V^*$ are called covectors. *Sometimes elements of V are referred to as* contravariant vectors *and elements of V^* as* covariant vectors.

We can associate to a a basis $\{e_\mu\}_{1\leq\mu\leq n}$ of V a *canonical* dual basis $\{f^\mu\}_{1\leq\mu\leq n}$ of V^*, which is defined via $f^\mu(e_\nu) = \delta^\mu_\nu$, with δ^μ_ν the Kronecker delta.[4]

We can also consider the *double dual space* of V: $V^{**} := (V^*)^*$. It turns out that $(V^*)^*$ is *isomorphic* to V if V is finite dimensional. This means that there exist a bijective (injective and surjective, or "onto" and "one-to one") linear map:

$$\Phi : V \to V^{**}.$$

Indeed, we define Φ as follows: let $f \in V^*$, then

$$\Phi(v)(f) := f(v)$$

is clearly linear. We will also use "$\cong$" to denote to indicate an isomorphism between vector spaces, so $V \cong V^{**}$.[5]

[Exercise: Show that Φ is injective, i.e. $\ker\Phi = \{0\}$. Then use the *rank-nullity theorem*, which says that general linear maps $L : V \to W$, with V, W vector spaces satisfy $\dim(\ker L) + \dim(\operatorname{ran} L) = \dim V$, to conclude that Φ is bijective.]

We can therefore also identify vectors $v \in V$ with linear maps of the form $v : V^* \to \mathbb{R}$.

[4] Exercise: Convince yourself that this prescription fixes f^μ uniquely.

[5] Since the map Φ does not depend on a choice of basis in V or V^{**}, V is also said to be naturally isomorphic to V^{**}. Since V and V^* have the same dimension, they are also isomorphic. However, in this case the isomorphism does depend on a choice of basis.

Definition 2.2. *An (r,s)-tensor in V is a* multilinear *(over $\mathbb{R}$) map of the form:*

$$T : \underbrace{V^* \times \ldots \times V^*}_{r \text{ times}} \times \underbrace{V \times \ldots \times V}_{s \text{ times}} \to \mathbb{R}.$$

The set of (r,s)-tensors in V forms a n^{r+s}-dimensional vector space, which we denote as follows:

$$T^{(r,s)}(V).$$

We also use the following alternative notation for $T^{(r,s)}(V)$:

$$\underbrace{V \otimes \ldots \otimes V}_{r \text{ times}} \otimes \underbrace{V^* \otimes \ldots \otimes V^*}_{s \text{ times}}.$$

The (r,s)-tensors of the form

$$\underbrace{e_{\mu_1} \otimes \ldots \otimes e_{\mu_r}}_{r \text{ times}} \otimes \underbrace{f^{\nu_1} \otimes \ldots \otimes f^{\nu_s}}_{s \text{ times}}$$

form a basis of $T^{(r,s)}(V)$. The above notation is defined is defined as follows: let $\omega_i \in V^*$, $i \in \{1, \ldots, r\}$, and $v_j \in V$, $j \in \{1, \ldots, s\}$, then:

$$(e_{\mu_1} \otimes \ldots \otimes e_{\mu_r} \otimes f^{\nu_1} \otimes \ldots \otimes f^{\nu_s})(\omega_1, \ldots, \omega_r, v_1, \ldots, v_s) := e_{\mu_1}(\omega_1) \ldots e_{\mu_r}(\omega_r) f^{\nu_1}(v_1) \ldots f^{\nu_s}(v_s).$$

In other words, we can expand $T \in T^{(r,s)}(V)$ as follows:

$$T = \sum_{\mu_1=1}^{n} \ldots \sum_{\mu_r=1}^{n} \sum_{\nu_1=1}^{n} \ldots \sum_{\nu_s=1}^{n} T^{\mu_1 \ldots \mu_r}{}_{\nu_1 \ldots \nu_s} e_{\mu_1} \otimes \ldots \otimes e_{\mu_r} \otimes f^{\nu_1} \otimes \ldots \otimes f^{\nu_s},$$

where $T^{\mu_1 \ldots \mu_r}{}_{\nu_1 \ldots \nu_s} \in \mathbb{R}$. To shorten notation, we will usually omit the summation symbols. This is called the *Einstein summation convention*:

$$T = T^{\mu_1 \ldots \mu_r}{}_{\nu_1 \ldots \nu_s} e_{\mu_1} \otimes \ldots \otimes e_{\mu_r} \otimes f^{\nu_1} \otimes \ldots \otimes f^{\nu_s}.$$

Example 2.1. *Consider the tensor $T : V^* \times V \to \mathbb{R}$. Then $A : V \to V$, defined as*

$$A(v) := \Phi^{-1}(T(\cdot, v)) \in V$$

is a linear map, which can be represented by a matrix after choosing a basis of V. The matrix coefficients of A are given by T^{μ}_{ν}.

To be able to define tensor *fields* on $\mathcal{M}$, we first need to introduce the relevant vector spaces that will take the role of V and V^*: the *tangent spaces* and *cotangent spaces* of the manifold $\mathcal{M}$.

Definition 2.3. *Let $\mathcal{M}$ be an n-dimensional manifold and let $x \in \mathcal{M}$. Suppose that $x \in U \subset \mathcal{M}$ is covered by the coordinate chart $\phi := (x^1, \ldots, x^n) : U \to \mathbb{R}^n$. Let $\gamma_1, \gamma_2 : \mathbb{R} \to U$ be smooth curves. Then we write $\gamma_1 \sim \gamma_2$ if $\gamma_1(0) = \gamma_2(0)$ and $(\phi \circ \gamma_1)'(0) = (\phi \circ \gamma_2)'(0)$.*[6]

Let $\gamma : \mathbb{R} \to U$ be smooth. Denote

$$[\gamma] := \{\widetilde{\gamma} : \mathbb{R} \to U \text{ smooth, with } \widetilde{\gamma} \sim \gamma\}$$

Then the tangent space at x, denoted $T_x\mathcal{M}$ is defined as follows:

$$T_x\mathcal{M} := \{[\gamma],\ \gamma : \mathbb{R} \to U \text{ smooth, with } \gamma(0) = x\}.$$

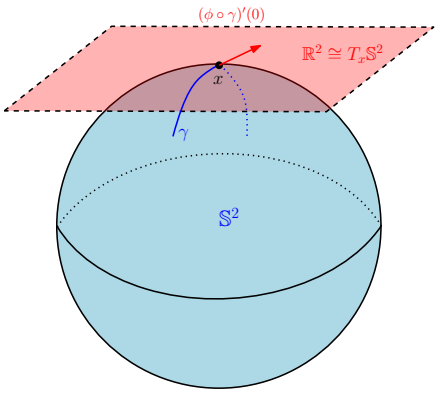


Figure 3: The tangent space $T_x\mathbb{S}^2$.

The set $T_x\mathcal{M}$ can be equipped with a vector space structure, by introducing "addition" and "scalar multiplication". This is not immediate, since we cannot simply "add" curves in $\mathcal{M}$ (unless $\mathcal{M} \cong \mathbb{R}^n$). Consider the following map:

$$d\phi_x : T_x\mathcal{M} \to \mathbb{R}^n$$
$$d\phi_x([\gamma]) := (\phi\circ\gamma)'(0).$$

[Exercise: Show that $d\phi_x$ is a bijection.]

Then we define for $\lambda \in \mathbb{R}$:

$$[\gamma_1] + \lambda\cdot[\gamma_2] := (d\phi_x)^{-1}(d\phi_x([\gamma_1]) + \lambda\cdot d\phi_x([\gamma_2])).$$

Definition 2.4. *We define the* coordinate basis vectors at $x \in \mathcal{M}$ as follows*:*

$$\left(\frac{\partial}{\partial x^\mu}\right)_x := d\phi_x^{-1}((\overbrace{0,\ldots,0,1}^{\mu},0,\ldots))$$

[Exercise: Convince yourself that the vectors $\left(\frac{\partial}{\partial x^\mu}\right)_x$ form a basis of $T_x\mathcal{M}$, so $T_x\mathcal{M}$ is an n-dimensional vector space.]

For example, $T_x\mathbb{R}^4 \cong \mathbb{R}^4$ and $T_x(\mathbb{R}^2\times\mathbb{S}^2) \cong \mathbb{R}^4$.

If $\psi = (y^1\ \ldots, y^n) : U \to \mathbb{R}^n$ is another choice of coordinates and $\gamma : \mathbb{R} \to U$ satisfies $\gamma(0) = p$ and $(\psi\circ\gamma)'(0) = (\overbrace{0,\ldots,0,1}^{\mu},0,\ldots)$. Then

$$(x^\nu\circ\gamma)'(0) = ((x^\nu\circ\psi^{-1})\circ(\psi\circ\gamma))'(0) = \sum_{\alpha=1}^{n}\frac{\partial(x^\nu\circ\psi^{-1})}{\partial y^\alpha}\Big|_{\psi(\gamma(0))}(y^\alpha\circ\gamma)'(0) = \frac{\partial(x^\nu\circ\psi^{-1})}{\partial y^\mu}\Big|_{\psi(p)}.$$

Hence, we can write

$$\left(\frac{\partial}{\partial y^\mu}\right)_p = \sum_{\nu=1}^{n}\frac{\partial(x^\nu\circ\psi^{-1})}{\partial y^\mu}\Big|_{\psi(p)}\left(\frac{\partial}{\partial x^\nu}\right)_p =: \frac{\partial x^\nu}{\partial y^\mu}(x)\left(\frac{\partial}{\partial x^\nu}\right)_p,$$

where we shortened the notation by writing $(\frac{\partial x^\nu}{\partial y^\mu})(p)$ instead of $(\frac{\partial(x^\nu\circ\psi^{-1})}{\partial y^\mu})(\psi(p))$.

[Exercise: Show that $T_x\mathcal{M}$ is independent of the choice of coordinate chart $\phi : U \to \mathbb{R}^n$.]

[6]The relation $\sim$ is an equivalence relation.

We can think of $\left(\frac{\partial}{\partial x^\mu}\right)_x$ as acting on functions $f \in C^\infty(\mathcal{M})$ in the following way: $\left(\frac{\partial}{\partial x^\mu}\right)_x(f) = \frac{\partial(f\circ\phi^{-1})}{\partial x^\mu}|_{\phi(x)}$. This leads to an alternative, but equivalent, characterization of $T_x\mathcal{M}$:

$T_x\mathcal{M} \cong \{v : C^\infty(\mathcal{M}) \to \mathbb{R} \text{ linear, such that } v(fg) = v(f)g(x) + f(x)v(g)\ \forall f, g \in C^\infty(\mathcal{M})\}$.[7]

We denote $T_x^*\mathcal{M} := (T_x\mathcal{M})^*$. Then $(dx^\mu)_x \in T_x^*\mathcal{M}$ is defined as follows:

$$(dx^\mu)_x \left(\left(\frac{\partial}{\partial x^\nu} \right)_x \right) = \delta^\mu_\nu,$$

so $\{(dx^\mu)_x\}$ is the canonical dual basis associated to the basis $\{\left(\frac{\partial}{\partial x^\mu}\right)_x\}$.

Definition 2.5. *A vector field X is a smooth map $X : \mathcal{M} \to T\mathcal{M}$, with $X(x) = (x, X_x)$, where $X_x \in T_x\mathcal{M}$ and $T\mathcal{M}$ is the* tangent bundle of $\mathcal{M}$*, which is defined as follows:*[8]

$$T\mathcal{M} := \bigcup_{x\in\mathcal{M}} \{x\} \times T_x\mathcal{M}.$$

We denote the space of all vector fields by $\mathcal{T}(\mathcal{M})$.

In a slight abuse of notation, we can also interpret $X \in \mathcal{T}(\mathcal{M})$ as a map $X : C^\infty(\mathcal{M}) \to C^\infty(\mathcal{M})$, where

$$X(f)(x) := X_x(f).$$

Note that on the right-hand side, we are interpreting vectors in $T_x\mathcal{M}$ as maps acting on functions.

Consider the coordinate chart $(U, \{x^\mu\})$. By construction of the differentiable structure on $T\mathcal{M}$, it follows that the following maps are smooth vector fields on U:

$$\frac{\partial}{\partial x^\mu} : U \to T\mathcal{M},$$
$$\frac{\partial}{\partial x^\mu}(x) := \left(x, \left(\frac{\partial}{\partial x^\mu} \right)_x \right).$$

Consider: $\frac{\partial}{\partial x^\mu} : C^\infty(\mathcal{M}) \to C^\infty(\mathcal{M})$, with

$$\frac{\partial f}{\partial x^\mu}(x) := \left(\frac{\partial}{\partial x^\mu} \right)_x (f) = \frac{\partial}{\partial x^\mu}(f \circ \phi^{-1})(\phi(x)).$$

We can express a general vector field $X \in \mathcal{T}(\mathcal{M})$ as follows when restricted to the domain U of the coordinate chart $\phi = (x^0, \ldots, x^n)$:

$$X|_U = X^\mu \frac{\partial}{\partial x^\mu},$$

with $X^\mu : U \to \mathbb{R}$ smooth functions. Vector fields need not be smooth, they can for example be C^k maps. This means that the functions X^μ are C^k. Unless specified otherwise, we will assume that the vector fields under consideration are smooth.

To shorten notation, we will write $\partial_\mu = \frac{\partial}{\partial x^\mu}$ when there is no ambiguity about the coordinate chart under consideration.

[7] Here "$\cong$" means: 'isomorphic as vector spaces".

[8] Note that the notion of smoothness of maps to $T\mathcal{M}$ makes sense since $T\mathcal{M}$ can be given the structure of a topological space and smooth manifold (it "locally looks like $\mathbb{R}^{2n}$). Indeed, given a coordinate chart $\phi : U \to \mathbb{R}^n$ on $\mathcal{M}$ with U open, $\psi : V \to \mathbb{R}^{2n}$, with $V = \{(x, v) \in T\mathcal{M}\,,\, x \in U\}$ is defined to be an open set and $\psi(x, v) := (\phi(x), v^1, \ldots, v^n)$, where $v = v^\mu(\frac{\partial}{\partial x^\mu})_x$, defines a coordinate chart.

Definition 2.6. *The* cotangent bundle of $\mathcal{M}$ *is the set:*

$$T^*\mathcal{M} := \bigcup_{x\in\mathcal{M}} \{x\} \times T_x^*\mathcal{M}.$$

Covector fields *or* 1-forms *are smooth maps* $\omega : \mathcal{M} \to T^*\mathcal{M}$*, with* $\omega(x) = (x, \omega_x)$*. We denote the space of all covector fields/*1*-forms by* $\Omega^1(\mathcal{M})$*.*

As in the case of vector fields, we can generalize the definition of 1-forms and consider C^k 1-forms.

We define $dx^\mu \in \Omega^1(\mathcal{M})$ as the 1-forms satisfying $dx^\mu(x) = (x, (dx^\mu)_x)$, with $\{(dx^\mu)_x\}$ denoting the dual basis in $T_x^*\mathcal{M}$ corresponding to the coordinate basis $\{\left(\frac{\partial}{\partial x^\mu}\right)_x\}$.

Now we are ready to define *tensor fields*. Note that are several different but equivalent approaches to defining tensor fields and we will only present the one that is most convenient for the purposes of this course.

Definition 2.7. *An* (r, s)-tensor field T *is a map*

$$T : \underbrace{\Omega^1(\mathcal{M}) \times \ldots \Omega^1(\mathcal{M})}_{r \text{ times}} \times \underbrace{\mathcal{T}(\mathcal{M}) \times \ldots \times \mathcal{T}(\mathcal{M})}_{s \text{ times}} \to C^\infty(\mathcal{M})$$

that is multilinear over $C^\infty(\mathcal{M})$*. This means that for all* $f \in C^\infty(\mathcal{M})$*,* $\omega^i, \theta \in \Omega^1(\mathcal{M})$ *with* $i \in \{1, \ldots, r\}$*,* $X_j, Y \in \mathcal{T}(\mathcal{M})$ *with* $j = \{1, \ldots, s\}$*:*

$$T(\omega^1, \ldots, (f\omega^i + \theta), \ldots, \omega^r, X_1, \ldots, X_s) = fT(\omega^1, \ldots, \omega^r, X_1, \ldots, X_s) + T(\omega^1, \ldots, \theta, \ldots, \omega^r, X_1, \ldots, X_s),$$
$$T(\omega^1, \ldots, \omega^r, X_1, \ldots, (fX_j + Y), \ldots, X_s) = fT(\omega^1, \ldots, \omega^r, X_1, \ldots, X_s) + T(\omega^1, \ldots, \omega^r, X_1, \ldots, Y, \ldots, X_s).$$

We denote the space of (r, s)*-tensor fields by* $\mathcal{T}^{(r,s)}(\mathcal{M})$*.*

If $C^\infty(\mathcal{M})$ *above is replaced with* $C^k(\mathcal{M})$*, with* $k \leq \infty$*, we say* T *is a* C^k *tensor field.*

Example 2.2. *Note that* $X \in \mathcal{T}(\mathcal{M})$ *can be interpreted as a* $(1, 0)$*-tensor field in the following way: for all* $\omega \in \Omega^1(\mathcal{M})$

$$(X(\omega))(x) := \omega_x(X_x).$$

Similarly, $\omega \in \Omega^1(\mathcal{M})$ *can be interpreted as a* $(0, 1)$*-tensor field in the following way: for all* $X \in \mathcal{T}(\mathcal{M})$

$$(\omega(X))(x) := \omega_x(X_x).$$

Example 2.3. *The following maps are special cases of* (r, s) *tensor fields: let* $X_i \in \mathcal{T}(\mathcal{M})$*,* $1 \leq i \leq r$*, and* $\omega^i \in \Omega^1(\mathcal{M})$*,* $1 \leq j \leq s$*, then for all* $Y_j \in \mathcal{T}(\mathcal{M})$ *and* $\theta^i \in \mathcal{T}(\mathcal{M})$*, we define* $X_1 \otimes \ldots \otimes X_r \otimes \omega^1 \otimes \ldots \omega^s \in \mathcal{T}^{(r,s)}(\mathcal{M})$ *as follows:*

$$(X_1 \otimes \ldots \otimes X_r \otimes \omega^1 \otimes \ldots \omega^s)(\theta^1, \ldots, \theta^r, Y_1, \ldots, Y_s) := X_1(\theta^1)X_2(\theta^2) \ldots X_r(\theta^r)\omega^1(Y_1)\omega^2(Y_2) \ldots \omega_s(Y_s).$$

We give a definition of $S \otimes T$ for more general tensor fields S, T:

Definition 2.8. *Let* $r, r', s, s' \geq 0$ *and* $S \in \mathcal{T}^{(r,s)}(\mathcal{M})$ *and* $T \in \mathcal{T}^{(r',s')}(\mathcal{M})$*. Then* $S \otimes T \mathcal{T}^{(r+s',s+s')}(\mathcal{M})$ *is defined as follows:*

$$(S \otimes T)(X_1, \ldots, X_{r+r'}, \omega^1, \ldots, \omega^{s+s'}) = S(X_1, \ldots, X_r, \omega^1, \ldots, \omega^s)T(X_{r+1}, \ldots, X_{r+r'}, \omega^{s+1}, \ldots, \omega^{s+s'}).$$

In the domain of a coordinate chart $(U, \{x^\mu\})$, we can express a general tensor field $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$ as follows: let $1 \leq \mu_i \leq n$ and $1 \leq \nu_j \leq n$, then

$$T|_U = T^{\mu_1 \ldots \mu_r}_{\nu_1 \ldots \nu_s} \frac{\partial}{\partial x^{\mu_1}} \otimes \ldots \otimes \frac{\partial}{\partial x^{\mu_r}} \otimes dx^{\nu_1} \otimes \ldots \otimes dx^{\nu_s}.$$

where $T^{\mu_1\cdots\mu_r}{}_{\nu_1\ldots\nu_s} \in C^\infty(U)$ for all $1 \leq \mu_i \leq n$ and $1 \leq \nu_1 \leq n$.

[Exercise: Let $\psi = (y^0, \ldots, y^n) : U \to \mathbb{R}^{n+1}$ be a different coordinate chart on U. Determine the relation between $T^{\mu_1\cdots\mu_r}{}_{\nu_1\ldots\nu_s}$ and $\widetilde{T}^{\mu_1\cdots\mu_r}{}_{\nu_1\ldots\nu_s}$, where

$$T|_U = \widetilde{T}^{\mu_1\cdots\mu_r}{}_{\nu_1\ldots\nu_s} \frac{\partial}{\partial y^{\mu_1}} \otimes \ldots \otimes \frac{\partial}{\partial y^{\mu_r}} \otimes dy^{\nu_1} \otimes \ldots \otimes dy^{\nu_s}|_U.]$$

A special class of tensor fields are the *alternating tensor fields* or *differential forms.*

Definition 2.9. *The space of s-forms on $\mathcal{M}$ (or differential forms of degree s, or alternating $(0,s)$-tensor fields) is denoted by $\Omega^s(\mathcal{M})$ and is defined as follows:*

$$\Omega^s(\mathcal{M}) := \left\{ T \in \mathcal{T}^{(0,s)}(\mathcal{M}) \,|\, T(X_{\sigma(1)}, X_{\sigma(2)}, \ldots, X_{\sigma(s)}) = \mathrm{sign}(\sigma) T(X_1, \ldots, X_s) \quad \forall \sigma \in S_s \right\},$$

with S_s the set of all permutations of the set $\{1, \ldots, s\}$ and $\mathrm{sign}(\sigma)$ the sign of the permutation σ which is $+1$ if σ can be written as an even number of transpositions (exchanges between two elements), for example, a cyclic permutation of $\{1,2,3\}$, and -1 if it can be written as an odd number of transpositions, for example, a cyclic permutation of $\{1,2,3,4\}$.

Example 2.4.

$$\Omega^2(\mathcal{M}) := \left\{ T \in \mathcal{T}^{(0,2)}(\mathcal{M}) \,|\, T(X,Y) = -T(Y,X) \right\}.$$

In particular, suppose $\omega, \theta \in \Omega^1(\mathcal{M})$. Then $\omega \otimes \theta$ or $\omega \otimes \theta + \theta \otimes \omega$ are not 2-forms, but $\omega \otimes \theta - \theta \otimes \omega$ is a 2-form.

With respect to a coordinate chart $(U, \{x^\mu\})$, $T \in \Omega^s(U)$ if and only if $T_{\nu_1\ldots\nu_s}$ is fully antisymmetric in its indices.

Definition 2.10. *The* wedge product *between an r-form ω and an s-form θ is defined as the following $(r+s)$-form:*

$$(\omega \wedge \theta)(X_1, \ldots, X_{r+s}) := \sum_{\sigma \in S_{r+s}} \mathrm{sign}(\sigma) (\omega \otimes \theta)(X_{\sigma(1)}, \ldots, X_{\sigma(r+s)})$$

By introducing the notation

$$T_{[\nu_1\ldots\nu_s]} := \frac{1}{s!} \sum_{\sigma \in S_s} \mathrm{sign}(\sigma) T_{\nu_{\sigma(1)}\ldots\nu_{\sigma(s)}},$$

we can write when restricted to the domain of a coordinate chart:

$$(\omega \wedge \theta)_{\nu_1\ldots\nu_{r+s}} = (r+s)! \omega_{[\nu_1\ldots\nu_r} \theta_{\nu_{r+1}\ldots\nu_{r+s}]}.$$

2.3 The exterior derivative

We now provide a definition of the exterior derivative d which can appear in front of a function $f : \mathcal{M} \to \mathbb{R}$ to obtain a 1-form df, but also in front of a general s-form ω to obtain an $s+1$-form $d\omega$.

Definition 2.11. *The exterior derivative $d : C^\infty(\mathcal{M}) \to \Omega^1(\mathcal{M})$ is defined as follows: for any $X \in \mathcal{T}(\mathcal{M})$*

$$(df)(X) := X(f).$$

With respect to a coordinate chart $(U, \{x^\mu\})$, we therefore have that

$$df|_U = \partial_\mu f dx^\mu = \partial_{x^\mu} f dx^\mu.$$

We can extend d as a map from $\Omega^s(\mathcal{M})$ to $\Omega^{s+1}(\mathcal{M})$ with $s \geq 0$ by imposing the following additional requirements:

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^r \omega \wedge d\eta \quad \text{for all } \omega \in \Omega^r(\mathcal{M}) \text{ and } \eta \in \Omega^s(\mathcal{M}),$$
$$d \circ d = 0.$$

For example, if $\omega|_U = \omega_\mu dx^\mu$ with respect to a coordinate chart, then

$$d\omega|_U = d(\omega_\nu) \wedge dx^\nu + 0 = \partial_\mu \omega_\nu dx^\mu \wedge dx^\nu = (\partial_\mu \omega_\nu - \partial_\nu \omega_\mu) dx^\mu \otimes dx^\nu = 2\partial_{[\mu}\omega_{\nu]} dx^\mu \otimes dx^\nu.$$

[Exercise: Show that, for $\omega \in \Omega^r(\mathcal{M})$, the following identity holds in the domain of a coordinate chart $(U, \{x^\mu\})$:

$$d\omega|_U = (r+1)\partial_{[\nu_1}\omega_{\nu_2 \ldots \nu_{r+1}]} dx^{\nu_1} \otimes \ldots \otimes dx^{\nu_{r+1}}.]$$

Let $\omega \in \Omega^s(\mathcal{M})$. If $d\omega = 0$, we say ω is *closed*. If $s \geq 1$ and $\omega = d\eta$ for some $\eta \in \Omega^{s-1}(\mathcal{M})$, then we say ω is *exact*. Note that in the latter case, $d\omega = d^2\eta = 0$, so ω must be closed. The converse need not be true: closed forms need not be exact.[9]

2.4 Pullbacks and pushforwards

Let $\psi : \mathcal{M} \to \mathcal{N}$ be a smooth diffeomorphism between two manifolds $\mathcal{M}$ and $\mathcal{N}$.

Remark 2.1. *Let $\phi : U \to \mathbb{R}^n$ be a coordinate chart on an n-manifold $\mathcal{M}$. Then we can use diffeomorphisms $\psi : U \to U$ to obtain a different coordinate chart $\widetilde{\phi} = \phi \circ \psi$.*

Conversely, given two coordinate charts $\phi, \widetilde{\phi} : U \to \mathbb{R}^n$, the map $\psi = \phi^{-1} \circ \widetilde{\phi}$ defines a diffeomorphism on U. The study of diffeomorphism on a single manifold $\mathcal{M}$ is therefore closely related to coordinate transformations. For example, let $\phi = \mathrm{id}$ be a Cartesian coordinate chart on $\mathbb{R}^3$ and $\widetilde{\phi}$ a spherical coordinate chart on $U \subset \mathbb{R}^3$. Then the change of coordinates from Cartesian to spherical corresponds to a diffeomorphism: $\psi = \widetilde{\phi}$ on U.

There is a natural way to associate to diffeomorphism ψ maps that act between spaces of tensor fields on $\mathcal{N}$ and $\mathcal{M}$. From the perspective of coordinate charts, this is closely related to determining how components of a tensor field with respect to one coordinate chart are related to its components with respect to another coordinate chart.

Definition 2.12. *Let $\psi : \mathcal{M} \to \mathcal{N}$ be a smooth map. The* pullback map $\psi^* : C^\infty(\mathcal{N}) \to C^\infty(\mathcal{M})$ *is defined as follows:*

$$\psi^* f := f \circ \psi,$$

for all $f \in C^\infty(\mathcal{N})$.

The pushforward map *on tangent spaces $\psi_* : T_x\mathcal{M} \to T_x\mathcal{N}$ with $x \in \mathcal{M}$, is defined as follows:*

$$(\psi_*(v))(f) := v(\psi^* f) = v(f \circ \psi).$$

If ψ is a diffeomorphism, then the pushforward map *on vector fields $\psi_* : \mathcal{T}(\mathcal{M}) \to \mathcal{T}(\mathcal{N})$ is defined as follows:*

$$(\psi_* X)(f) := X(f \circ \psi) \circ \psi^{-1},$$

for all $X \in \mathcal{T}(\mathcal{M})$ and $f \in C^\infty(\mathcal{N})$.

[9] When restricted to a suitably small neighbourhood of any point $x \in \mathcal{M}$, all closed forms are exact. Under a topological condition on the manifold $\mathcal{M}$, namely *simply connectedness*, all closed forms are globally exact. Simply connectedness roughly means that any loop in $\mathcal{M}$ can be continuously contracted to a single point.

The pullback map *on 1-forms* $\psi^* : \Omega^1(\mathcal{N}) \to \Omega^1(\mathcal{M})$ *is defined as follows:*

$$(\psi^*\omega)(X) := \omega(\psi_* X) \circ \psi,$$

for all $X \in \mathcal{T}(\mathcal{M})$*. Note that the pullback on 1-forms is well-defined, even if* ψ *is not a diffeomorphism, but merely a smooth map, since:*

$$((\psi^*\omega)(X))_x = \omega_{\psi(x)}((\psi_* X)_{\psi(x)}) = \omega_{\psi(x)}(\psi_*(X_x)).$$

The pullback map *on general tensor fields* $\psi^* : \mathcal{T}^{(r,s)}(\mathcal{N}) \to \mathcal{T}^{(r,s)}(\mathcal{M})$ *is defined as follows (with* ψ *a diffeomorphism if* $r > 0$*):*

$$(\psi^*T)(\omega^1, \dots, \omega^r, X_1, \dots, X_s) := T((\psi^{-1})^*\omega^1, \dots, (\psi^{-1})^*\omega^r, \psi_* X_1, \dots, \psi_* X_s) \circ \psi,$$

for all $\omega^i \in \Omega^1(\mathcal{M})$*, with* $i \in \{1, \dots, r\}$ *and* $X_j \in \mathcal{T}(\mathcal{M})$*, with* $j \in \{1, \dots, s\}$*.*[10]

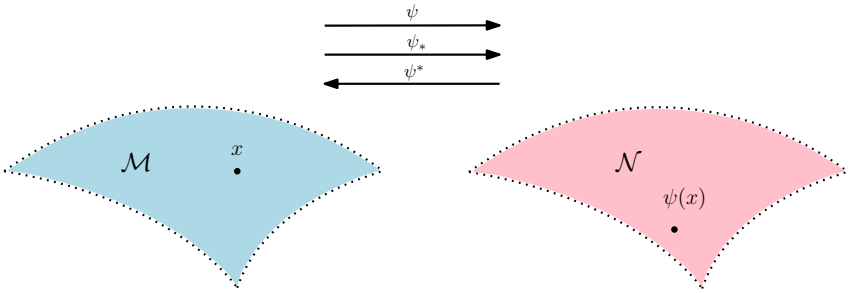


Figure 4: The pushforward and pullback maps.

The behaviour of the pullback map to linear order is determined by the *Lie derivative*.

Definition 2.13.

- *An* integral curve *of a vector field* $X \in \mathcal{T}(\mathcal{M})$ *is a map* $\gamma : \mathbb{R} \supseteq I \to \mathcal{M}$ *satisfying* $\gamma'(s) = X_{\gamma(s)} \in T_{\gamma(s)}\mathcal{M}$*. Here,* $\gamma'(s) = [\delta] \in T_{\gamma(s)}\mathcal{M}$*, with* $\delta : \mathbb{R} \to \mathcal{M}$ *satisfying* $\delta(0) = \gamma(s)$ *and* $(\phi \circ \delta)'(0) = (\phi \circ \gamma)'(s)$*, where* $\phi : U \to \mathcal{M}$ *is a coordinate chart and* $\gamma(s) \in U$*.*

- *A* global flow *is a map* $\psi : \mathbb{R} \times \mathcal{M} \to \mathcal{M}$ *satisfying* $\psi(s, p) = \gamma(s)$*, with* $\gamma : \mathbb{R} \to \mathcal{M}$ *an integral curve of* X *such that* $\gamma(0) = p$*, assuming that the domain of all integral curves* γ *is* $\mathbb{R}$*. Denote* $\psi_s = \psi(s, \cdot)$*. Then the following properties hold (*Exercise*)*

$$\psi_0 = \mathrm{id}, \tag{2.1}$$

$$\psi_{s+\tau} = \psi_s \circ \psi_\tau \quad \text{for all } s, \tau \in \mathbb{R}. \tag{2.2}$$

  For general $X \in \mathcal{T}(\mathcal{M})$*,* ψ *need not be defined for all* $s \in \mathbb{R}$*. In this case, we simply refer to* ψ *as a* (local) flow*.*

- *Conversely, given a smooth map* $\psi : \mathbb{R} \times \mathcal{M} \to \mathcal{M}$ *satisfying* (2.1) *and* (2.2)*, the map* $X(f) = \frac{d}{ds}|_{s=0}(f \circ \psi_s)$*, with* $f \in C^\infty(\mathcal{M})$ *defines a vector field* $X \in \mathcal{T}(\mathcal{M})$*, such that* ψ *is the global flow corresponding to* X *(*Exercise*).*

- *The* Lie derivative *in the direction of* X *of a tensor field* $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$ *is defined as follows:*

$$\mathcal{L}_X T := \frac{d}{ds}\Big|_{s=0} \psi_s^* T,$$

  with ψ_s *corresponding to the flow of* X*.*

[10] As in the case of 1-forms, a pullback on $(0, s)$-tensor field is well-defined, even if ψ is not invertible.

For $X, Y \in \mathcal{T}(\mathcal{M})$, we can alternatively express:

$$\mathcal{L}_X Y = \frac{d}{ds}\Big|_{s=0} (\psi_{-s})_* Y = [X, Y],$$

with $[X, Y](f) = X(Y(f)) - Y(X(f))$ the commutator.

It can be shown that the Lie derivative satisfies the following additional properties: for all $X, X_i \in \mathcal{T}(\mathcal{M})$, with $i \in \{i, \dots, s\}$, $\omega^j \in \Omega^1(\mathcal{M})$, with $j \in \{1, \dots, r\}$, $f \in C^\infty(\mathcal{M})$, $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$, $S \in \mathcal{T}^{(r',s')}(\mathcal{M})$

$$\begin{aligned}
\mathcal{L}_X(f) &= X(f), \\
\mathcal{L}_X(S \otimes T) &= (\mathcal{L}_X S) \otimes T + S \otimes (\mathcal{L}_X T), \\
\mathcal{L}_X(T(\omega^1, \dots, \omega^r, X_1, \dots, X_s)) &= (\mathcal{L}_X T)(\omega^1, \dots, \omega^r, X_1, \dots, X_s) \\
&\quad + T(\mathcal{L}_X \omega^1, \dots, \omega^r, X_1, \dots, X_s) + \dots + T(\omega^1, \dots, \mathcal{L}_X \omega^r, X_1, \dots, X_s) \\
&\quad + T(\omega^1, \dots, \omega^r, \mathcal{L}_X X_1, \dots, X_s) + \dots + T(\omega^1, \dots, \omega^r, X_1, \dots, \mathcal{L}_X X_s) \\
[\mathcal{L}_X, d] f &= 0.
\end{aligned}$$

2.5 Metrics and causality

The main mathematical object in Einstein's theory of gravity is a *metric*. We recall the definition here.

Definition 2.14. *A* metric *(tensor field) is a* $(0,2)$*-tensor field* $g \in \mathcal{T}^{(0,2)}(\mathcal{M})$ *that satisfies the additional two properties:*

- *(symmetry)* $g(X, Y) = g(Y, X)$ *for all* $X, Y \in \mathcal{T}(\mathcal{M})$*,*
- *(non-degeneracy) if* $g(X, Y) = 0$ *for all* $Y \in \mathcal{T}(\mathcal{M})$*, then* $X = 0$*.*

We say g *is a* Riemannian metric *(tensor field) if moreover*

- *(positive-definiteness) for all* $X \in \mathcal{T}(\mathcal{M})$*,* $g(X, X) \geq 0$ *with equality if and only if* $X = 0$*.*

To define a Lorentzian metric (tensor field) we first make the following observation. The following map is bilinear and non-degenerate:

$$\begin{aligned}
g_x : T_x\mathcal{M} \times T_x\mathcal{M} &\to \mathbb{R}, \\
g_x(v, w) &:= g(V, W)(x),
\end{aligned}$$

with $V, W \in \mathcal{T}(\mathcal{M})$ such that $V_x = v, W_x = w$.[11]

Pick a basis $\{e_\mu\}$ of $T_x\mathcal{M}$. Let G be the matrix with coefficients $G_{\mu\nu} = g_x(e_\mu, e_\nu)$. Then, by the symmetry and non-degeneracy of g, G is invertible and symmetric. This means we can diagonalize G and that its eigenvalues are non-zero.

We define the *signature* (n_+, n_-) of g_x as follows: n_+ is the total number of positive eigenvalues and n_- is the total number of negative eigenvalues of G. This definition only makes sense if the signature is a quantity that is invariant under a choice of basis. That is to say, if we define $\tilde{G}$ as the matrix with coefficients $\tilde{G}_{\mu\nu} = g_x(\tilde{e}_\mu, \tilde{e}_\nu)$, then we need to show that n_+ and n_- of $\tilde{G}$ are the same as n_+ and n_- of G. This follows from *Sylvester's Law of Inertia* (Exercise).[12]

For a Riemannian metric g, we have that $n_- = 0$ for all $x \in \mathcal{M}$.

[11] Exercise: Convince yourself of the existence of V, W and the fact that $g(V, W)(x)$ only depends on V_x and $W_x = v$.

[12] Sylvester's Law of Inertia: if A is an $n \times n$ symmetric matrix with n_+ positive eigenvalues and n_- negative eigenvalues, then, for any $n \times n$ matrix B, the symmetric matrix BAB^T will also have n_+ positive eigenvalues and n_- negative eigenvalues.

Definition 2.15. *A metric g is* Lorentzian *if $n_- = 1$ for all $x \in \mathcal{M}$.*[13]

Not every manifold $\mathcal{M}$ admits a Lorentzian metric. For example, the sphere $\mathbb{S}^4$ does not admit a Lorentzian metric.[14]

[Exercise: Manifolds of the form $\mathcal{M} = \mathbb{R} \times \Sigma$ always admit a Lorentzian metric. *Hint*: You may use that Σ always admits a Riemannian metric.]

2.5.1 Basic concepts of causality

The structure of a Lorentzian metric allows us to talk about *causality* and make sense of whether spacetime points are in the future or past of other spacetime points. We will assume below that the manifold $\mathcal{M}$ is $(n+1)$-dimensional.

Definition 2.16. *A vector $v \in T_x\mathcal{M}$ is:*

- timelike *if $g_x(v,v) < 0$,*
- null *if $g_x(v,v) = 0$ and $v \neq 0$,*
- spacelike *if $g_x(v,v) > 0$.*

Via the isometric identification $T_x\mathcal{M} \cong \mathbb{R}^{n+1}$,[15] *we can identify the union of all null vectors at $x \in \mathcal{M}$ with a double cone in $\mathbb{R}^{n+1}$ minus the vertex:*

$$C_x = \left\{ v \in \mathbb{R}^{n+1} \setminus \{0\} \,|\, v_0^2 = \sum_{i=1}^{n} (v^i)^2 \right\}.$$

The cone C_x is called the lightcone.

The union of all timelike vectors at $x \in \mathcal{M}$ can be identified with the interior of a double cone in $\mathbb{R}^{n+1}$:

$$I_x = \left\{ v \in \mathbb{R}^{n+1} \,|\, v_0^2 > \sum_{i=1}^{n} (v^i)^2 \right\}.$$

The union of all spacelike vectors is the interior of the following double cone:

$$S_x = \left\{ v \in \mathbb{R}^{n+1} \,|\, v_0^2 < \sum_{i=1}^{n} (v^i)^2 \right\}.$$

A vector field $X \in \mathcal{T}(\mathcal{M})$ is timelike/spacelike/null if X_x satisfies exactly one of the three above three conditions for all $x \in \mathcal{M}$.

We similarly say that a curve $\gamma : \mathbb{R} \supseteq I \to \mathcal{M}$ is timelike/spacelike/null if its tangent vector $\gamma'(s) \in T_{\gamma(s)}\mathcal{M}$ is everywhere timelike/spacelike/null, respectively.[16] *We say a curve is causal if $\gamma'(s)$ is timelike or null everywhere and* achronal *if $\gamma'(s)$ is spacelike or null.*

We will frequently refer to timelike curves of timelike vector fields as (idealized) *observers*.

[13] Exercise: If $\mathcal{M}$ connected, $n_- = 1$ at some $x_0 \in \mathcal{M}$ implies that $n_- = 1$ at all $x \in \mathcal{M}$.

[14] By the so-called Hairy Ball Theorem, the sphere $\mathbb{S}^4$ does not admit a vector field that is non-vanishing everywhere. If we could equip $\mathbb{S}^4$ with a Lorentzian metric, then once can show that there must exist a vector field that does not vanish everywhere, which is a contradiction. Conversely, if a manifold admits a non-vanishing vector field X, we can construct the following Lorentzian metric: $g = -2\frac{X^\flat \otimes X^\flat}{g(X,X)} + h$, with h a Riemannian metric on $\mathcal{M}$ and where $X^\flat$ is a 1-form dual to X that we define below.

[15] This amounts to a choice of orthonormal basis $\{e_\mu\}$ of $T_x\mathcal{M}$.

[16] Here, we denoted $\gamma'(s) := [\tilde{\gamma}]$, with $\tilde{\gamma}(\tau) = \gamma(\tau + s)$.

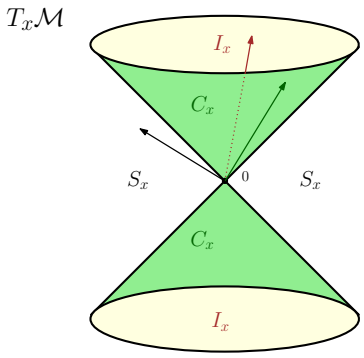


Figure 5: The double lightcone C_x (drawn as a subset of $\mathbb{R}^{n+1}$) and its interior I_x and exterior S_x.

Definition 2.17. *A (smooth)* hypersurface *is a subset $\Sigma \subset \mathcal{M}$, which has the structure of an n-dimensional manifold, such that the inclusion map $\iota : \Sigma \to \mathcal{M}$, $\iota(x) = x$ is a smooth embedding.*[17]

- *If $\iota^* g$* [18] *is a Riemannian metric, we say Σ is a* spacelike hypersurface. *Equivalently, there exists a timelike vector field $N \in \mathcal{T}(\mathcal{M})$ such that $g(N, X)|_{\iota(\Sigma)} = 0$ for all $X \in \mathcal{T}(\mathcal{M})$ with $X_x \in \iota_*(T_x\Sigma)$ (i.e. N is normal to Σ).*
- *If $\iota^* g$ is a Lorentzian metric, we say Σ is a* timelike hypersurface. *Equivalently, there exists a spacelike vector field $N \in \mathcal{T}(\mathcal{M})$ such that $g(N, X)|_{\iota(\Sigma)} = 0$ for all $X \in \mathcal{T}(\mathcal{M})$ with $X_x \in \iota_*(T_x\Sigma)$.*
- *If $\iota^* g$ is a degenerate (i.e. <u>not</u> non-degenerate) tensor field, we say that Σ is a* null hypersurface. *Equivalently, there exists a vector field $L \in \mathcal{T}(\mathcal{M})$, such that $L_{\iota(x)}$ is null for all $x \in \Sigma$, such that $g(L, X)|_{\iota(\Sigma)} \equiv 0$ for all $X \in \mathcal{T}(\mathcal{M})$ with $X_x \in \iota_*(T_x\Sigma)$.*

[Exercise: Show that in the case of a null hypersurface, $L_x \in \iota_*(T_x\Sigma)$. The normal L to Σ is therefore also *tangent* to Σ!]

Definition 2.18. *The* arc length *of a spacelike curve $\gamma : I \to \mathcal{M}$ between $s = s_1$ and $s = s_2$, with $s_1 < s_2$ and $(s_1, s_2) \in I$, is defined as follows:*

$$\ell[\gamma](s_1, s_2) := \int_{s_1}^{s_2} \sqrt{g_{\gamma(s)}(\gamma'(s), \gamma'(s))}\, ds.$$

In the case of a timelike curve γ, we define the proper time *of γ between $s = s_1$ and $s = s_2$, with $s_1 < s_2$ and $(s_1, s_2) \in I$, is defined as follows:*

$$\tau[\gamma](s_1, s_2) := \int_{s_1}^{s_2} \sqrt{-g_{\gamma(s)}(\gamma'(s), \gamma'(s))}\, ds.$$

We need to equip our manifold with an additional structure to be able to define the notion of a spacetime.

[17] This means that $\iota : \Sigma \to \mathcal{M}$ has an injective derivative map at every point and is a homeomorphism to its image with respect to the subspace topology. More generally, a subset $\Sigma \subseteq \mathcal{M}$ is called a (smooth) *embedded k-dimensional submanifold* if it is a k-dimensional manifold and the inclusion map is a smooth embedding.

[18] If $\iota : \Sigma \to \mathcal{M}$ is the inclusion map, which just maps $x \in \Sigma$ to $x \in \Sigma \subset \mathcal{M}$, then the induced metric g_Σ is defined as $g_\Sigma = \iota^* g$ (the pullback makes sense, even though ι is not a diffeomorphism!). In practice, you may use that Σ can locally be expressed as the level set of a function with respect to a coordinate chart. For example $x^0 - h(x^1, \dots, x^n) \equiv 0$ along Σ, so you can use that $\iota^*(dx^0) = \iota^*(\partial_i h dx^i)$, $i = 1, \dots, n$ on Σ to write $g_\Sigma = g_{00}(\partial_i h dx^i) \otimes (\partial_i h dx^i) + g_{j0} dx^j \otimes (\partial_i h dx^i) + g_{ij} dx^i \otimes dx^j$, with $j = 1, \dots, n$ and with the notational convention : $\partial_i h dx^i := \iota^*(\partial_i h dx^i)$.

Definition 2.19. *A Lorentzian manifold $(\mathcal{M}, g)$ is said to be* time-orientable *if there exists a global timelike vector field T. We define a* spacetime *to be the triple*

$$(\mathcal{M}, g, [T]),$$

with $[T] := \{X \in \mathcal{T}(\mathcal{M}) \text{ causal} \,|\, g(X,T) < 0\}$ the time orientation*. We will refer to a spacetime as $(\mathcal{M}, g)$ and omit $[T]$ from the notation.*

Remark 2.2. *In our definition of a "spacetime", we will also include "manifolds with boundary" $\mathcal{M}$. These, strictly speaking, are not "manifolds" because in a neighbourhood of points on their boundary, they look like a half-space $\{(x^1, \ldots, x^n) | x^n \geq 0\} \subset \mathbb{R}^{n+1}$ rather than $\mathbb{R}^{n+1}$. For example, the set $\{x^n \geq 0\} \subset \mathbb{R}^{n+1}$ itself is a manifold-with-boundary, or the set $[0,1]$.*

A time-orientation $[T]$ allows us to define the future and past lightcones C_x^+ and C_x^-:

$$C_x^+ = \{v \in C_x \,|\, g_x(v, T_x) < 0\},$$
$$C_x^- = \{v \in C_x \,|\, g_x(v, T_x) > 0\},$$

with $C_x = C_x^+ \cup C_x^-$.

We similarly define:

$$I_x^+ = \{v \in I_x \,|\, g_x(v, T_x) < 0\},$$
$$I_x^- = \{v \in I_x \,|\, g_x(v, T_x) > 0\},$$

with $I_x = I_x^+ \cup I_x^-$.

Definition 2.20. *Give a time-orientation $[T]$, a causal curve $\gamma : I \to \mathcal{M}$ is* future-directed *if $g_{\gamma(s)}(\gamma'(s), T_{\gamma(s)}) < 0$ for all $s \in I$ and* past-directed *if $g_{\gamma(s)}(\gamma'(s), T_{\gamma(s)}) > 0$.*

Let $\gamma : [a,b] \to \mathcal{M}$ be a curve with $a < b$. Then γ is extendible *if there exist $c \leq a$ and $d \geq b$, with $c < a$ or $d > b$, such that $\widetilde{\gamma} : [c,d] \to \mathcal{M}$ and*

$$\widetilde{\gamma}(s) = \gamma(s) \quad \text{for all } s \in [a,b].$$

If γ is not extendible, we say it is inextendible*.*

If γ is causal, we say it is future-extendible *if $\widetilde{\gamma}$ is causal and $d > b$. We say it is* future-extendible *if $\widetilde{\gamma}$ is causal and $c < a$.*

The structures of a Lorentzian metric and time orientation allow us to determine the regions of $\mathcal{M}$ that can be influenced by a given subset S of $\mathcal{M}$. This is a notion that is fundamentally Lorentzian; it does not have a Riemannian analogue.

Definition 2.21. *Let $S \subseteq \mathcal{M}$. The* causal future/past $J^{\pm}(S)$ *is the following subset of $\mathcal{M}$:*

$$J^{\pm}(S) := S \cup \{p \in \mathcal{M}\,,\, \exists \textbf{ causal} \text{ future-/past-directed curve } \gamma : [0,1] \to \mathcal{M}, \text{ with } \gamma(0) \in S \text{ and } \gamma(1) = p\}\,.$$

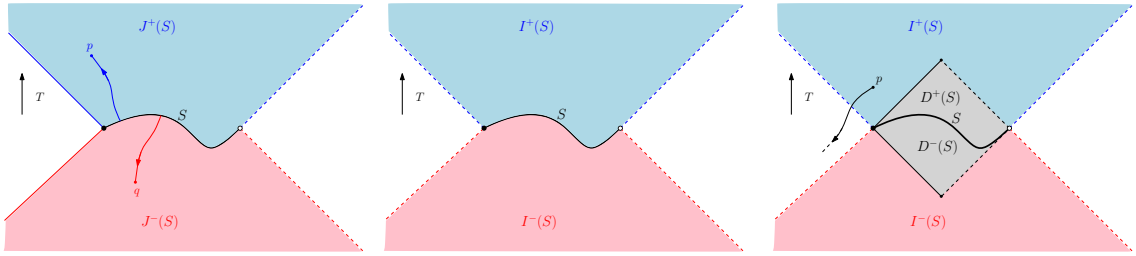
The chronological future/past $I^{\pm}(S)$ *of S is the following subset of $\mathcal{M}$:*

$$I^{\pm}(S) := \{p \in \mathcal{M}\,,\, \exists \textbf{ timelike} \text{ future-/past-directed curve } \gamma : [0,1] \to \mathcal{M}, \text{ with } \gamma(0) \in S \text{ and } \gamma(1) = p\}\,.$$

Note that the constant curve $\gamma(s) = p$ for all s is not timelike, null or spacelike.

The future/past domain of dependence $D^+(S)$ *of S is the following subset of $\mathcal{M}$:*

$$D^{\pm}(S)$$
$$:= \{p \in \mathcal{M}\,,\, \underline{\text{all}} \text{ causal past/future directed, past/future inextendible curves containing } p \text{ intersect } S\}\,.$$



(a) The causal future and past $J^{\pm}(S)$ of a set $S \subset \mathbb{R}^{1+1}$, with $(\mathbb{R}^{1+1}, m)$ the 1+1-dimensional Minkowski spacetime.

(b) The chronological future and past $I^{\pm}(S)$ of a set $S \subset \mathbb{R}^{1+1}$, with $(\mathbb{R}^{1+1}, m)$ the 1+1-dimensional Minkowski spacetime.

(c) The future/past domains of dependence $D^{\pm}(S)$ of a set $S \subset \mathbb{R}^{1+1}$, with $(\mathbb{R}^{1+1}, m)$ the 1+1-dimensional Minkowski spacetime. Here, $p \in J^+(S)$ but $p \notin D^+(S) \cup D^-(S)$.

Figure 6: Examples of chronological/causal futures/pasts and future/past domains of dependence in the 2-dimensional Minkowski spacetime $(\mathbb{R}^{1+1}, m)$, with $m = -dt \otimes dt + dx \otimes dx$.

Observe that $S \subseteq J^{\pm}(S)$, by definition and we also have that $S \subseteq D^{\pm}(S)$. However, we do not necessarily have that $S \subseteq I^{\pm}(S)$. Indeed, we could even have that $I^{\pm}(S) \cap S = \emptyset$:

Definition 2.22. *A set $S \subset \mathcal{M}$ is* achronal *if $I^+(S) \cap S = \emptyset$.*

[Exercise: Show that spacelike or null hypersurfaces are achronal, but timelike hypersurfaces are not.]

A particularly important class of spacetimes are *globally hyperbolic* spacetimes. As we will later see, their relevance is motivated by the fact that they constitute the spacetimes that can be obtained from the time evolution of initial data in the context of the initial value problem for the Einstein equations.

Definition 2.23. *A* Cauchy hypersurface Σ *is an achronal hypersurface of a spacetime $(\mathcal{M}, g, [T])$ such that*

$$D^+(\Sigma) \cup D^-(\Sigma) = \mathcal{M}.$$

If a spacetime admits a Cauchy hypersurface, we say it is globally hyperbolic.

If $D^+(\Sigma) \cup D^-(\Sigma) \neq \mathcal{M}$, then the future boundary of $D^+(\Sigma)$ in $\mathcal{M}$,

$$\mathcal{CH}^+(\Sigma) := \overline{D^+(\Sigma)} \setminus \left(I^-(D^+(\Sigma)) \cap \overline{D^+(\Sigma)} \right)$$

is called the future Cauchy horizon of Σ *and the past boundary of $D^-(\Sigma)$ in $\mathcal{M}$,*

$$\mathcal{CH}^-(\Sigma) := \overline{D^-(\Sigma)} \setminus \left(I^+(D^-(\Sigma)) \cap \overline{D^-(\Sigma)} \right)$$

is called the past Cauchy horizon of Σ.

[Exercise: Show that $\mathcal{CH}^+(\Sigma)$ must be achronal. *Hint*: Show that the boundary ∂S of $S = D^+(\Sigma) \cup I^-(D^+(\Sigma))$ is achronal by using that $I^-(S)$ is an open set so $I^-(S) \subset S \setminus \partial S$.]

The presence of a future Cauchy horizon of a achronal hypersurface Σ indicates the *end of predictability* of the spacetime arising from initial data on Σ.

Lemma 2.1. *If Σ is a Cauchy hypersurface of a spacetime $(\mathcal{M}, g)$, then any inextendible timelike curve in $\mathcal{M}$ must intersect Σ exactly once.*

Proof. Let γ be an inextendible timelike curve. Suppose γ intersects Σ at times $s, t \in \Sigma$, with $s \neq t$. Then $\gamma(s) \in I^+(\gamma(t))$ or $\gamma(t) \in I^+(\gamma(s))$, which is in contradiction with achronality. □

[Exercise: Explain whether any inextendible null curve needs to intersect a Cauchy hypersurface exactly once.]

It turns out for globally hyperbolic spacetimes, we must have: $\mathcal{M} \cong \mathbb{R} \times \Sigma$, with Σ a Cauchy hypersurface.

Theorem 2.2 (A spacetime splitting theorem)**.** *Let $(\mathcal{M}, g, [T])$ be a globally hyperbolic spacetime, with $\mathcal{M}$ a smooth manifold and g a smooth Lorentzian metric. Let Σ be a Cauchy hypersurface.*

Then $\mathcal{M}$ is diffeomorphic to $\mathbb{R} \times \Sigma$.

Proof. (*non-examinable*) By the time-orientability property, there exists a smooth timelike vector field T. At each $x \in \Sigma$, consider the inextendible curve $\tilde{\gamma}_x : I_x \to \mathcal{M}$ satisfying $\tilde{\gamma}_x(0) = x$ and $\dot{\tilde{\gamma}} = T$. We can then reparametrize $\tilde{\gamma}$ to obtain $\gamma_x : \mathbb{R} \to \mathcal{M}$, with $\gamma_x(0) = x$ and $\dot{\gamma}$ proportional to T. Since $\mathcal{M}$ is smooth, $x \mapsto \gamma_x(t)$ is smooth for all $t \in \mathbb{R}$.

Now let $\Phi : \mathbb{R} \times \Sigma \to \mathcal{M}$ be the smooth map defined as $\Phi(t, x) = \gamma_x(t)$. We will now show that this map is bijective.

- "*Surjectivity*": For any $p \in \mathcal{M}$, the timelike curve obtained by flowing along T must intersect Σ at some $x \in \Sigma$, using the Cauchy hypersurface property of Σ. After possible reparametrization, we therefore have that $p = \gamma_x(t)$ for some $t \in \mathbb{R}$.
- "*Injectivity*": Let $p = \gamma_x(t) = \gamma_y(s)$, with $s, t \in \mathbb{R}$ and $x, y \in \Sigma$. By Lemma 2.1, we must have that $x = y$. Suppose $s \neq t$, then there exists a closed timelike curve emanating from p. Suppose that this curve does not intersect Σ. Then this is in contradiction with the Cauchy surface property of Σ. Suppose the curve intersects Σ at two points. Then this is in contradiction with Lemma 2.1. Finally, suppose the curve touches Σ at once point. Then we can perturb it a little bit to obtain a closed timelike curve that either does not intersect Σ, which is a case that leads to a contradiction, as we have shown above.

  The inverse map Φ^{-1} corresponds to flowing along the integral curves of a rescaled T and therefore also is smooth. Hence, Φ is a diffeomorphism between $\mathbb{R} \times \Sigma$ and $\mathcal{M}$.

[Exercise: Suppose Σ and Σ' are two Cauchy hypersurfaces. Show that Σ is diffeomorphic to Σ'.] □

2.5.2 Musical isomorphisms: "raising and lowering indices"

On a manifold equipped with a metric $(\mathcal{M}, g)$, we can use the metric g to identify vector fields and 1-forms.

Let $X \in \mathcal{T}(\mathcal{M})$. Then we define

$$X^\flat := g(X, \cdot) \in \Omega^1(\mathcal{M}).$$

The symbol $\flat$ is called "flat" and is inspired by musical notation. Indeed $g(X, \cdot) : \mathcal{T}(\mathcal{M}) \to C^\infty(\mathcal{M})$ is linear over $C^\infty(\mathcal{M})$ so it defines a 1-form. We therefore have the existence of a map:

$$\flat : \mathcal{T}(\mathcal{M}) \to \Omega^1(\mathcal{M})$$

such that $\flat(X) = X^\flat$. Note that the associated maps $\flat_x : T_x\mathcal{M} \to T_x^*\mathcal{M}$, $\flat_x(v) = g_x(v, \cdot)$, with $x \in \mathcal{M}$, are linear maps between two vector spaces of equal dimension. Furthermore, by the non-degeneracy of g, $\flat_x$ must be injective. By the rank–nullity theorem of linear algebra, we therefore have that:

$$\dim \ker \flat_x + \dim \operatorname{ran} \flat_x = \dim T_x\mathcal{M}$$

and hence $\dim \operatorname{ran} \flat_x = \dim T_x^*\mathcal{M}$, so $\flat_x$ must also be surjective. It therefore has a well-defined inverse $\sharp_x : T_x^*\mathcal{M} \to T_x\mathcal{M}$ at each $x \in \mathcal{M}$ and we can make sense of the associated map:

$$\sharp : \Omega^1(\mathcal{M}) \to \mathcal{T}(\mathcal{M}).$$

For $\omega \in \Omega^1(\mathcal{M})$, we denote $\omega^\sharp := \sharp(\omega)$. The maps $\flat$ and $\sharp$ are called *musical isomorphisms*.

We can extend $\flat$ and $\sharp$ to act on $\mathcal{T}^{(r,s)}(\mathcal{M})$ as follows:

Definition 2.24. *The maps $\flat^{(i)} : \mathcal{T}^{(r,s)}(\mathcal{M}) \to \mathcal{T}^{(r-1,s+1)}(\mathcal{M})$ (i-th index lowering map) and $\sharp^{(j)} : \mathcal{T}^{(r,s)}(\mathcal{M}) \to \mathcal{T}^{(r+1,s-1)}(\mathcal{M})$ (j-th index raising map) can be defined as follows:*

$$(\flat^{(i)}T)(\omega^1, \ldots, \omega^{r-1}, X_1, \ldots, X_{s+1}) := T(\omega^1, \ldots, \omega^{i-1}, \flat(X_i), \omega^i, \ldots, \omega^{r-1}, X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_{s+1}),$$
$$(\sharp^{(j)}T)(\omega^1, \ldots, \omega^{r+1}, X_1, \ldots, X_{s-1}) := T(\omega^1, \ldots, \omega^{j-1}, \omega^{j+1}, \ldots, \omega^{r+1}, X_1, \ldots, X_{j-1}, \sharp(\omega_j), X_j, \ldots, X_{s-1}).$$

It is instructive to investigate the above maps with respect to a coordinate chart $(U, \{x^\mu\})$. We can then express:

$$g|_U = g_{\mu\nu} dx^\mu \otimes dx^\nu =: g_{\mu\nu} dx^\mu dx^\nu,$$

with $g_{\mu\nu} \in C^\infty(U)$. Let $X = X^\alpha \partial_{x^\alpha} \in \mathcal{T}(U)$ and $Y = Y^\alpha \partial_{x^\alpha} \in \mathcal{T}(U)$. Then we can express in U:

$$X^\flat(Y) = g_{\mu\nu} X^\mu Y^\nu.$$

hence, $X^\flat = (g_{\nu\mu} X^\nu) dx^\mu = (g_{\mu\nu} X^\nu) dx^\mu$. In particular,

$$\left(\frac{\partial}{\partial x^\mu}\right)^\flat = g_{\mu\beta} dx^\beta.$$

We will denote the components of $X^\flat$ with an index in the subscript, i.e.

$$X_\mu := g_{\mu\nu} X^\nu.$$

For this reason, it is said that the map $\flat$, "lowers indices".

We now define $\mathfrak{g} \in \mathcal{T}^{(2,0)}(\mathcal{M})$ in the following way: for all $\omega, \theta \in \Omega^1(\mathcal{M})$,

$$\mathfrak{g}(\omega, \theta) := g(\omega^\sharp, \theta^\sharp).$$

We can express in U:

$$\mathfrak{g} = \mathfrak{g}^{\mu\nu} \partial_{x^\mu} \otimes \partial_{x^\nu} =: \mathfrak{g}^{\mu\nu} \partial_{x^\mu} \partial_{x^\nu}.$$

By the definition of $\mathfrak{g}$, we have that:

$$g_{\mu\nu} = g(\partial_{x^\mu}, \partial_{x^\nu}) = \mathfrak{g}((\partial_{x^\mu})^\flat, (\partial_{x^\nu})^\flat) = g_{\mu\alpha} g_{\nu\beta} \mathfrak{g}(dx^\alpha, dx^\beta) = g_{\mu\alpha} g_{\nu\beta} \mathfrak{g}^{\alpha\beta}$$

and hence

$$g_{\nu\beta} \mathfrak{g}^{\alpha\beta} = \delta^\alpha_\nu.$$

This means that the matrix with components $\mathfrak{g}^{\mu\nu}$ is the inverse of the matrix with components $g_{\mu\nu}$. We will therefore use the notation:

$$g^{-1} := \mathfrak{g}.$$

We can use g^{-1} to "raise indices". Let $\omega, \theta \in \Omega^1(\mathcal{M})$. We have that

$$\omega^\sharp(\theta)|_U = g^{-1}(\omega, \theta) = (g^{-1})^{\mu\nu} \omega_\mu \theta_\nu.$$

Define $\omega^\mu := (g^{-1})^{\mu\nu} \omega_\nu$.

Let $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$. With respect to the coordinate chart $(U, \{x^\mu\})$, we can write:

$$T = T^{\mu_1 \dots \mu_r}{}_{\nu_1 \dots \nu_s} \frac{\partial}{\partial x^{\mu_1}} \otimes \dots \otimes \frac{\partial}{\partial x^{\mu_r}} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_s}.$$

We then define:

$$(\flat^{(i)} T)^{\mu_1 \dots \mu_{r-1}}{}_{\nu_1 \dots \nu_{s+1}} = g_{\alpha \nu_i} T^{\mu_1 \dots \mu_{i-1} \alpha \mu_i \dots \mu_r}{}_{\nu_1 \dots \nu_{i-1} \nu_{i+1} \dots \nu_s} =: T^{\mu_1 \dots \mu_{i-1}}{}_{\nu_i}{}^{\mu_i \dots \mu_{r+1}}{}_{\nu_1 \dots \nu_{i-1} \nu_{i+1} \dots \nu_s}$$

and

$$(\sharp^{(j)} T)^{\mu_1 \dots \mu_{r+1}}{}_{\nu_1 \dots \nu_{s-1}} = (g^{-1})^{\alpha \mu_j} T^{\mu_1 \dots \mu_{j-1} \mu_{j+1} \dots \mu_{r+1}}{}_{\nu_1 \dots \nu_{j-1} \alpha \nu_j \dots \nu_{s-1}} =: T^{\mu_1 \dots \mu_{j-1} \mu_{j+1} \dots \mu_{r+1}}{}_{\nu_1 \dots \nu_{j-1}}{}^{\mu_j}{}_{\nu_{j+1} \dots \nu_{s-1}}.$$

2.5.3 The Levi–Civita connection and covariant derivatives

A priori, there is no way to compare vectors in different tangent spaces on a general manifold $\mathcal{M}$. The Levi–Civita connection is a construction that provides a natural way to *transport* vectors from one tangent space to the other in the setting of manifolds equipped with a metric.

Definition 2.25. *An* affine connection *is a bilinear (over $\mathbb{R}$) map $\nabla : \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \to \mathcal{T}(\mathcal{M})$ with $(X, Y) \mapsto \nabla_X Y$, such that for all $X, Y \in \mathcal{T}(\mathcal{M})$ and $f \in C^\infty(\mathcal{M})$*

1. *(linearity over $C^\infty(\mathcal{M})$) $\nabla_{fX} Y = f \nabla_X Y$,*
2. *(Leibniz rule) $\nabla_X (fY) = X(f) Y + f \nabla_X Y$.*

A Levi-Civita connection *with respect to a metric g is an affine connection that satisfies additionally: for all $X, Y, Z \in \mathcal{T}(\mathcal{M})$:*

1. *(metric-preserving) $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$,*
2. *(torsion-free) $\nabla_X Y - \nabla_Y X = [X, Y]$.*

We define (with some abuse of notation) the corresponding covariant derivative *as the map:*

$$\nabla : \mathcal{T}(\mathcal{M}) \to \mathcal{T}^{(1,1)}(\mathcal{M}),$$
$$(\nabla Y)(\omega, X) := \omega(\nabla_X Y).$$

Note that ∇Y is a well-defined tensor field because it is bilinear over $C^\infty(\mathcal{M})$.

[Exercise: Verify this by considering $(\nabla Y)(fX + Z, h\omega + \alpha)$ for $f, h \in C^\infty(\mathcal{M})$, $X, Y, Z \in \mathcal{T}(\mathcal{M})$ and $\omega, \alpha \in \Omega^1(\mathcal{M})$.]

We state without proof (the proof can be found in several of the suggested texts in the first section of these notes):

Theorem 2.3. *Let $(\mathcal{M}, g)$ be a manifold equipped with a metric. Then there exists a unique Levi–Civita connection with respect to g and it satisfies the following identity: for all $X, Y, Z \in \mathcal{T}(\mathcal{M})$.*

$$g(\nabla_X Y, Z) = \frac{1}{2} \left[X(g(Y, Z)) + Y(g(Z, X)) - Z(g(Y, X)) + g([X, Y], Z) - g([X, Z], Y) - g([Y, Z], X) \right]. \tag{2.3}$$

We can express the Levi–Civita connection as follows with respect to a coordinate chart $(U, \{x^\mu\})$:

$$(\nabla_{\partial_\mu} \partial_\nu)^\sigma = \frac{1}{2} (g^{-1})^{\sigma \rho} \left(\partial_\mu g_{\nu \rho} + \partial_\nu g_{\mu \rho} - \partial_\rho g_{\mu \nu} \right) =: \Gamma^\sigma_{\mu \nu}.$$

The expressions $\Gamma^\sigma_{\mu \nu}$ are called the *Christoffel symbols*. Hence,

$$(\nabla_X Y)^\sigma = (\nabla_{X^\mu \partial_\mu} (Y^\nu \partial_\nu))^\sigma = X^\mu \partial_\mu Y^\sigma + \Gamma^\sigma_{\mu \nu} X^\mu Y^\nu,$$
$$(\nabla Y)^\sigma_\mu = \partial_\mu Y^\sigma + \Gamma^\sigma_{\mu \nu} Y^\nu.$$

We would like to extend ∇ to act on more general tensor fields: $\nabla : \mathcal{T}^{(r,s)}(\mathcal{M}) \to \mathcal{T}^{(r,s+1)}(\mathcal{M})$.

Definition 2.26. *Define $\nabla f := df$ for $f \in C^\infty(\mathcal{M}) = T^{(0,0)}(\mathcal{M})$. This implies that $C^\infty(\mathcal{M}) \ni \nabla_X f = df(X) = X(f)$. Write $\nabla_\mu f := \partial_\mu f$.*

Consider $\omega \in \Omega^1(\mathcal{M}) = \mathcal{T}^{(0,1)}(\mathcal{M})$ and define $\nabla\omega$ as follows:

$$(\nabla\omega)(X,Y) := X(\omega(Y)) - \omega(\nabla_X Y)$$

for all $X, Y \in \mathcal{T}(\mathcal{M})$. Note that linearity over $C^\infty(\mathcal{M})$ in the first argument follows immediately, and linearity in the second argument follows from:

$$(\nabla\omega)(X,fY) = \nabla_X(\omega(fY)) - \omega(\nabla_X(fY)) = f(\nabla\omega)(X,Y) + (\nabla_X f)\omega(Y) - (\nabla_X f)\omega(Y) = f(\nabla\omega)(X,Y).$$

This definition ensures that ∇ is compatible with tensor contraction. With respect to a coordinate chart $(U, \{x^\mu\})$, we obtain for all $0 \le \nu \le n+1$

$$\nabla_\nu(\omega_\rho Y^\rho) = (\nabla\omega)_{\nu\rho} Y^\rho + \omega_\rho (\nabla Y)^\rho{}_\nu.$$

We can therefore also express:

$$(\nabla\omega)_{\nu\mu} = \partial_\nu \omega_\mu - \Gamma^\rho_{\mu\nu}\omega_\rho.$$

We will use the following notational conventions:

$$\nabla_\nu \omega_\mu := (\nabla\omega)_{\nu\mu},$$
$$\nabla_\nu Y^\mu := (\nabla Y)^\mu{}_\nu$$

We now extend ∇ as a map on general tensor fields: $\nabla : \mathcal{T}^{(r,s)}(\mathcal{M}) \to \mathcal{T}^{(r,s+1)}(\mathcal{M})$ is defined as

$$(\nabla T)(X, \omega^1, \ldots, \omega^r, Y_1, \ldots, Y_s) := (\nabla_X T)(\omega^1, \ldots, \omega^r, Y_1, \ldots, Y_s),$$

with

$$\begin{aligned}(\nabla_X T)(\omega^1, \ldots, \omega^r, Y_1, \ldots, Y_s) &= X(T(\omega^1, \ldots, \omega^r, Y_1, \ldots, Y_s)) \\ &\quad - T(\nabla_X\omega^1, \ldots, \omega^r, Y_1, \ldots, Y_s) - \ldots - T(\nabla_X\omega^1, \ldots, \nabla_X\omega^r, Y_1, \ldots, Y_s) \\ &\quad - T(\omega^1, \ldots, \omega^r, \nabla_X Y_1, \ldots, Y_s) - \ldots - T(\omega^1, \ldots, \omega^r, Y_1, \ldots, \nabla_X Y_s).\end{aligned}$$

[Exercise: Show that $\nabla g = 0$.]

With respect to a coordinate chart $(U, \{x^\mu\})$, we obtain:

$$\begin{aligned}\nabla_\mu T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} &= \partial_\mu T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} - \Gamma^\rho_{\mu\mu_1} T^{\nu_1\ldots\nu_r}{}_{\rho\ldots\mu_s} - \Gamma^\rho_{\mu\mu_2} T^{\nu_1\ldots\nu_r}{}_{\mu_1\rho\mu_3\ldots\mu_s} - \ldots - \Gamma^\rho_{\mu\mu_s} T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_{s-1}\rho} \\ &\quad + \Gamma^{\nu_1}_{\mu\rho} T^{\rho\nu_2\ldots\nu_r}{}_{\mu_1\ldots\mu_s} + \Gamma^{\nu_2}_{\mu\rho} T^{\nu_1\rho\nu_3\ldots\nu_r}{}_{\mu_1\ldots\mu_s} + \ldots + \Gamma^{\nu_r}_{\mu\rho} T^{\nu_1\ldots\nu_{r-1}\rho}{}_{\mu_1\ldots\mu_s}.\end{aligned}$$

Proposition 2.4. *Covariant derivatives and Lie derivatives are related in the following way. Let $(U, \{x^\mu\})$ be an arbitrary coordinate chart. Let $X \in \mathcal{T}(\mathcal{M})$ and $T \in \mathcal{T}^{(r,s)}(\mathcal{M})$, then*

$$\begin{aligned}(\mathcal{L}_X T)^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} &= X^\alpha \nabla_\alpha T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_s} - T^{\alpha\nu_2\ldots\nu_r}{}_{\mu_1\ldots\mu_s}\nabla_\alpha X^{\nu_1} - \ldots - T^{\nu_1\ldots\nu_{r-1}\alpha}{}_{\mu_1\ldots\mu_s}\nabla_\alpha X^{\nu_r} \\ &\quad + T^{\nu_1\ldots\nu_r}{}_{\alpha\mu_2\ldots\mu_s}\nabla_{\mu_1} X^\alpha + \ldots + T^{\nu_1\ldots\nu_r}{}_{\mu_1\ldots\mu_{s-1}\alpha}\nabla_{\mu_s} X^\alpha\end{aligned}$$

The Lie derivative plays an important role when considering *isometries*.

Definition 2.27. *An isometry between manifolds equipped with metric, $(\mathcal{M}, g)$ and $(\widetilde{\mathcal{M}}, \widetilde{g})$, is a diffeomorphism $\psi : \mathcal{M} \to \widetilde{\mathcal{M}}$ satisfying*

$$\psi^* \widetilde{g} = g.$$

Definition 2.28. *A vector field $X \in \mathcal{T}(\mathcal{M})$ is called a* Killing vector field *if it satisfies:*

$$\mathcal{L}_X g = 0.$$

In particular, if $X(f) = \frac{d}{ds}|_{s=0}(f \circ \psi_s)$ for a 1-parameter group of diffeomorphisms $\psi_s : \mathcal{M} \to \mathcal{M}$ that are isometries, i.e. $\psi_s^* g = g$, then X is a Killing vector field.

[Exercise: Show that $\mathcal{L}_X g = 0$ implies the following equation with respect to an arbitrary coordinate chart:

$$\nabla_\mu X_\nu + \nabla_\nu X_\mu = 0.]$$

2.5.4 Geodesics

Geodesics can be defined compactly using the covariant derivative.

Definition 2.29. *A curve $\gamma : \mathbb{R} \supseteq I \to \mathbb{R}$, with I an interval, is an* affinely-parametrized geodesic*, if an extension $X \in \mathcal{T}(\mathcal{M})$ of the tangent vectors to the curve, $\gamma'(s) \in T_{\gamma(s)}\mathcal{M}$, away from the curve γ:*

$$\nabla_X X = 0.$$

We will refer to curves which arise as reparametrizations of affinely-parametrized geodesics as simply geodesics.

[Exercise: Convince yourself that the above definition is invariant under the choice of extension X.]

With respect to a coordinate chart $(U, \{x^\mu\})$, we can write

$$(\gamma'')^\sigma + \Gamma^\sigma_{\mu\nu} \gamma'^\mu \gamma'^\nu = 0,$$

where we can express $\gamma'^\mu(s) = (x^\mu \circ \gamma)'(s)$ and we define $(\gamma'')^\mu(s) = (x^\mu \circ \gamma)''(s)$.

[Exercise: 1) Show that for affinely parametrized spacelike geodesics γ there exists an $a \in \mathbb{R} \setminus \{0\}$ such that $\ell[\gamma](s_1, s_2) = a(s_2 - s_2)$. This explains the term "affinely parametrized".] 2) Show that for affinely parametrized timelike geodesics γ there exists an $a \in \mathbb{R} \setminus \{0\}$ such that $\tau[\gamma](s_1, s_2) = a(s_2 - s_2)$.]

It can be shown that spacelike geodesics are *locally arc-length minimizing* and timelike geodesics are *locally proper time maximizing*. Null geodesics do not admit such a variational interpretation.

Note that the spacelike/timelike/null character of a geodesic is determined by the spacelike/timelike/null character of a single tangent vector. Indeed, let $\gamma : I \to \mathcal{M}$ be an affinely-parametrized geodesics, then

$$\frac{d}{ds} g_{\gamma(s)}(\gamma'(s), \gamma'(s)) = [X(g(X, X)](\gamma(s)) = 2g(\nabla_X X, X)(\gamma(s)) = 0,$$

so the "inner product" $g_{\gamma(s)}(\gamma'(s), \gamma'(s))$ is conserved along γ.

The "inner product" of a Killing vector field with the tangent to an affinely-parametrized geodesic is also conserved along the geodesic:

Proposition 2.5. *Let Y be a Killing vector field. Let X be a vector field such that $X|_{\gamma(s)} = \gamma'(s)$, with $\gamma : I \to \mathcal{M}$ an affinely parametrized geodesic. Then*

$$X(g(X, Y))|_{\gamma(I)} = 0.$$

Proof. Exercise □

Definition 2.30. *Let $\gamma : I \to \mathcal{M}$ be a curve and let $X \in \mathcal{T}(\mathcal{M})$ satisfy $X_{\gamma(s)} = \gamma'(s)$. We say a vector field $V \in \mathcal{T}(\mathcal{M})$ is the* parallel transport *of a vector $v \in T_{\gamma(0)}\mathcal{M}$ along γ if V satisfies:*

$$\begin{aligned} V_{\gamma(0)} &= v, \\ (\nabla_X V)_{\gamma(s)} &= 0 \quad \text{for all } s \in I. \end{aligned}$$

2.6 The Riemann tensor, Ricci tensor and Ricci scalar

The Riemann tensor field is a geometric object that captures the "intrinsic curvature" of a manifold. That is to say, it provides a notion of curvature that is independent of how the manifold may be embedded inside some bigger manifold. It is the central object of study in general relativity.

Definition 2.31. *We define the* Riemann tensor field *as the following map:*

$$\begin{aligned}\mathrm{Riem}[g] &: \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \times \mathcal{T}(\mathcal{M}) \times \Omega^1(\mathcal{M}) \to C^\infty(\mathcal{M}) \\ \mathrm{Riem}[g](Z, X, Y, \omega) &= \omega\left(\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z\right).\end{aligned}$$

We will also refer to this map as the Riemann (curvature) tensor.

[Exercise: Show that $\mathrm{Riem}[g]$ is multilinear over $C^\infty(\mathcal{M})$ and hence, it is a smooth tensor field: $\mathrm{Riem}[g] \in \mathcal{T}^{(1,3)}(\mathcal{M})$.]

With respect to a coordinate chart $(U, \{x^\mu\})$ we obtain:

$$\begin{aligned}R_{\mu\nu\rho}{}^\sigma := \mathrm{Riem}[g](\partial_\mu, \partial_\nu, \partial_\rho, dx^\sigma) &= dx^\sigma(\nabla_{\partial_\nu}\nabla_{\partial_\rho}\partial_\mu - \nabla_{\partial_\rho}\nabla_{\partial_\nu}\partial_\mu) \\ &= dx^\sigma(\nabla_{\partial_\nu}(\Gamma^\alpha_{\rho\mu}\partial_\alpha) - \nabla_{\partial_\rho}(\Gamma^\alpha_{\nu\mu}\partial_\alpha)) \\ &= dx^\sigma(\partial_\nu\Gamma^\alpha_{\rho\mu}\partial_\alpha - \partial_\rho\Gamma^\alpha_{\nu\mu}\partial_\alpha + \Gamma^\alpha_{\rho\mu}\Gamma^\beta_{\nu\alpha}\partial_\beta - \Gamma^\alpha_{\nu\mu}\Gamma^\beta_{\rho\alpha}\partial_\beta) \\ &= \partial_\nu\Gamma^\sigma_{\rho\mu} - \partial_\rho\Gamma^\sigma_{\nu\mu} + \Gamma^\alpha_{\rho\mu}\Gamma^\sigma_{\nu\alpha} - \Gamma^\alpha_{\nu\mu}\Gamma^\sigma_{\rho\alpha}.\end{aligned}$$

By acting with $\flat$, we obtain a $(0,4)$-tensor field with components $R_{\mu\nu\rho\sigma} = g_{\sigma\alpha}R_{\mu\nu\rho}{}^\alpha$. We will also refer to this tensor field as "the Riemann tensor".

Definition 2.32. *The* Ricci tensor *(field) is a* $(0,2)$*-tensor field* $\mathrm{Ric}[g]$ *with components* $R_{\mu\nu}$ *with respect to an arbitrary coordinate chart satisfying:*[19]

$$R_{\mu\nu} = R_{\mu\sigma\nu}{}^\sigma = (g^{-1})^{\sigma\rho}R_{\mu\sigma\nu\rho}.$$

Definition 2.33. *The* Ricci scalar $R[g] \in C^\infty(\mathcal{M})$ *is defined as follows:*

$$R := \mathrm{tr}\,\mathrm{Ric}[g] := (g^{-1})^{\mu\nu}R_{\mu\nu}.$$

The Riemann tensor and its tensor contractions ("summing over upper and lower indices") are natural quantities to consider on $(\mathcal{M}, g)$ because they are invariant under isometries. Indeed, consider a diffeomorphism $\psi : \mathcal{M} \to \widetilde{\mathcal{M}}$ where $\widetilde{\mathcal{M}}$ is equipped with a metric $\widetilde{g}$. Then it can be shown that

$$\mathrm{Riem}[\psi^*\widetilde{g}] = \psi^*(\mathrm{Riem}[\widetilde{g}]), \tag{2.4}$$
$$\mathrm{Ric}[\psi^*\widetilde{g}] = \psi^*(\mathrm{Ric}[\widetilde{g}]), \tag{2.5}$$
$$R[\psi^*\widetilde{g}] = \psi^*(R[\widetilde{g}]) = R[\widetilde{g}] \circ \psi. \tag{2.6}$$

If in addition ψ is an isometry, then we can use that $\psi^*\widetilde{g} = g$ to obtain.

$$\begin{aligned}\mathrm{Riem}[g] &= \psi^*(\mathrm{Riem}[\widetilde{g}]), \\ \mathrm{Ric}[g] &= \psi^*(\mathrm{Ric}[\widetilde{g}]), \\ R[g] &= \psi^*(R[\widetilde{g}]) = R[\widetilde{g}] \circ \psi.\end{aligned}$$

The Riemann tensor enjoys various symmetry properties.

[19] Exercise: Show that $\mathrm{Ric}[g]$ is independent of the particular choice of coordinate chart and is therefore uniquely defined on the whole of $\mathcal{M}$.

Proposition 2.6. *With respect to an arbitrary coordinate chart, the components of the Riemann tensor satisfy:*

(i) $R_{(\mu\nu)\rho}{}^{\sigma} = 0$,

(ii) $R_{[\mu\nu\rho]}{}^{\sigma} = 0$,

(iii) $R_{\mu\nu\rho\sigma} = R_{\rho\sigma\mu\nu}$,

(iv) $\nabla_{[\alpha} R_{\sigma\mu]\nu\rho} = 0$ *(Bianchi identity).*

Here, we take the round brackets to mean a symmetrization over the indices: for example, $T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$.

From the above symmetry properties, we can deduce symmetry properties at the level of the Ricci tensor and Ricci scalar. We first introduce the Einstein tensor.

Definition 2.34. *The* Einstein tensor *(field)* $G[g] \in \mathcal{T}^{(0,2)}(\mathcal{M})$ *is defined as follows:*

$$G[g] = \mathrm{Ric}[g] - \frac{1}{2} R[g] g.$$

Corollary 2.7. *The Ricci and Einstein tensors are symmetric and, with respect to an arbitrary coordinate chart,* $G[g]$ *satisfies the* contracted Bianchi identity

$$\nabla^{\mu} G_{\mu\nu} = 0.$$

Proof. Symmetry of $G[g]$ follows immediately from symmetry of $\mathrm{Ric}[g]$. Recall,

$$R_{\mu\nu} = R_{\mu\sigma\nu}{}^{\sigma} = (g^{-1})^{\sigma\alpha} R_{\mu\alpha\nu\sigma}.$$

By property (iii) of Proposition 2.6, we have that

$$(g^{-1})^{\sigma\alpha} R_{\mu\alpha\nu\sigma} = (g^{-1})^{\sigma\alpha} R_{\nu\sigma\mu\alpha} = R_{\nu\mu}.$$

To derive the contracted Bianchi identity, we combine (i) and (iv) to obtain:

$$\nabla_{\alpha} R_{\sigma\mu\nu\rho} + \nabla_{\sigma} R_{\mu\alpha\nu\rho} + \nabla_{\mu} R_{\alpha\sigma\nu\rho} = 0.$$

Now we contract the above identity with $(g^{-1})^{\sigma\nu}(g^{-1})^{\mu\rho}$ and use that $\nabla g = 0$ to obtain:

$$\begin{aligned} 0 &= (g^{-1})^{\sigma\nu}(g^{-1})^{\mu\rho}(\nabla_{\alpha} R_{\sigma\mu\nu\rho} + \nabla_{\sigma} R_{\mu\alpha\nu\rho} + \nabla_{\mu} R_{\alpha\sigma\nu\rho}) \\ &= \nabla_{\alpha} R - \nabla^{\nu} R_{\alpha\nu} - \nabla^{\rho} R_{\alpha\rho} \\ &= \nabla_{\alpha} R - 2\nabla^{\nu} R_{\alpha\nu} \\ &= -2\nabla^{\nu} G_{\alpha\nu}. \end{aligned}$$

□

2.7 The Einstein equations

The Einstein equations in $(\mathcal{M}, g)$ (with respect to physical units in which $c = G = 1$):

$$G[g] = 8\pi \mathbb{T}.$$

Here, $\mathbb{T} \in \mathcal{T}^{(0,2)}(\mathcal{M})$ is the *stress-energy tensor* (field) or *energy-momentum tensor* (field), whose expression depends on the physical matter that one is interested in studying. By the contracted Bianchi identity $\nabla^{\mu} G_{\mu\nu} = 0$, the stress-energy tensor must be divergence-free:

$$\nabla^{\mu} \mathbb{T}_{\mu\nu} = 0.$$

The Einstein equations are coupled with appropriate equations for the matter model of interest.

Let $\psi : \mathcal{M} \to \mathcal{M}$ be a diffeomorphism. By (2.5) and (2.6), $G[g] = 8\pi\mathbb{T}$ if and only if

$$G[\psi^* g] = 8\pi(\psi^*\mathbb{T}).$$

If $\mathbb{T} = \mathbb{T}[\phi, g]$, with $\phi \in \mathcal{T}^{(r,s)}(\mathcal{M})$ a field, such that $\psi^*(\mathbb{T}[\phi, g]) = \mathbb{T}[\psi^*\phi, \psi^* g]$, then we in fact have:

$$G[\psi^* g] = 8\pi\mathbb{T}[\psi^*\phi, \psi^* g].$$

So the Einstein equations are satisfied with respect to (g, ϕ) if and only if they are satisfied with respect to $(\psi^*\phi, \psi^* g)$. This property is sometimes also referred to as *general covariance* ("the laws of general relativity take the same form when expressed with respect to a different coordinate chart").

As we will see later in the course, it will be useful to restrict to stress-energy tensors satisfying additional *energy conditions*, which are supposed to describe necessary conditions on "physically reasonable" matter.

Definition 2.35. *A stress-energy tensor $\mathbb{T}$ satisfies:*

- *the* null energy condition *(NEC) if for all* null *$X \in \mathcal{T}(M)$*

$$\mathbb{T}(X, X) \geq 0.$$

- *the* weak energy condition *(WEC) if for all* causal *$X \in \mathcal{T}(M)$*

$$\mathbb{T}(X, X) \geq 0.$$

  If we interpret the integral curve of X as representing an observer, this means that the energy density measured by the observer is non-negative.

- *the* dominant energy condition *(DEC) if for all* future-directed timelike *$X \in \mathcal{T}(M)$*

$$P := -\mathbb{T}(X, \cdot)^\sharp$$

  is a future-directed causal vector field. We can interpret P as as a momentum 4-vector field according to the observer represented by the integral curve of X. Hence, the "speed of matter" is less or equal to the speed of light.

- *the* strong energy condition *(SEC) if for all* causal *$X \in \mathcal{T}(M)$*

$$\mathbb{T}(X, X) - \frac{1}{2}\operatorname{tr}\mathbb{T} g(X, X) \geq 0.$$

  In the context of the Einstein equations, this means that $\mathrm{Ric}[g](X, X) \geq 0$. *By considering the integral curves of a timelike X, it can be shown that this implies that observers get closer together towards the future, i.e.* gravity is attractive. *For null curves, this is already encoded in the null energy condition.*

[Exercise: Show that

$$\begin{aligned} \text{DEC} &\Rightarrow \text{WEC} \Rightarrow \text{NEC}, \\ \text{SEC} &\Rightarrow \text{NEC}.] \end{aligned}$$

Despite what its name might suggest, SEC $\not\Rightarrow$ WEC and also WEC $\not\Rightarrow$ SEC . For example, consider the following example of a valid stress-energy tensor:

$$\mathbb{T} = g.^{20}$$

Then $\mathbb{T} - \frac{1}{2}\operatorname{tr}\mathbb{T} g = g - \frac{1}{2}\cdot 4g = -2g$, so the SEC is satisfied, but the WEC fails. If we instead take $\mathbb{T} = -g$, then WEC holds but SEC fails.[21]

2.7.1 Vacuum

If $\mathbb{T} = 0$, we refer to the corresponding Einstein equations as the *vacuum Einstein equations*. In that case, they reduce to

$$\mathrm{Ric}[g] = 0.$$

Let $\psi : \mathcal{M} \to \mathcal{M}$ be a diffeomorphism. By (2.5), $\mathrm{Ric}[g] = 0$ iff

$$\mathrm{Ric}[\psi^* g] = \psi^*(\mathrm{Ric}[g]) = 0,$$

so $\psi^* g$ is automatically also a solution to the vacuum Einstein equations. Nevertheless, we would like to think of g and $\psi^* g$ as the "same" solution. This becomes relevant in the context of the initial value problem, where "uniqueness" of solutions corresponding to the same initial data can only hold "up to diffeomorphism".

2.7.2 Lagrangian field theories

Given a choice of matter model, how do we determine the stress-energy tensor? In the context of matter models described by *Lagrangian field theories*, one can follow the prescription that we outline below.

Suppose the equations of motion of the matter is described by a Lagrangian $\mathcal{L} \circ (\phi, \nabla^g \phi, g^{-1})$,[22] with a *field* $\phi \in \mathcal{T}^{(r,s)}(\mathcal{M})$, where ∇^g denotes the Levi–Civita covariant derivative with respect to g.

For example, in the case of a Klein–Gordon scalar field $\phi \in C^\infty(\mathcal{M})$

$$\mathcal{L}_{KG} \circ (\phi, \nabla^g \phi, g^{-1}) = \frac{1}{2} g^{-1}(\nabla^g \phi, \nabla^g \phi) + \frac{\mathfrak{m}^2}{2}\phi^2,$$

or in the case of an electromagnetic potential $A \in \Omega^1(\mathcal{M})$

$$\mathcal{L}_{EM} \circ (A, \nabla^g A, g^{-1}) = \frac{1}{16\pi} F_{\alpha\beta} F^{\alpha\beta},$$

with $F = dA$.

Then the stress energy tensor $\mathbb{T}$ can be defined as follows in the domain U of a coordinate chart:

$$\mathbb{T}[\phi, g] := 2\left(\frac{\partial \mathcal{L}}{\partial (g^{-1})^{\mu\nu}} \circ (\phi, \nabla^g \phi, g^{-1}) - \frac{1}{2} g_{\mu\nu} \mathcal{L} \circ (\phi, \nabla^g \phi, g^{-1})\right) dx^\mu \otimes dx^\nu.$$

It can be shown that this expression remains valid in U when expressed with respect to a any coordinate chart on U, so $\mathbb{T}$ is in fact globally well-defined on $\mathcal{M}$.

[20] Note that g is symmetric and $\nabla g = 0$, so $\nabla_\mu \mathbb{T}^{\mu\nu} = 0$.

[21] The latter is also the case for the Einstein equations in vacuum, but with a positive cosmological constant Λ: $G[g] + \Lambda g = 0$.

[22] The Lagrangian is a map $\mathcal{L} : \mathcal{N} \to \mathbb{R}$, where $\mathcal{N}$ is manifold that can be chosen appropriately so that the composition $\mathcal{L} \circ (\phi, \nabla^g \phi, g^{-1}) : \mathcal{M} \to \mathbb{R}$ is a well-defined smooth map and we can make sense of the partial derivatives of $\mathcal{L}$.

Hence,

$$\mathbb{T}^{\mathrm{KG}}_{\mu\nu}[\phi, g] = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}((g^{-1})^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + \mathfrak{m}^2\phi^2),$$
$$\mathbb{T}^{\mathrm{EM}}_{\mu\nu}[A, g] = \frac{1}{4\pi}\left[(g^{-1})^{\alpha\beta}F_{\mu\alpha}F_{\nu\beta} - \frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}g_{\mu\nu}\right].$$

It can be shown that in general, $\nabla_\mu\mathbb{T}^{\mu\nu}[\phi, g] = 0$ for <u>all</u> metrics g if ϕ satisfies the Euler–Lagrange equations corresponding to $\mathcal{L}$ and is diffeomorphism-invariant, i.e. for any diffeomorphism $\Psi : \mathcal{M} \to \mathcal{M}$:[23]

$$\mathcal{L} \circ (\Psi^*\phi, \nabla^{\Psi^*g}\Psi^*\phi, \Psi^*g^{-1}) = \mathcal{L} \circ (\phi, \nabla^g\phi, g) \circ \Psi.$$

If we do not have diffeomorphism-invariance of $\mathcal{L}$, but g is a solution to the Einstein equations with respect to $\mathbb{T}$, then $\nabla_\mu\mathbb{T}^{\mu\nu}[\phi, g] = 0$ is also true.

Note that the equation of motion for ϕ (which corresponds to the Euler–Lagrange equations) can be expressed as follows:

$$\mathfrak{m}^2\phi = \Box_g\phi := (g^{-1})^{\alpha\beta}(\nabla^2\phi)_{\alpha\beta} = \frac{1}{\sqrt{-\det g}}\partial_\alpha(\sqrt{-\det g}(g^{-1})^{\alpha\beta}\partial_\beta\phi).$$

The differential operator on the very right-hand side is called the *Laplace–Beltrami operator* associated to g. When $\mathfrak{m} = 0$, we refer to this equation as the (geometric) wave equation. Combining it with the Einstein equations, we obtain the *Einstein-scalar field system* of equations.

The equations of motion for F correspond to the Maxwell equations on a curved background with no charges or currents:

$$dF = 0,$$
$$\nabla_\mu F^{\mu\nu} = 0.$$

Exercise: Show that the stress-energy tensors corresponding to the Klein–Gordon equation and the Maxwell equations satisfy the dominant energy condition. Do they satisfy the strong energy condition?]

2.7.3 Perfect fluids

Suppose we would like to describe the gravitational properties of fluids, modelling for example stars. If we ignore properties of fluids like their viscosity and heat conduction, they are described by so-called *perfect fluids*.

The equations of motion of perfect fluids can be obtained by starting with the following stress-energy tensor: let $U \in \mathcal{T}(\mathcal{M})$ with $g(U, U) = -1$ and $\rho, p \in C^\infty(\mathcal{M})$. Then

$$\mathbb{T}^{\mathrm{fluid}}_{\mu\nu}[U, p, \rho] := (\rho + p)U_\mu U_\nu + pg_{\mu\nu}.$$

With respect to an orthononormal basis $\{e_\mu\}$ of $T_x\mathcal{M}$ with $x \in \mathcal{M}$, such that $e_0 = U_x$, we can represent $\mathbb{T}$ by the following matrix with components $\mathbb{T}^{\mathrm{fluid}}_x(e_\mu, e_\nu)$:

$$\begin{pmatrix} \rho(x) & 0 & 0 & 0 \\ 0 & p(x) & 0 & 0 \\ 0 & 0 & p(x) & 0 \\ 0 & 0 & 0 & p(x) \end{pmatrix}.$$

[23] Strictly speaking, this follows from considering the linearization around the identity of a flow of diffeomorphisms ("*infinitesimal diffeomorphisms*").

Then it can be shown that the equation $\nabla_\mu \mathbb{T}^{\mu\nu}_{\text{fluid}} = 0$ gives the *relativistic Euler equations* on $(\mathcal{M}, g)$:

$$\begin{aligned} U^\alpha \nabla_\alpha \rho + (\rho + p)\nabla_\alpha U^\alpha &= 0, \\ (\rho + p)U^\alpha \nabla_\alpha U^\mu &= -\left((g^{-1})^{\mu\nu} + U^\mu U^\nu\right)\nabla_\nu p, \end{aligned}$$

where ρ can be interpreted as the mass-energy density of a fluid, p can be interpreted as the fluid pressure and U is the 4-velocity of the fluid.[24]

Using that $g(U, U) = -1$, U has three independent components, so the relativistic Euler equations constitute four equations for five variables, three components of U^μ, p and ρ. To obtain a closed system of equations, one needs to prescribe an *equation of state* relating p and ρ. For example, one can assume that p is a function of ρ, i.e. $p = f \circ \rho$, for some $f : [0, \infty) \to \mathbb{R}$. To obtain a locally well-posed initial value problem, one moreover requires $f' > 0$. For example, for an ideal polytropic gas $f(\rho) = A\rho^\gamma$, with $A \in \mathbb{R}$ and $\gamma \in (0, \infty)$ constants.

The special case $p \equiv 0$ is called *dust*. Note that in this case $\nabla_U U = 0$, so the integral curves of U are affinely-parametrized timelike geodesics. [Exercise: Show that the perfect fluid stress-energy tensor satisfies:

- the null energy condition if and only if $\rho + p \geq 0$,
- the weak energy condition if and only if $\rho + p \geq 0$ and $\rho \geq 0$,
- the dominant energy condition if and only if $\rho \geq |p|$,
- the strong energy condition if and only if $\rho + p \geq 0$ and $\rho + 3p \geq 0$.]

[24] If we first take $g = m$ (the Minkowski metric), assuming moreover that $U = \gamma(1, \mathbf{u})$, with $\gamma = (1 - |\mathbf{u}|^2)^{\frac{1}{2}}$, and we then restore factors of c to guarantee all quantities have the correct dimensions (for example, ρ becomes ρc^2), then the *compressible Euler equations* follow from the relativistic Euler equations with respect to the Minkowski metric by neglecting terms with factors of c^{-1} as they vanish in the limit $c \to \infty$.

3 The Minkowski spacetime

Before we introduce examples of black hole spacetimes, we will first review the Minkowski spacetime solution to the vacuum Einstein equations (Minkowski 1908). This is sometimes also referred to as the "Minkowski space". This can be thought of as the "trivial solution" to the vacuum Einstein equations (which, as we will see, admit highly non-trivial solutions!). The study of causal curves on the Minkowski spacetime encompasses Einstein's theory of special relativity.

The Minkowski spacetime $(\mathbb{R}^{3+1}, m)$ is defined as follows:

$$m = -dt^2 + dx^2 + dy^2 + dz^2,$$

with (t, x, y, z) Cartesian coordinates on $\mathbb{R}^{3+1}$, which cover the spacetime globally.

In order to obtain a compact, two-dimensional representation of Minkowski, it will be useful to switch to spherical coordinates (r, θ, φ) with

$$\begin{aligned} x &= r \sin\theta \cos\varphi, \\ y &= r \sin\theta \sin\varphi, \\ z &= r \cos\theta, \end{aligned}$$

where $\theta \in (0, \pi)$, $\varphi \in (0, 2\pi)$ and $r \in (0, \infty)$. Note that these coordinates do not cover $\mathbb{R}^{3+1}$ globally due to a degeneracy at the origin $0 \in \mathbb{R}^{n+1}$, as well as at a great circle segment on $\mathbb{S}^2$ connecting the north and south poles.

[Exercise: Show that with respect to standard spherical coordinates (t, r, θ, φ), we can express:

$$m = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2).]$$

We denote:

$$\mathring{g} = d\theta^2 + \sin^2\theta d\varphi^2.$$

And observe that $\mathring{g}$ is the induced Riemannian metric of the unit round sphere $\mathbb{S}^2$ in $\mathbb{R}^3$.

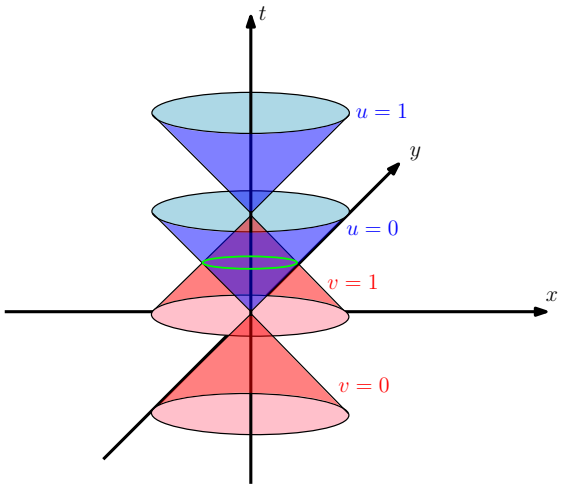


Figure 7: A depiction of the u- and v-level sets for $(\mathbb{R}^{2+1}, m)$. The intersections of the level sets are in this case round circles.

We now introduce the ingoing null coordinate u and the outgoing null coordinate v, which are defined as follows:

$$u = t - r,$$

$$v = t + r.$$

Then:

$$m = -\frac{1}{2}(du \otimes dv + dv \otimes du) + r^2 \mathring{\not{g}} =: -dudv + r^2 \mathring{\not{g}}.$$

Note that the level sets $\{u = u_0\}$ and $\{v = v_0\}$ are outgoing and ingoing cones with vertices at the t-axis and an opening angle of 45° or $\frac{\pi}{2}$ rad. The ingoing null coordinate is also called the *retarded time function* and the outgoing null coordinate v is called the *advanced time function*.

We can depict $(\mathbb{R}^{3+1}, m)$ as a half plane $\{u - v > 0\}$ by suppressing the spheres of intersection of u- and v-level sets.

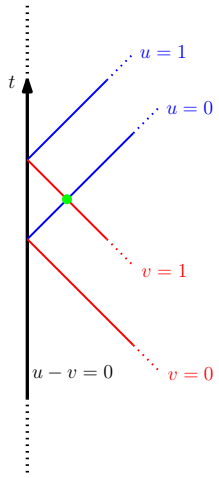


Figure 8: A depiction of the u- and v-level sets in $(\mathbb{R}^{3+1}, m)$ supressing the spherical directions. Every point on away from the t-axis represents a round sphere of radius $r(u, v) = \frac{v-u}{2}$.

To be able to depict $(\mathbb{R}^{3+1}, m)$ more compactly, we introduce the following rescaled double null coordinates.

$$\tilde{u} = \arctan u,$$
$$\tilde{v} = \arctan v.$$

[Exercise: Show that the metric takes the form $m = -\frac{1}{\cos^2 \tilde{u} \cos^2 \tilde{v}} d\tilde{u} d\tilde{v} + \frac{(\tan \tilde{v} - \tan \tilde{u})^2}{4} \mathring{\not{g}}$.]

We can represent $(\mathbb{R}^{3+1}, m)$ by the following subset of $\mathbb{R}^2$:

$$\mathcal{Q} = \left\{ (\tilde{u}, \tilde{v}) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \ , \ \tilde{v} - \tilde{u} > 0 \right\}$$

We denote

$$\Gamma := \left\{ (\tilde{u}, \tilde{v}) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \ , \ \tilde{v} - \tilde{u} = 0 \right\}.$$

Note that Γ represent the t-axis and it is a centre of spherical symmetry, i.e. it consists of all points that stay fixed with respect to spatial rotations around the t-axis.

The closure $\overline{\mathcal{Q}}$ of $\mathcal{Q}$ in $\mathbb{R}^2$ is called a *(Carter–)Penrose diagram* of the Minkowski spacetime. Changing the function arctan to a different function results in a different diagram. However, lightcones emanating from the t-axis will always be represented by straight lines at a 45° angle with the vertical. Note that Γ need not not be a straight line in all Penrose diagrams! We will give a definition of these diagrams in a more general setting later in the course.

We label elements of the boundary $\partial\mathcal{Q} \setminus \mathcal{Q}$ in the following way:

$$\begin{aligned}
\mathcal{I}^+ &:= \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left\{\frac{\pi}{2}\right\}, \\
\mathcal{I}^- &:= \left\{-\frac{\pi}{2}\right\} \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \\
i^+ &:= (\frac{\pi}{2}, \frac{\pi}{2}), \\
i^- &:= (-\frac{\pi}{2}, -\frac{\pi}{2}), \\
i^0 &:= (-\frac{\pi}{2}, \frac{\pi}{2}).
\end{aligned}$$

We refer to $\mathcal{I}^\pm$ as *future/past null infinity*, $i^\pm$ as *future/past timelike infinity* and i^0 as *spacelike infinity*. This nomenclature is motivated by the fact that null cones in Minkowski emanating from the t-axis are represented by lines at 45° with Γ (or with the lines $v - u = \text{const.}$) with future/past limit points on $\mathcal{I}^\pm$, timelike curves γ with uniform bounds $-g(\gamma, \gamma) > c > 0$ are represented by curves whose tangent is at an angle uniformly smaller than 45° with Γ with future/past limit points at $i^\pm$ and spacelike curves with uniform bounds $g(\gamma, \gamma) > c > 0$ are represented by curves whose tangent is at an angle uniformly larger than 45° with Γ and which have endpoints at i^0.

Similarly, spherically symmetric null/ timelike/ spacelike hypersurfaces are represented by curves at/below/above 45° with γ.

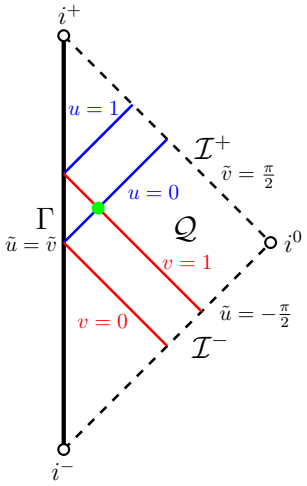


Figure 9: A Penrose diagram of the Minkowski spacetime.

[Exercise: Prove that all geodesics in Minkowski must be straight lines.]

[Exercise: Draw the null line $\{x - t = 1, z = 0, y = 1\}$ in a Minkowski Penrose diagram.]

[Exercise: Draw the hyperboloids $\{t^2 - r^2 = s^2\}$ and the hyperboloids $\{(t + s)^2 - r^2 = 1\}$ in a Minkowski Penrose diagram, for different values of $s \in \mathbb{R}$.]

[Exercise: Consider the sphere $S^2_{u_0,v_0} := \{u = u_0\} \cap \{v = v_0\}$ in Minkowski, with $u_0, v_0 \in \mathbb{R}$. Draw $J^+(S^2_{u_0,v_0})$, $J^-(S^2_{u_0,v_0})$ in a Penrose diagram.]

[Exercise: Let $\Sigma = \{t = 0\} \cap \{r \leq 1\}$ be a subset of Minkowski. Draw Σ, $D^+(\Sigma)$ and $D^-(\Sigma)$ in a Penrose diagram.]

4 Schwarzschild black hole spacetimes

Let $M \in (0, \infty)$. Then the Schwarzschild spacetimes (SCHWARZSCHILD 1915) are pairs $(\mathcal{M}_{\text{ext}}, g_M)$, with $\mathcal{M}_{\text{ext}} = \mathbb{R} \times (2M, \infty) \times \mathbb{S}^2$ and

$$g_M = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 \mathring{\not{g}}, \tag{4.1}$$

where $t \in \mathbb{R}$, $r \in (0, 2M)$ and $\theta \in (0, \pi)$ and $\varphi \in (0, 2\pi)$. Note that this expression features the usual degeneration of spherical coordinates at a great circle segment connecting the north and south pole and hence multiple coordinate charts are required to fully cover the spheres of constant t and r.[25]

It can be shown that the Schwarzschild spacetimes are solutions to the vacuum Einstein equations (we will revisit this later). We will also refer to them as *Schwarzschild exteriors*, since we will show in the next section that they can be extended to obtain a Schwarzschild *interior*.

The parameter M is called the *mass* of the Schwarzschild spacetime. Somewhat paradoxically, it is sensible to assign a notion of mass or energy to spacetimes even when they solve the vacuum Einstein equations, i.e. *matter is absent*.

We will encounter the concepts of mass and energy in more generality later in the course. Loosely speaking, you may think of this as a measure of the total energy contained in the "gravitational field" at any fixed time.

It may seem like the Schwarzschild metric is singular at $r = 2M$. Indeed, the metric components in (4.1) become ill-defined at $r = 2M$. This fact was a source of significant confusion after Schwarzschild wrote down his metric. It was LEMAÎTRE who first realized in 1932 that the spacetime $(\mathcal{M}_{\text{ext}}, g_M)$ can actually be extended across $r = 2M$ and any problems at $r = 2M$ are purely an artefact of the particular choice of coordinates (t, r, θ, φ). They can therefore be compared to the issues at $\theta = 0, \pi$ in spherical coordinates.

To see that nothing goes wrong at $r = 2M$, we will first introduce null coordinates (u, v) that generalize the null coordinates that we already encounter in the Minkowski spacetime. First, we introduce the *tortoise coordinate* $r_* : (2M, \infty) \to \mathbb{R}$, which satisfies

$$\frac{dr_*}{dr} = \frac{1}{1 - \frac{2M}{r}}.$$

[EXERCISE: Show that we can write

$$r_*(r) = c_0 + r + 2M \log\left(\frac{r - 2M}{2M}\right)$$

for an arbitrary constant $c_0 \in \mathbb{R}$ and hence the range of r_* is the full real line $\mathbb{R}$. Furthermore, since $\frac{dr_*}{dr} \neq 0$, r_* is also injective and therefore bijective, with a smooth inverse.]

For the sake of convenience, we will take $c_0 = 0$ and define:

$$r_* = r + 2M \log\left(\frac{r - 2M}{2M}\right). \tag{4.2}$$

With respect to $(t, r_*, \theta, \varphi)$ coordinates, we can express:

$$g_M = \left(1 - \frac{2M}{r}\right)\left[-dt^2 + dr_*^2\right] + r^2 \mathring{\not{g}}.$$

[25] EXERCISE: Show that there exist coordinates $(t, r, \tilde{\theta}, \tilde{\varphi})$ that cover the complement of the domain of the coordinates (t, r, θ, φ), with $g_M = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\tilde{\theta}^2 + \sin^2\tilde{\theta} d\tilde{\varphi}^2)$, so that (t, r, θ, φ) and $(t, r, \tilde{\theta}, \tilde{\varphi})$ cover the full manifold $\mathcal{M}_{\text{ext}}$.

We now define the null functions u and v as follows:

$$u = t - r_*,$$
$$v = t + r_*.$$

Note that u, v can take on any value in $\mathbb{R}$ and: [Exercise: Show that $du^\sharp$ and $dv^\sharp$ are null vector fields.]

Then

$$g_M = -\left(1 - \frac{2M}{r}\right) dudv + r^2 \mathring{g},$$

where we view r as a function of u and v in the following way: $r = r(r_*) = r\left(\frac{1}{2}(v-u)\right)$. The coordinates (u, v, θ, φ) are called *Eddington–Finkelstein double null coordinates*.

[Exercise: Show that g_M takes the following form with respect to (v, r) coordinates (*ingoing Eddington–Finkelstein coordinates*) and (u, r) coordinates (*outgoing Eddington–Finkelstein coordinates* or *Bondi coordinates*):

$$g_M = -\left(1 - \frac{2M}{r}\right) dv^2 + 2dvdr + r^2 \mathring{g},$$
$$g_M = -\left(1 - \frac{2M}{r}\right) du^2 - 2dudr + r^2 \mathring{g}.]$$

As in the Minkowski case, we can obtain a Penrose diagram of the Schwarzschild spacetime by rescaling:

$$\tilde{u} = \arctan u,$$
$$\tilde{v} = \arctan v.$$

Then $\mathcal{M}_{\text{ext}}$ is represented by the bounded subset $\mathcal{Q} = (-\frac{\pi}{2}, \frac{\pi}{2}) \times (-\frac{\pi}{2}, \frac{\pi}{2}) \subset \mathbb{R}^2$.

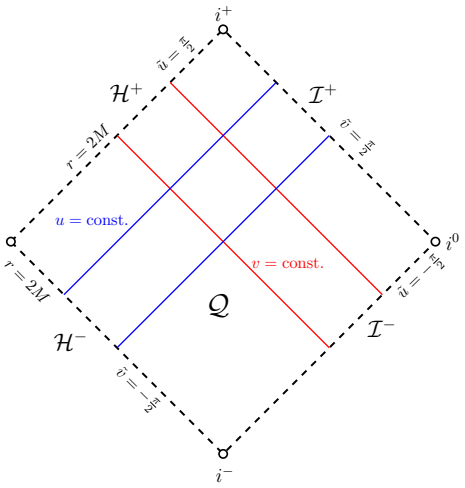


Figure 10: A Penrose diagram of the Schwarzschild exterior spacetime.

Note that, we can define, as in the Minkowski case, future/past null infinity $\mathcal{I}^\pm$, future/past timelike infinity $i^\pm$ and spacelike infinity i^0. We moreover have two additional boundary components $\mathcal{H}^\pm$ and their intersection, which we will give a name to later.

$$\mathcal{I}^+ := \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left\{\frac{\pi}{2}\right\},$$

$$\mathcal{H}^+ := \left\{\frac{\pi}{2}\right\} \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$
$$\mathcal{I}^- := \left\{-\frac{\pi}{2}\right\} \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right),$$
$$\mathcal{H}^- := \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left\{-\frac{\pi}{2}\right\},$$
$$i^+ := \left\{\frac{\pi}{2}\right\} \times \left\{\frac{\pi}{2}\right\},$$
$$i^- := \left\{-\frac{\pi}{2}\right\} \times \left\{-\frac{\pi}{2}\right\},$$
$$i^0 := \left\{-\frac{\pi}{2}\right\} \times \left\{\frac{\pi}{2}\right\}.$$

4.1 Maximally-extended Schwarzschild

Now we will show that we can extend the spacetime smoothly across the boundaries $\mathcal{H}^+$ and $\mathcal{H}^-$ of $\mathcal{Q}$ in $\mathbb{R}^2$, where r approaches the value $2M$. We will do so by first switching from the ingoing null coordinate u to an affine parameter of appropriate ingoing null geodesics.

Consider the coordinate chart $\phi = (u, v, \theta, \varphi)$ and note first that ∂_u is a null vector field. Furthermore: [Exercise: Show that

$$\nabla_{\partial_u}\partial_u = \Gamma^u_{uu}\partial_u = \Omega^{-2}(\partial_u\Omega^2)\partial_u,$$

with $\Omega^2(u, v) := 1 - \frac{2M}{r(u,v)}$.]

Consider an integral curve $\gamma : \mathbb{R} \to \mathcal{M}_{\text{ext}}$ of ∂_u, with

$$\begin{pmatrix}\gamma^u\\ \gamma^v\\ \gamma^\theta\\ \gamma^\varphi\end{pmatrix}(u) := (\phi \circ \gamma)(u) = \begin{pmatrix}u\\ v_0\\ \theta_0\\ \varphi_0\end{pmatrix},$$

for some $v_0 \in \mathbb{R}$, $\theta_0 \in (0, \pi)$ and $\varphi_0 \in (0, 2\pi)$.

To show that γ is a (non-affinely parametrized) null geodesic, we rescale:

$$\frac{dU'}{du} = \Omega^2(u, v_0) = 1 - \frac{2M}{r(u, v_0)}.$$

Let $X = \Omega^{-2}(u, v_0)\partial_u$. Then along γ:

$$\nabla_X X|_\gamma = \nabla_{\Omega^{-2}\partial_u}(\Omega^{-2}\partial_u)|_\gamma = -\Omega^{-6}\partial_u\Omega^2|_\gamma\partial_u|_\gamma + \Omega^{-4}\nabla_{\partial_u}\partial_u|_\gamma = 0,$$

so $X|_\gamma$ satisfies the defining property of a tangent vector field to an affinely-parametrized null geodesic. We will show below that $U'(\infty) = 0$, so γ reaches the boundary $\mathcal{H}^+$ at a finite affine time.

By (4.2), we can express:

$$e^{\frac{v-u}{4M}} = e^{\frac{r_*}{2M}} = e^{\frac{r}{2M}}\left(\frac{r - 2M}{2M}\right),$$

so that

$$\frac{dU'}{du} = 1 - \frac{2M}{r(u, v_0)} = \frac{2M}{r(u, v_0)}e^{-\frac{r(u,v_0)}{2M}}e^{\frac{v_0-u}{4M}} = e^{-1+\frac{v_0}{4M}}e^{-\frac{u}{4M}}(1 + O(e^{-\frac{u}{4M}}))$$

and therefore, as $u \to \infty$,

$$U'(u) = -4Me^{-1+\frac{v_0}{4M}}e^{-\frac{u}{4M}}(1 + O(e^{-\frac{u}{4M}})).$$

Note the exponential relation between U and u and $U'(\infty) = 0$. To simplify matters a bit, we consider instead the following ingoing null coordinate:

$$U(u) = -e^{-\frac{u}{4M}}.$$

We can similarly define

$$\frac{dV'}{dv} = 1 - \frac{2M}{r(u_0, v)},$$

to obtain

$$\frac{dV'}{dv} = \frac{2M}{r(u_0, v)} e^{-\frac{r(u_0,v)}{2M}} e^{\frac{v-u_0}{4M}} = e^{-1-\frac{u_0}{4M}} e^{\frac{v}{4M}} (1 + O(e^{\frac{v}{4M}}))$$

and therefore, as $v \to -\infty$,

$$V'(v) = 4M e^{-1-\frac{u_0}{4M}} e^{\frac{v}{4M}} (1 + O(e^{\frac{v}{4M}})),$$

so $V'(-\infty) = 0$.

Again, we consider the simplified outgoing null coordinate:

$$V(v) = e^{\frac{v}{4M}}.$$

We refer to the coordinates (U, V, θ, φ) as Kruskal–Szekeres coordinates. Note that

$$U \cdot V = -e^{\frac{v-u}{4M}} = -e^{\frac{r_*}{2M}} = -e^{\frac{r}{2M}} \frac{r - 2M}{2M}. \tag{4.3}$$

We can therefore express

$$\begin{aligned} g_M &= -r^{-1}(r - 2M) \frac{du}{dU} \frac{dv}{dV} dU dV + r^2 \mathring{\not{g}} \\ &= 16M^2 r^{-1}(r - 2M)(UV)^{-1} dU dV + r^2 \mathring{\not{g}} \\ &= -32M^3 r^{-1} e^{-\frac{r}{2M}} dU dV + r^2 \mathring{\not{g}}, \end{aligned}$$

with $U < 0$ and $V > 0$.

Since nothing "singular" occurs at $U = 0$ or $V = 0$, we can *extend* $(\mathcal{M}_{\text{ext}}, g_M)$ to a bigger spacetime $(\mathcal{M}_{\text{Krus}}, g_M)$, which is called a *maximally-extended Schwarzschild spacetime* or *Kruskal spacetime* by extending the domain of validity of the Kruskal coordinates U and V. More precisely, our construction will imply the existence of an *isometric embedding* $\iota : (\mathcal{M}_{\text{ext}}, g_M) \to (\mathcal{M}_{\text{Krus}}, \widetilde{g})$, which is a smooth embedding that satisfies $\iota^* \widetilde{g} = g_M$, such that $\iota(\mathcal{M}_{\text{ext}}) \neq \mathcal{M}_{\text{Krus}}$, i.e. $\iota(\mathcal{M}_{\text{ext}})$ is a proper subset of $\mathcal{M}_{\text{Krus}}$. In this case, we can think of ι as being simply the inclusion map.

We proceed as follows:

- First, we will extend $r(U, V) = r(U \cdot V)$ to $U \geq 0$ and $V \leq 0$ by requiring (4.3) to hold, i.e. let $f : (0, \infty) \to (-\infty, 1)$ with $f(x) = -e^{\frac{x}{2M}} \frac{x-2M}{2M}$, then:
  - $\star$ f is smooth (in fact, analytic),
  - $\star$ $\frac{df}{dx}(x) = -\frac{1}{2M} \left(\frac{x-2M}{2M} + 1 \right) e^{\frac{x}{2M}} < 0$ for all $x \in (0, \infty)$,
  - $\star$ f is bijective (injective by $\frac{df}{dx}(x) < 0$ and surjective by the fact that $f(x) \to 1$ as $x \downarrow 0$ and $f(x) \to -\infty$ as $x \to \infty$ and the intermediate value theorem.

  This implies that the inverse $f^{-1} : (-\infty, 1) \to (0, \infty)$ is well-defined and, by the Inverse Function Theorem, f^{-1} is smooth (in fact, analytic). We can therefore extend: $r(U, V) := f^{-1}(U \cdot V)$ smoothly to the set $\{(U, V) \in \mathbb{R}^2 \, , \, UV < 1\}$, where it will take on values in $(0, 2M]$.

- We will define the *maximally-extended Schwarzschild* or *Kruskal* manifold by

$$\mathcal{M}_{\text{Krus}} := \{(U,V) \in \mathbb{R}^2 \, , \, UV < 1\} \times \mathbb{S}^2$$

and equip it with the metric

$$\tilde{g} = -32M^3 r^{-1} e^{-\frac{r}{2M}} dU dV + r^2 \mathring{g}.$$

It is clear that $(\mathcal{M}_{\text{Krus}}, g_M)$ agrees with the Schwarzschild exterior spacetime in the subset $\{U < 0, V > 0\}$ and that it remains well-defined in the region $\{0 \leq UV < 1\}$. We will use the shorthand notation $\tilde{g} = g_M = -\Omega^2_{\text{Krus}}(U,V) dU dV + r^2 \mathring{g}$, with

$$\Omega^2_{\text{Krus}}(U,V) = -32M^3 r^{-1} e^{-\frac{r}{2M}}.$$

[Exercise: Show that the region $\{0 \leq UV < 1\}$ can be covered by coordinates (t', r, θ, φ), such that

$$g_M = -\left(\frac{2M}{r} - 1\right)^{-1} dr^2 + \left(\frac{2M}{r} - 1\right) dt'^2 + r^2 \mathring{g}.$$

for a suitable choice of $t'(U,V)$.]

We can represent the maximally-extended Schwarzschild spacetime $(\mathcal{M}_{\text{Krus}}, g_M)$ via a Penrose diagram. We redefine:

$$\tilde{U} = \arctan U,$$
$$\tilde{V} = \arctan V.$$

Note that $UV = 1$ corresponds to $\tan\tilde{U} \cdot \tan\tilde{V} = 1$. Hence $(\tilde{U}, \tilde{V})$ have the following range:

$$\{(\tilde{U}, \tilde{V}) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \times \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \, , \, \tan\tilde{U} \cdot \tan\tilde{V} < 1\}.$$

The set $\{r = 0\}$ is therefore represented by the curves:

$$\left\{\tilde{U} = \arctan\left(\frac{1}{\tan\tilde{V}}\right)\right\} = \left\{\tilde{U} = \frac{\pi}{2} - \tilde{V} \,\middle|\, 0 < \tilde{V} < \frac{\pi}{2}\right\} \cup \left\{\tilde{U} = -\frac{\pi}{2} - \tilde{V} \,\middle|\, -\frac{\pi}{2} < \tilde{V} < 0\right\},$$

where we used that

$$\begin{aligned}\arctan\left(\frac{1}{\tan x}\right) = \arctan\left(\frac{\cos x}{\sin x}\right) = \arctan\left(\frac{\sin(x+\frac{\pi}{2})}{-\cos(x+\frac{\pi}{2})}\right) &= \arctan\left(-\tan(x+\frac{\pi}{2})\right) \\ &= -\arctan\left(\tan(x+\frac{\pi}{2})\right) \\ &= \begin{cases} -(x+\frac{\pi}{2}) & (-\frac{\pi}{2} < x < 0), \\ -(x+\frac{\pi}{2}-\pi) & (0 < x < \frac{\pi}{2}). \end{cases}\end{aligned}$$

We can immediately see from the Penrose diagram that future-directed causal curves in the region where $\{U \geq 0, V \geq 0\}$ cannot enter the region $\{U < 0, V \geq 0\}$, since $J^-(\{U < 0, V \geq 0\}) \cap \{U \geq 0, V \geq 0\} = \emptyset$. We refer to the region $\mathcal{BH} := \{U \geq 0, V \geq 0\}$ as the *Schwarzschild black hole region*. The intersection of $\mathcal{H}^+_L$ and $\mathcal{H}^+_R$ at $(U,V) = (0,0)$ is called the *bifurcation sphere*.

The lines $\mathcal{S}_+$ and $\mathcal{S}_-$ in $\partial\mathcal{Q}$ represents the Schwarzschild singularity, where $r = 0$. It is a spacelike curve in $\mathbb{R}^{1+1}$, so we say the Schwarzschild singularity is spacelike.

[Exercise: 1) Construct a Penrose diagram of Schwarzschild spacetimes with $M < 0$. 2) Are these spacetimes globally hyperbolic?]

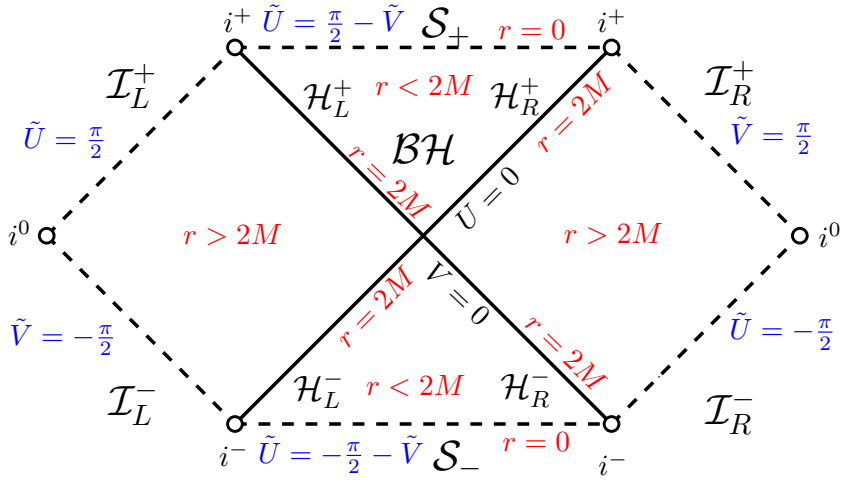


Figure 11: A Penrose diagram of the maximally-extended Schwarzschild spacetime.

One can show that the following contraction of the Riemann tensor, the so-called *Kretschmann scalar* $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ satisfies [EXCERCISE: Show this (at your own peril!).]:

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48M^2}{r^6}.$$

and hence $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ blows up as r approaches 0, or equivalently, as we approach the level set $\{UV = 1\}$. Using this fact, we will now show that we cannot extend the spacetime any further with a C^2-metric.

Proposition 4.1. *The spacetime $(\mathcal{M}_{\text{Krus}}, g_M)$ is C^2-inextendible, which means that there does not exist an isometric embedding $\iota : (\mathcal{M}_{\text{Krus}}, g_M) \to (\widetilde{\mathcal{M}}, \tilde{g})$, with $\tilde{g}$ a C^2 metric and $\iota(\mathcal{M}_{\text{Krus}}) \neq \widetilde{\mathcal{M}}$.*

Proof. We will prove this statement by contradiction. Suppose that there exists a spacetime $(\widetilde{\mathcal{M}}, \tilde{g})$ with an isometric embedding $\iota : (\mathcal{M}_{\text{Krus}}, g_M) \to (\widetilde{\mathcal{M}}, \tilde{g})$, such that $\iota(\mathcal{M}_{\text{Krus}}) \neq \widetilde{\mathcal{M}}$. Then $\partial(\iota(\mathcal{M}_{\text{Krus}})) \neq \emptyset$. Let $p \in \partial(\iota(\mathcal{M}_{\text{Krus}}))$. By Lemma 4.2 below, there exists an affinely-parametrized timelike geodesic $\tilde{\gamma} : (-\epsilon, \epsilon) \to \widetilde{\mathcal{M}}$, satisfying $\tilde{\gamma}(0) = p$ and $\tilde{\gamma}(s) \in \iota(\mathcal{M}_{\text{Krus}})$ for all $s < 0$.

By injectivity of ι, there exists a unique curve $\gamma : (-\epsilon, 0) \to \mathcal{M}_{\text{ext}}$ such that $\iota \circ \gamma = \tilde{\gamma}|_{(-\epsilon,0)}$. By $\iota^*\tilde{g} = g$, γ must also be an affinely-parametrized timelike geodesic.[26] Without loss of generality, we may assume that γ is future directed.

In our analysis of timelike geodesics on maximally-extended Schwarzschild in §4.3, we will see that future-directed, future-inextendible timelike geodesics either have an affine parameter that is unbounded to the future, or $r|_{\gamma(s)} \to 0$ as s approaches its supremum. Since the affine parameter s of $\tilde{\gamma}$ approaches 0 as we approach p, γ must fall in the latter category: $r|_{\gamma(s)} \to 0$ as $s \uparrow 0$.

Now we use that

$$(R_{\mu\nu\rho\sigma}[\tilde{g}]R^{\mu\nu\rho\sigma}[\tilde{g}]) \circ \iota = R_{\mu\nu\rho\sigma}[\iota^*\tilde{g}]R^{\mu\nu\rho\sigma}[\iota^*\tilde{g}] = R_{\mu\nu\rho\sigma}[g_M]R^{\mu\nu\rho\sigma}[g_M] = \frac{48M^2}{r^6},$$

to conclude that the Kretschmann scalar of $\tilde{g}$ blows up along $\tilde{\gamma}$ as $s \uparrow 0$. But this is a contradiction, since $\tilde{g}$ is C^2 at p and must therefore have a well-defined Riemann tensor and associated Kretschmann scalar. □

[26] EXCERCISE: Convince yourself of this.

In fact, it can be shown (SBIERSKI 2016) that $(\mathcal{M}_{\text{Krus}}, g_M)$ is C^0-inextendible: it cannot even be isometrically embedded into a larger spacetime with a continuous metric (continuity would be a necessary requirement for the metric to make sense as a (weak) solution to the Einstein equations). The limiting set $\{r = 0\}$ or $\{UV = 1\}$ may therefore be thought of as a *singularity* in the strongest sense.

A compact, small matter distribution approaching $r = 0$ (idealized by nearby radial timelike geodesics) can be shown to experience an *infinite tidal deformation* and not just an infinite tidal force: the volume encompassed by the matter particles will be infinitely extended in one direction and infinitely squeezed in the remaining two directions ("spaghettified") and the total volume will shrink to zero.

[EXERCISE: Show that the Minkowski spacetime is C^2-inextendible.]

Lemma 4.2. *Let $(\widetilde{\mathcal{M}}, g)$ be a C^2-extension of a Lorentzian manifold $(\mathcal{M}, g)$ with isometric embedding $\iota : \mathcal{M} \to \widetilde{\mathcal{M}}$. Then there exists a timelike geodesic $\gamma : [0, 1] \to \widetilde{\mathcal{M}}$ satisfying $\gamma((0, 1]) \subset \iota(\mathcal{M})$ and $\gamma(0) \in \partial\iota(\mathcal{M})$.*

Proof. (*non-examinable, credit: Benedikt Miethke*) We will first show that there exists a timelike *curve* with the properties stated in the lemma. We will later comment on the existence of a timelike geodesic. Let $p \in \partial\iota(\mathcal{M})$. Let $\tau : [0, 1] \to \widetilde{\mathcal{M}}$ be a timelike curve satisfying $\tau(0) = p$. Suppose that $\tau([0, 1]) \cap \iota(\mathcal{M}) \neq \emptyset$ and let $s_1 \in (0, 1]$ satisfy $\tau(s_1) \in \iota(\mathcal{M})$. Then there exists an s_2 such that $\tau(s_2) \in \partial\iota(\mathcal{M})$ and either $\tau([s_1, s_2)) \subset \iota(\widetilde{\mathcal{M}})$ (if $s_2 > s_1$) or $\tau((s_2, s_1]) \subset \iota(\widetilde{\mathcal{M}})$ (if $s_2 < s_1$). By reparametrizing τ and restricting its domain, we obtain the desired timelike curve γ.

Now suppose that $\tau([0, 1]) \subset \widetilde{\mathcal{M}} \setminus \iota(\mathcal{M})$ and let $p = \tau(0)$. Then we can also connect any point in a suitably small neighbourhood U of p, with a timelike curve emanating from $\tau(1)$, via a small, smooth deformation of τ. Since $p \in \partial\iota(\widetilde{\mathcal{M}})$, we have that $U \cap \iota(\mathcal{M}) \neq \emptyset$, so there exists a timelike curve $\widetilde{\tau} : [0, 1] \to \widetilde{\mathcal{M}}$, such that $\widetilde{\tau}(0) \in \iota(\mathcal{M})$ and $\widetilde{\tau}(1) = \tau(1) \notin \iota(\mathcal{M})$. There must in particular exist a $s_* \in (0, 1)$ such that $\tilde{\tau}(s_*) \in \partial\iota(\mathcal{M})$. After reparametrizing $\widetilde{\tau}$, we obtain the desired timelike curve γ.

Now let $\gamma : [0, 1] \to \widetilde{\mathcal{M}}$ be a timelike curve satisfying the properties stated in the lemma, with $p = \gamma(0)$. By taking $\epsilon > 0$ suitably small, $q = \gamma(\epsilon)$ lies in a normal neighbourhood U around p and can therefore be connected to p via a unique geodesic (up to reparametrization), by applying the bijectivity of the exponential map $\exp_p : T_p\widetilde{\mathcal{M}} \to U$. One can also show that this geodesic must in fact be timelike, i.e. $\exp_p^{-1}(q) \in T_p\widetilde{\mathcal{M}}$ is timelike; we refer the interested reader to the Proposition 4.5.1 of HAWKING AND ELLIS. Then we can reparametrize γ to obtain the desired timelike geodesic. □

4.2 Isometries and Killing vector fields

Consider the region $(\mathcal{M}_{\text{ext}}, g_M)$. Since the metric coefficients of g_M do not depend on t, it follows that

$$(\mathcal{L}_{\partial_t} g_M)_{\mu\nu} = \partial_t (g_M)_{\mu\nu} = 0,$$

so ∂_t is a Killing vector field. With respect to (U, V) coordinates, we can express

$$\partial_t = \partial_t U \partial_U + \partial_t V \partial_V = \partial_t u \partial_u U \partial_U + \partial_t v \partial_v V \partial_U = \frac{1}{4M}(-U\partial_U + V\partial_V).$$

We can therefore define the vector field $T = \frac{1}{4M}(-U\partial_U + V\partial_V)$ in the full spacetime $(\mathcal{M}_{\text{Krus}}, g_M)$ and observe that it agrees with ∂_t in $(\mathcal{M}_{\text{ext}}, g_M)$.

Along $\mathcal{H}_R^+ = \{U = 0\} \cap \{V > 0\}$, we have that $T = \frac{1}{4M} V\partial_V$, so T is an outgoing null vector field that is tangent to $\mathcal{H}_R^+$.

Note that we can cover the region $\{V > 0\} \cap \mathcal{M}_{\text{Krus}}$ by ingoing Eddington–Finkelstein coordinates (v, r, θ, φ), with

$$g_M = -\left(1 - \frac{2M}{r}\right) dv^2 + 2dvdr + r^2 \mathring{\not{g}}.$$

[Exercise: Show that $g_M^{-1} = (\partial_v \otimes \partial_r + \partial_r \otimes \partial_v) + \left(1 - \frac{2M}{r}\right) \partial_r \otimes \partial_r + r^{-2} \mathring{\not{g}}^{-1}$.]

Note that in ingoing Eddington–Finkelstein coordinates, we can express:

$$T = \frac{1}{4M}(-U\partial_U + V\partial_V) = \frac{1}{4M} V(v)(\partial_V v)\partial_v = \partial_v.$$

Since g_M in (v, r) coordinates is also v-independent in the region $r \leq 2M$, we conclude that T must remain a Killing vector field when $r \leq 2M$. It can similarly be shown that T is Killing in the remaining regions of $\mathcal{M}_{\text{Krus}}$.

The vector field ∂_φ is clearly also a Killing vector field. In fact:

[Exercise: Show that the following *angular momentum vector fields* are all Killing vector fields of $(\mathcal{M}_{\text{Krus}}, g_M)$:

$$\begin{aligned} O_1 &= -\sin\varphi\partial_\theta - \cot\theta\cos\varphi\partial_\varphi, \\ O_2 &= \cos\varphi\partial_\theta - \cot\theta\sin\varphi\partial_\varphi, \\ O_3 &= \partial_\varphi.] \end{aligned}$$

Definition 4.1. *A spacetime $(\mathcal{M}, g)$ is* static *if it admits a timelike Killing vector field T, which satisfies:*

$$T^\flat \wedge dT^\flat = 0.$$

This is equivalent[27] *to the statement that around any point $p \in \mathcal{M}$, there exist local coordinates $(\tau, x^1, \ldots, x^n)$, such that we can express:*

$$g = -f d\tau^2 + h_{ij} dx^i dx^j,$$

where $T = \partial_\tau$, f, h_{ij} are functions of $x^1, \ldots, x^n$ and $h_{ij}dx^i dx^j$ is a Riemannian metric on the level sets of τ.

Clearly, the Schwarzschild exteriors $(\mathcal{M}_{\text{ext}}, g_M)$ are static.

The existence of angular momentum Killing vector fields L_1, L_2, L_3 is a consequence of the *spherical symmetry* of the (extended) Schwarzschild spacetimes.

4.3 Dynamics of geodesics

In Einstein theory of general relativity, idealized observers in a spacetime are represented by timelike geodesics and the path of photons is represented by null geodesics. This is known as the *geodesic hypothesis*.

Independently of its relation with observers, the behaviour of causal geodesics also serves as a convenient tool for probing important geometric properties of the spacetime.

4.3.1 Black hole exterior

Let $\gamma : I \to \mathcal{M}_{\text{ext}}$ be an affinely-parametrized causal geodesic, which is either timelike or null. In the case of timelike geodesics, we will take the affine parameter to be the proper time along the geodesic. Then $g(\dot\gamma, \dot\gamma) = -\sigma$, with $\sigma = 0$ if γ is null and $\sigma = 1$ if γ is timelike.

[27] This equivalence follows from the Frobenius theorem.

Consider the coordinate chart $\phi = (t, r, \theta, \varphi)$. Then

$$(\phi \circ \gamma)(s) = \begin{pmatrix} t(s) \\ r(s) \\ \theta(s) \\ \varphi(s) \end{pmatrix}, \qquad (\dot{\phi \circ \gamma})(s) = \begin{pmatrix} \dot{t} \\ \dot{r} \\ \dot{\theta} \\ \dot{\varphi} \end{pmatrix}(s).$$

Then $g(\dot\gamma, \dot\gamma) = -\sigma$ implies that:

$$-\sigma = -\left(1 - \frac{2M}{r}\right)\dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\varphi}^2. \tag{4.4}$$

We can simplify this equation a bit by redefining our spherical coordinates (θ, φ) so that we are guaranteed that $\theta(0) = \frac{\pi}{2}$ and $\dot\theta(0) = 0$. We can now apply the transformation $\theta' : \theta \mapsto \pi - \theta$, which leaves the Schwarzschild metric and hence the geodesic equation invariant, and which also leaves the geodesic initial data $(\gamma(s), \dot\gamma(s))$ invariant. But uniqueness of solutions to the geodesic equation, we must have that $\theta'(s) = \theta(s)$ for all $s \in I$, so that necessarily $\theta(s) = \frac{\pi}{2}$. That is to say, the geodesic will stay restricted to the equator.

We can therefore simplify (4.4) to obtain:

$$-\sigma = -\left(1 - \frac{2M}{r}\right)\dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\varphi}^2.$$

Now we use the Killing property of ∂_t and ∂_φ to conclude that $g_{\gamma(s)}((\partial_t)_{\gamma(s)}, \dot\gamma(s))$ and $g_{\gamma(s)}((\partial_\varphi)_{\gamma(s)}, \dot\gamma(s))$ are conserved in s (Proposition 2.5). We define:

$$\begin{aligned} E &= -g_{\gamma(s)}((\partial_t)_{\gamma(s)}, \dot\gamma(s)) = \left(1 - \frac{2M}{r(s)}\right)\dot{t}(s), \\ L &= g_{\gamma(s)}((\partial_\varphi)_{\gamma(s)}, \dot\gamma(s)) = r^2(s)\dot{\varphi}(s) \end{aligned}$$

and refer to E as the energy of the geodesic with respect to ∂_t and to L as the angular momentum of the geodesic (in the z-direction). Then we obtain:

$$-\sigma = -\left(1 - \frac{2M}{r}\right)^{-1}E^2 + \left(1 - \frac{2M}{r}\right)^{-1}\dot{r}^2 + r^{-2}L^2$$

Rearranging the above equation, we obtain:

$$E^2 = \dot{r}^2 + \left(1 - \frac{2M}{r}\right)(L^2 r^{-2} + \sigma) =: \dot{r}^2 + V_\sigma(r).$$

The above equation resembles the conservation of energy property of a 1-D dynamical system, where V_σ plays the role of a potential energy term and $\dot{r}^2$ plays the role of a kinetic energy term.

By studying the qualitative properties of the potential function V_σ, we can determine the qualitative properties of the dynamics of causal geodesics.

Null geodesics We consider first the case $\sigma = 0$. Then $V_0(r) = L^2\left(1 - \frac{2M}{r}\right)r^{-2}$. If $L = 0$, then $V_0 \equiv 0$. Suppose that $L \neq 0$.

Then we have that $V_0(2M) = 0$ and $V_0(r) \to 0$ as $r \to \infty$. Furthermore,

$$V_0'(r) = -2L^2 r^{-4}(r - 3M)$$

Hence, V_0 has one extremum at $r = 3M$, which is a maximum.

Suppose $r(0) > 3M$, $\dot{r}(0) < 0$. Then we distinguish three cases:

- $E^2 > V(3M)$: the geodesic enters the black hole region;
- $E^2 < V(3M)$: $r(s)$ decreases until $V_0(r(s)) = E^2$ and then $r(s)$ increases and $r(s) \to \infty$;
- $E^2 = V(3M)$: the geodesic approaches $r = 3M$.

The timelike hypersurface $\{r = 3M\}$ is called the *photon sphere* or *light ring* and it admits solutions $\gamma_{3M}(s)$ with constant radius (circular orbits):

$$\gamma_{3M}(s) \cong \begin{pmatrix} 3Es \\ 3M \\ \frac{\pi}{2} \\ \frac{1}{9}M^{-2}Ls \mod 2\pi \end{pmatrix}.$$

We refer to γ_{3M} as *trapped null geodesics*.

Consider a geodesic with $E^2 = V(3M)$. Since a generic small perturbation of the initial data $(\gamma(0), \dot{\gamma}(0))$ for the geodesic equation results in $E^2 \neq V(3M)$ and therefore corresponds to a geodesic that either enters the black hole region or escapes to infinity, we have that the (asymptotically) circular orbits are *unstable*.

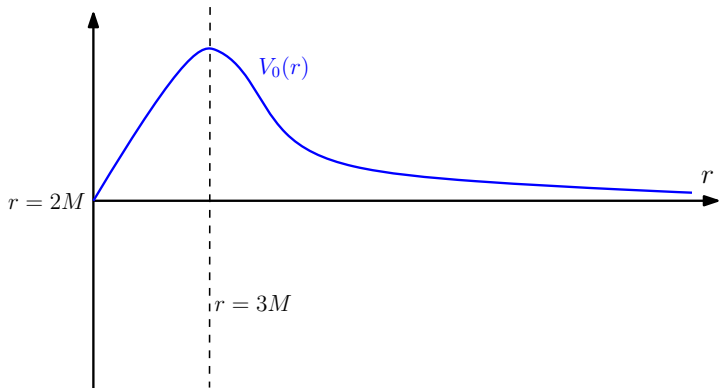


Figure 12: The potential $V_0(r)$.

Timelike geodesics Now consider the case $\sigma = 1$. Then

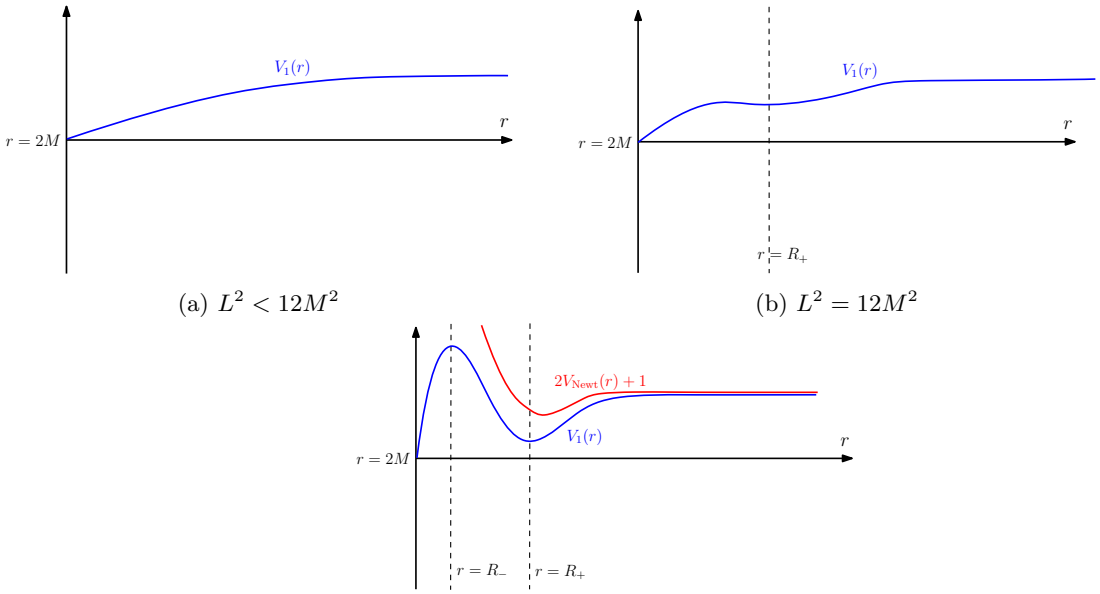
$$V_1(r) = 1 - \frac{2M}{r} + \frac{L^2}{r^2} - \frac{2L^2M}{r^3}.$$

We can compare $V_1(r)$ to the potential appearing in the 1-dimensional reduction of the following problem: a test particle of mass 1 in a gravitational force field created by a body of mass M in Newton's theory of gravity. In that case, the distance r between the test particle and the body satisfies $\frac{1}{2}\dot{r}^2 + V_{\text{Newt}}(r) = \tilde{E}$, with $\tilde{E}$ the total energy and

$$2V_{\text{Newt}}(r) + 1 = 1 - \frac{2M}{r} + \frac{L^2}{r^2}.$$

Label $E^2 = 2\tilde{E} + 1$. Then $\dot{r}^2 + (2V_{\text{Newt}}(r) + 1) = E^2$. The key difference with $V_1(r)$ is the term $-\frac{2L^2M}{r^3}$, which becomes dominant for small r. We have that $V_1(2M) = 0$ and $V_1(r) \to 1$ as $r \to \infty$. Furthermore,

$$V_1'(r) = \frac{2M}{r^2} - \frac{2L^2}{r^3} + \frac{6L^2M}{r^4} = 2r^{-4}(Mr^2 - L^2r + 3L^2M).$$



(a) $L^2 < 12M^2$ (b) $L^2 = 12M^2$

(c) $L^2 > 12M^2$

Figure 13: The potential $V_1(r)$.

Hence, $V_1(r)$ admits at most two extrema at radii $r = R_+$ and $r = R_+$, with

$$R_\pm = \frac{L^2}{2M}(1 \pm \sqrt{1 - 12M^2L^{-2}}).$$

If $L^2 < 12M^2$, then there are no extrema. If $L^2 = 12M^2$, then there is exactly one extremum at $r = \frac{L^2}{2M} = 6M$. Since

$$V_1''(r) = 2r^{-5}(-2Mr^2 + 3L^2r - 12L^2M),$$

one can verify that in the case $L^2 = 12M^2$:

$$V_1''(6M) = 0.$$

Hence, the extremum is a saddle point.

If $L^2 > 12M^2$, then there are exactly two extrema: R_- is a maximum and R_+ is a minimum.

As in the case of null geodesics, when $E = V_1(R_\pm)$, there are geodesics that stay fixed at $r = R_\pm$. Since $r = R_-$ is a maximum of the potential, these geodesics correspond to unstable circular orbits. As $r = R_+$ is a maximum, for any $\epsilon > 0$, all sufficiently small perturbations of the initial data $(\gamma(0), \dot\gamma(0))$ will result in geodesics which stay in the region $R_+ - \epsilon \leq r \leq R_+ + \epsilon$ and therefore correspond to bounded orbits (that need not be elliptical!).[28]

4.3.2 Red-shift

Consider the region of the maximally-extended Schwarzschild spacetime that is spanned by ingoing Eddington–Finkelstein coordinates (v, r, θ, φ).

[28] In the Newtonian two-body problem, such orbits are closed and form ellipses. In the Schwarzschild case, this is no longer the case. It can be shown that for $R_+M^{-1} \gg 1$, the orbits trace to leading-order ellipses whose periapsis (the line segment corresponding to the shortest distance between the two massive bodies) precesses (rotates). Historically, this deviation from Newtonian theory was used to explain the anomalous precession of the perihelion (the periapsis where the larger body is the sun) of Mercury orbiting around the sun, which could not be accounted for using solely Newtonian theory and the effects of the other planets in the solar system.

The global red-shift effect There is a *global red-shift effect* present in the Schwarzschild black hole exterior. Consider an observer, Alice, entering the black hole and an observer Bob, who stays outside of the black hole. Suppose Alice emits a photon at the spacetime point $(u, v_A, \theta_0, \varphi_0)$ moving in the radial direction, which is intercepted by Bob at the spacetime point $(u, v_B, \theta_0, \varphi_0)$.

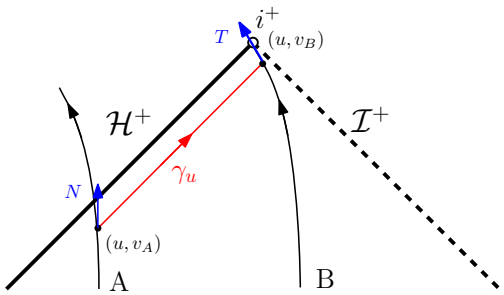


Figure 14: Alice emitting a photon travelling at constant (θ, φ), represented by γ_u, that is intercepted by Bob, who stays in the black hole exterior. Bob measures a red-shift of the energy of the photon.

We describe the photon by an outgoing null geodesic γ_u. As we already saw, V', defined via:

$$\frac{dV'}{dv} = 1 - \frac{2M}{r(u,v)},$$

defines an affine parameter along γ_u. Hence,

$$\dot{\gamma}_u = \partial_{V'} = \frac{1}{1 - \frac{2M}{r(u,v)}} \partial_v.$$

Let us assume for the sake of simplicity that the timelike curve describing Alice's spacetime path is an integral curve of the vector field

$$N = \partial_v - \partial_r$$

in (v, r) coordinates, in a neighbourhood of the event horizon.

[Exercise: Show that N is timelike and that with respect to Eddington–Finkelstein double null coordinates (u, v):

$$N = \left(1 + \frac{2}{(1 - 2Mr^{-1})}\right) \partial_u + \partial_v = \frac{1}{(1 - 2Mr^{-1})} \left(3 - 2Mr^{-1}\right) \partial_u + \partial_v.]$$

Then the energy of the photon, according to Alice is given by the expression:

$$E_{\gamma_u, A} = -g(\dot{\gamma}_u, N)(u, v_A, \theta_0, \varphi_0) = -\frac{1}{(1 - \frac{2M}{r})^2} \left(3 - 2Mr^{-1}\right) g_{uv}\Big|_{r=r(u,v_A)}$$
$$= \frac{1}{2(1 - \frac{2M}{r(u,v_A)})} \left(3 - 2Mr^{-1}\right)(u, v_A).$$

We assume also, for the sake of simplicity that the timelike curve describing Bob's spacetime path is tangent to $T = \partial_u + \partial_v$. Then the energy of the photon, according to Bob is:

$$E_{\gamma_u, B} = -g(\dot{\gamma}_u, T)(u, v_B, \theta_0, \varphi_0) = -\frac{1}{1 - \frac{2M}{r}} g_{uv}\Big|_{r=r(u,v_B)} = \frac{1}{2}.$$

The ratio of the two energies is given by

$$\frac{E_{\gamma_u,B}}{E_{\gamma_u,A}} = \frac{1-\frac{2M}{r}}{3-\frac{2M}{r}}(u, v_A).$$

Observe:

1. $E_{\gamma_u,B} < E_{\gamma_u,A}$,
2. $\frac{E_{\gamma_u,B}}{E_{\gamma_u,A}} \to 0$ as $u \to \infty$.

We say the energy of the photon is *red-shifted* and there is an *infinite red-shift* in the limit of the photon emission occurring as Alice crosses the event horizon of the black hole.

This terminology can motivated by the following relation between the energy E of a photon and its frequency ω (in radians/second): $E = \hbar\omega$, with $\hbar$ the reduced Planck constant. That is to say, a decrease in energy corresponds to a lowering in frequency, which in turn means a shift to the red part of the spectrum.

Bob will eventually see Alice frozen in place, turning redder and redder, until the light emanating from her enters the infrared spectrum. Alice on the other hand, will reach $r = 0$ in finite time. For this reason, black holes arising from a star gravitationally collapsing to a Schwarzschild black hole were also called "frozen stars" (before the term "black hole" was coined).[29]

Remark 4.1. *We do not need to invoke the concept of photons (or quantum physics) and the formula $E = \hbar\omega$ to describe the red-shift of light. From the perspective of the Maxwell equations, the dynamics of the electromagnetic tensor F on a spacetime $(\mathcal{M}, g)$ are governed by wave equations. For simplicity, we consider the scalar wave equation $\Box_g\phi = 0$ as a toy model.*

The statement that null geodesics describe the propagation of light can be phrased as follows. Let γ be a null geodesic in $(\mathcal{M}, g)$. Then there exist solutions ϕ to $\Box_g\phi = 0$ that can be approximated for an arbitrarily long amount of time by functions of the form $\phi_\lambda = a \cdot e^{-i\lambda\psi}$ ("Gaussian beams"), with a, ψ complex-valued functions and λ a positive parameter, such that $\psi|_\gamma, d\psi|_\gamma$ are real-valued and $(d\psi)^\sharp|_\gamma$ can be approximated by $-\dot{\gamma}$, provided we take λ to be sufficiently large. Furthermore, for suitably large λ, $|\phi|$ can also be made very small away from γ. One may think of the above as a more robust analogue of the "geometric optics approximation" and interpret ϕ as "high-frequency" solutions that are centred around γ.

Note that the relevant length scale on Schwarzschild is set by the choice of M. "High frequency" ω in that context means: $\omega M \gg 1$. Note that if we take M to be the mass of the sun and ω to be the frequency of the same order as visible light, we would obtain ωM of order 10^{10}.

The energy of ϕ is the integral of $\mathbf{T}[\phi](N, N)$ (the energy density) along appropriate spacelike slices and it can then be shown to be close to $-g(\dot{\gamma}, N)$, with N a timelike vector field whose integral curves represent a family of observers. Furthermore, since $\lambda\psi|_\gamma$ is the phase of wave ϕ at γ, its frequency according to N satisfies: $N(\lambda\psi)|_\gamma = \lambda d\psi(N)|_\gamma \approx -\lambda g(\dot{\gamma}, N)$, which is therefore approximately proportional to its the energy of the wave.

The horizon red-shift effect Recall that the Killing vector field $T = \partial_v$ in (v, r) coordinates is tangential to $\mathcal{H}^+_R$.

Since T is null along $\mathcal{H}^+_R$ and independent of the angular coordinates, we must have that for all $p \in_{\mathcal{H}^+_R}$: $(\nabla_T T)_p = f(v(p))T_p$ with $f \in C^\infty(\mathbb{R})$ (Exercise/problem sheet). Furthermore, in (v, r) coordinates, using that $(g_M)_{vr} = 1$, the Levi-Civita connection is metric-preserving and $\nabla_X Y - \nabla_Y X = [X, Y]$ for $X, Y \in \mathcal{T}(\mathcal{M})$:

[29] We will discuss gravitational collapse in a highly idealized model later in the course.

$$f(v) = g_M(f(v)\partial_v, \partial_r)|_{\mathcal{H}^+_R} = g_M(\partial_r, \nabla_{\partial_v}\partial_v)|_{\mathcal{H}^+_R} = \overbrace{\partial_v(g_M(\partial_v, \partial_r))|_{\mathcal{H}^+_R}}^{=0} - g(\nabla_{\partial_v}\partial_r, \partial_v)|_{\mathcal{H}^+_R}$$
$$= -\overbrace{g_M([\partial_v, \partial_r], \partial_v)|_{\mathcal{H}^+_R}}^{=0} - g_M(\nabla_{\partial_r}\partial_v, \partial_v)|_{\mathcal{H}^+_R} = -\frac{1}{2}\partial_r((g_M)_{vv})|_{\mathcal{H}^+_R} = \frac{M}{r^2}|_{\mathcal{H}^+_R} = \frac{1}{4M}.$$

In particular, for $p \in \mathcal{H}^+_R$: $(\nabla_T T)_p = \kappa_+ T_p$, where $\kappa_+ = \frac{1}{4M}$ is called the *surface gravity* of the Schwarzschild event horizon.

Since $\kappa_+ \neq 0$, T_p with $p \in \mathcal{H}^+_R$ does not correspond to $\dot{\gamma}_{\gamma(s)}$ with γ an affinely-parametrized null geodesic. However, we have that

$$\nabla_{e^{-\kappa_+ v}\partial_v}(e^{-\kappa_+ v}\partial_v)|_{\mathcal{H}^+_R} = 0.$$

Hence, we can reparametrize the integral curves γ of ∂_v along $\mathcal{H}^+_R$ to obtain an affinely-parametrized null geodesic γ_{2M} with parameter $s(v) = \kappa_+^{-1}(e^{\kappa_+ v} - 1)$. More explicitly, we obtain the following null geodesics $\gamma_{2M} : [0, \infty) \to \mathcal{M}_{\text{Krus}}$ that are tangent to the future-event horizon $\mathcal{H}^+$: with respect to the coordinate chart $\phi = (v, r, \theta, \varphi)$:

$$(\phi \circ \gamma_{2M})(s) = \begin{pmatrix} \kappa_+^{-1}\log(\kappa_+ s + 1) \\ 2M \\ \theta_0 \\ \varphi_0 \end{pmatrix},$$

with $\theta_0 \in (0, \pi)$ and $\varphi_0 \in (0, 2\pi)$ arbitrary constants.

Although the geodesics γ_{2M} are "trapped" at $r = 2M$, it turns out their energy decays exponentially with respect to v. We define energy with respect to the timelike vector field $N = \partial_v - \partial_r$.

Note that $\mathcal{L}_T N = [T, N] = 0$, so N is time-translation invariant. We can think of the integral curves of N as representing timelike observers crossing $\mathcal{H}^+$. The energy of γ_{2M} with respect to N is as follows:

$$E_{\gamma_{2M}}(v) = -g_{\gamma_{2M}(s(v))}(\dot{\gamma}_{2M}(s(v)), N_{\gamma_{2M}(s(v))}) = -g_{\gamma_{2M}(s(v))}(e^{-\kappa_+ v}(\partial_v)_{\gamma_{2M}(s(v))}, (\partial_v - \partial_r)_{\gamma_{2M}(s(v))})$$
$$= e^{-\kappa_+ v}.$$

The exponential decay of the above energy is called the *horizon red-shift effect* and it is closely related to the non-vanishing of the surface gravity κ_+. We will later see that there are black hole spacetimes that have a vanishing κ_+ and a vanishing horizon red shift effect, but, due to the presence of an event horizon, they *do* have a non-vanishing global red-shift effect.

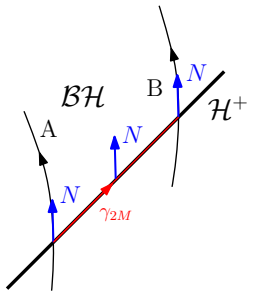


Figure 15: Alice emitting a photon travelling along the event horizon, which is intercepted by Bob as he crosses the event horizon, who measures a red-shift of the energy of the photon.

4.3.3 Black hole interior

Consider the black hole interior region $\mathcal{M}_{\text{int}} := \{U > 0, V > 0\} \subset \mathcal{M}_{\text{Krus}}$, where $r < 2M$. With respect to (t', r, θ, φ) coordinates, we can express:

$$g_M = -\left(\frac{2M}{r} - 1\right)^{-1} dr^2 + \left(\frac{2M}{r} - 1\right) dt'^2 + r^2 \mathring{g}.$$

As in the black hole exterior, we obtain the following equation for timelike curves γ, parametrized by their proper time s:

$$-1 = (g_M)_{\gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)) = \left(\frac{2M}{r(s)} - 1\right) \dot{t'}^2(s) - \left(\frac{2M}{r(s)} - 1\right)^{-1} \dot{r}^2(s) + r^2(s)\dot{\theta}^2(s) + r^2(s)\sin^2\theta(s)\dot{\varphi}^2(s), \tag{4.5}$$

[Exercise: Show that ∂_r, with respect to (t', r, θ, φ) coordinates is a past-directed timelike vector field.]

Since ∂_r past-directed and timelike, future-directed causal curves must satisfy the following inequality:

$$0 < (g_M)_{\gamma(s)}((\partial_r)_{\gamma(s)}, \dot{\gamma}(s)) = -\left(\frac{2M}{r(s)} - 1\right)^{-1} \dot{r}(s)$$

and therefore, $\dot{r}(s) < 0$. We will now show that $r = 0$ is attained at a finite value of s denoted $s_{\max}$.

By (4.5), we have that

$$\dot{r}^2 \geq \left(\frac{2M}{r} - 1\right),$$

with equality if the curve is an integral curve of ∂_r. Since moreover $\dot{r} \leq 0$, we in fact have that:

$$\dot{r} \leq -\sqrt{\frac{2M}{r} - 1}$$

and therefore

$$s(r) = -\int_r^{r(0)} \frac{ds}{dr}(r)\, dr' \leq \int_r^{r(0)} \frac{1}{\sqrt{2Mr'^{-1} - 1}}\, dr \leq \frac{1}{\sqrt{2Mr^{-1}(0) - 1}}(r(0) - r),$$

so $s_{\max} \leq \frac{r(0)}{\sqrt{2Mr^{-1}(0)-1}}$.

We conclude that observers, represented by timelike curves, must reach the Schwarzschild singularity at $r = 0$ in finite proper time, bounded above by $s_{\max}$. No matter how much they try to accelerate away from the singularity, they are doomed to reach the end of the spacetime in finite time!

4.4 Reissner–Nordström spacetimes

Schwarzschild spacetimes can be embedded into the larger Reissner–Nordström family of spacetimes (Reissner 1916, Weyl 1917, Nordström 1918), which solve the Einstein–Maxwell equations with $F_{Q,P} = \frac{Q}{r^2} dt \wedge dr + P \sin\theta d\theta \wedge d\varphi$, where $Q \in \mathbb{R}$ is called the *electric charge* of the spacetime and $P \in \mathbb{R}$ is called the *magnetic charge*. We define $e = \sqrt{Q^2 + P^2}$. Reissner–Nordström spacetimes are pairs $(\mathcal{M}_{\text{ext}}, g_{M,e})$, with $\mathcal{M}_{\text{ext}} = \mathbb{R} \times (r_+, \infty) \times \mathbb{S}^2$ and

$$g_{M,e} = -D dt^2 + D^{-1} dr^2 + r^2 \mathring{g},$$

$$D(r) = 1 - \frac{2M}{r} + \frac{e^2}{r^2},$$

with $r_\pm = M \pm \sqrt{M^2 - e^2}$ the roots of the polynomial $r^2 D(r)$ in the case $M \geq e$, otherwise $r_+ = 0$. If $M = e$ (so $r_+ = r_-$) we say the spacetime is **extremal**, if $M < e$, we say the spacetime is **sub-extremal**, if $M > e$, then we says the spacetime is **super-extremal**.

Just like Schwarzschild spacetimes, Reissner–Nordström spacetimes are static and the angular momentum vector fields are Killing.

In order to extend the spacetime from $r_+ < r < \infty$ to $r_- < r < \infty$, we will apply the same strategy as in Schwarzschild. First, we can introduce double null coordinates (u, v) with respect to which:

$$g_{M,e} = -\Omega^2 du dv + r^2 \mathring{\not{g}},$$

with $\Omega^2 = 1 - \frac{2M}{r} + \frac{e^2}{r^2}$ by defining

$$\begin{aligned} u =& t - r_*, \\ v =& t + r_*, \\ \frac{dr_*}{dr} =& \frac{1}{D(r)} = \frac{1}{1 - \frac{2M}{r} + \frac{e^2}{r^2}}. \end{aligned}$$

[Exercise: Verify that r_* is given by the following expression (up to the addition of a constant):

$$\begin{aligned} r_*(r) =& r + \frac{r_+^2}{2\sqrt{M^2 - e^2}} \log\left(\frac{r - r_+}{r_+}\right) - \frac{r_-^2}{2\sqrt{M^2 - e^2}} \log\left(\frac{r - r_-}{r_-}\right) \quad \text{if} \quad e < M, \\ r_*(r) =& r - \frac{M^2}{r - M} + 2M \log(r - M) \quad \text{if} \quad e = M, \\ r_*(r) =& r + \frac{\left(2M^2 - e^2\right) \arctan\left(\frac{r - M}{\sqrt{e^2 - M^2}}\right)}{\sqrt{e^2 - M^2}} + M \log\left(r^2 - 2Mr + e^2\right) \quad \text{if} \quad e > M.] \end{aligned}$$

Before defining Kruskal coordinates, we first switch to ingoing Eddington–Finkelstein coordinates (v, r, θ, φ) to obtain:

$$g_{M,e} = -D dv^2 + 2 dv dr + r^2 \mathring{\not{g}}.$$

for $0 < r < \infty$.

Let $M \geq e$. As in Schwarzschild, we define in these coordinates the null hypersurface $\mathcal{H}_R^+ := \{r = r_+\}$ and $T = \partial_v$, and we observe that for all $p \in \mathcal{H}_R^+$, $(\nabla_T T)_p = \kappa_+ T_p$, with the surface gravity κ_+ taking on the values:

$$\kappa_+ = -\frac{1}{2} \partial_r (g_{vv})|_{\mathcal{H}_R^+} = \frac{1}{2r_+^2}(2M - 2e^2 r_+^{-1}) \overbrace{=}^{e^2 = 2Mr_+ - r_+^2} \frac{1}{2r_+^2}(2r_+ - 2M) = \frac{\sqrt{M^2 - e^2}}{r_+^2} = \frac{r_+ - r_-}{2r_+^2}.$$

Note in particular, the in the extremal case, $\kappa_+ = 0$. Defining moreover $\kappa_- = \frac{r_- - r_+}{2r_-^2}$, we can rewrite r_* as follows in the sub-extremal case:

$$r_*(r) = r + \frac{1}{2\kappa_+} \log\left(\frac{r - r_+}{r_+}\right) + \frac{1}{2\kappa_-} \log\left(\frac{r - r_-}{r_-}\right).$$

As in the Schwarzschild case, we consider the null curves $\{v = v_0, \theta = \theta_0, \varphi = \varphi_0\}$ and observe that the affine parameter is given by: $\frac{dU'}{du}(u) = \Omega^2(u, v_0)$. We will now restrict ourselves to the sub-extremal case with non-vanishing e: $0 < e < M$. We can express: Furthermore,

$$e^{\kappa_+(v-u)} = e^{2\kappa_+ r_*} = e^{2\kappa_+ r} \left(\frac{r - r_+}{r_+}\right) \left(\frac{r - r_-}{r_-}\right)^{\frac{\kappa_+}{\kappa_-}}. \tag{4.6}$$

Hence

$$\frac{dU'}{du}(u) = \Omega^2(u, v_0) = r^{-2}(r-r_+)(r-r_-) = \left[r^{-2} r_+ e^{-\kappa_+ r} \left(\frac{r-r_-}{r_-} \right)^{-\frac{\kappa_+}{\kappa_-}} (r-r_-) \right] (u, v_0) e^{\kappa_+(v_0-u)}$$
$$= r_+^{-1}(r_+ - r_-) e^{-\kappa_+ r_+} \left(\frac{r_+ - r_-}{r_-} \right)^{-\frac{\kappa_+}{\kappa_-}} e^{\kappa_+ v_0} e^{-\kappa_+ u} + O(e^{-2\kappa_+ u}).$$

This motivates the following Kruskal coordinate:

$$U(u) = -e^{-\kappa_+ u}.$$

Similarly, we define $V(v) = e^{\kappa_+ v}$. Note that

$$-UV = e^{\kappa_+(v-u)} = r^2\Omega^2(u,v) r_+^{-1} r_-^{-1} e^{\kappa_+ r} \left(\frac{r-r_-}{r_-} \right)^{\frac{\kappa_+}{\kappa_-}-1} = r_+^{-1} r_-^{-1} e^{\kappa_+ r} (r-r_+) \left(\frac{r-r_-}{r_-} \right)^{\frac{\kappa_+}{\kappa_-}}$$

and $\frac{dU}{du} = -\kappa_+ U$, $\frac{dV}{dv} = \kappa_+ V$.

We can then express:

$$g_{M,e} = -\Omega^2(u,v) \frac{du}{dU} \frac{dv}{dV} dU dV + r^2 \mathring{\not{g}} = -\frac{r_+ r_-}{\kappa_+^2 r^2 e^{\kappa_+ r}} \left(\frac{r-r_-}{r_-} \right)^{1-\frac{\kappa_+}{\kappa_-}} dU dV + r^2 \mathring{\not{g}}.$$

We can extend $r(U,V)$ analytically to $U \geq 0$ and $V \leq 0$ and therefore extend $(\mathcal{M}_{\text{ext}}, g_{M,e})$ analytically to the manifold:

$$\mathcal{M}_{\text{Krus}} = \mathbb{R}_U \times \mathbb{R}_V \times \mathbb{S}^2_{\theta,\varphi}.$$

Rescaling $\tilde{U} = \arctan U$ and $\tilde{V} = \arctan V$, we obtain a Penrose diagram of Reissner–Nordström spacetimes with $0 < e < M$. Compared to the Schwarzschild diagram, we have the following

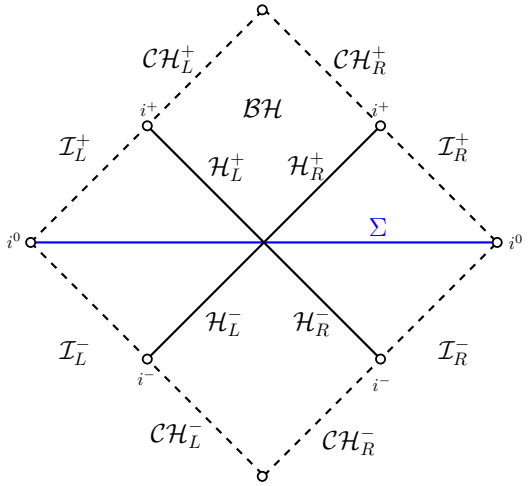


Figure 16: Sub-extremal Reissner–Nordström in the case $0 < e < M$, with Cauchy hypersurface $\Sigma = \{U + V = 0\}$.

additional boundary components:

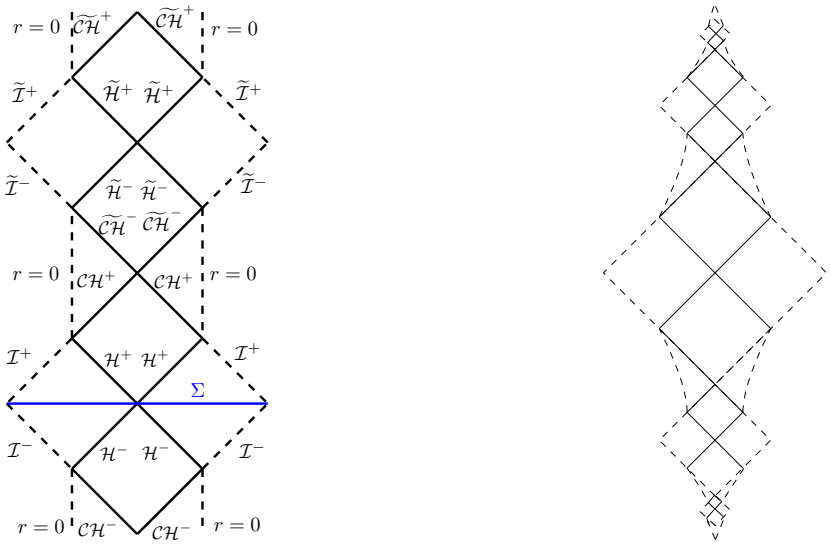
$$\mathcal{CH}_L^{\pm} := \{\tilde{U} = \pm\frac{\pi}{2}, \pm U > 0\},$$

$$\mathcal{CH}_R^{\pm} := \{\tilde{V} = \pm\frac{\pi}{2}, \pm V > 0\}.$$

Note that $\mathcal{CH}_L^{\pm}$ and $\mathcal{CH}_R^{\pm}$ are the Cauchy horizons corresponding to the Cauchy hypersurface $\Sigma = \{U + V = 0\}$. The spacetime $(\mathcal{M}_{\text{Krusk}}, g_{M,e})$ is therefore globally hyperbolic. Furthermore, since they correspond to $\pm U \to \infty$ and constant V or $\pm V \to \infty$ and constant U, we can apply (4.6) to conclude that $r \to r_-$ as we approach the Cauchy horizons.

In (v, r, θ, φ) coordinates, we were able to consider also $0 < r < r_-$. Indeed, in these coordinates $\mathcal{CH}_L^+ = \{r = r_-\}$ is a smooth (in fact analytic) hypersurface of an extended manifold. Similarly, in (u, r, θ, φ) coordinates $\mathcal{CH}_R^+ = \{r = r_-\}$ is a smooth (in fact analytic) hypersurface of an extended manifold obtained by considering (u, r, θ, φ) coordinates in the region $r < r_+$.

We can similarly draw a (partial) Penrose diagram for the analytic extensions of $\mathcal{M}_{\text{Krus}}$ across $\mathcal{CH}_L^{\pm}$ and $\mathcal{CH}_R^{\pm}$.



(a) A partial Penrose diagram of the maximal analytic extension of sub-extremal Reissner–Nordström spacetimes.

(b) A full Penrose diagram of the maximal analytic extension of sub-extremal Reissner–Nordström spacetimes.

Figure 17: The boundary in the full Penrose diagram has a fractal character as it consists of an infinite sequence of null corners and timelike segments. The un-dashed null lines are copies of the horizons $\mathcal{H}^+ \cup \mathcal{H}^-$ and $\mathcal{CH}^+ \cup \mathcal{CH}^-$.

We will later see that a globally hyperbolic spacetime, like $(\mathcal{M}_{\text{Krusk}}, g_{M,e})$, arises uniquely from the time evolution of appropriate initial data posed along a Cauchy hypersurface, like Σ. Extensions across the Cauchy horizons are independent of the initial data are therefore highly non-unique.[30] Uniqueness can be obtained by restricting the analytic extensions, but this is a very unphysical regularity class to restrict to as it is incompatible with the finite speed of propagation property embedded in the Einstein equations. Indeed, knowledge of the metric on a open subset $S \subset \mathcal{M}$, would completely determine the spacetime globally, rather than only affect the part of the spacetime in the causal future $J^+(S)$.

[30]Note that in the particular case of Reissner–Nordström imposing the additional restriction of spherical symmetry of smooth extensions actually singles out the unique analytic extension as the only possibility satisfying the Einstein–Maxwell equations. We will encounter a similar "rigidity" property in vacuum in the context of Birkhoff's theorem. When allowing for extensions that are not spherically symmetric, however, we can construct infinitely many extensions satisfying the Einstein–Maxwell equations.

[Exercise: Consider the null hypersurface $H = \{v = v_0, 0 < r < \infty\}$ in (v, r, θ, φ) coordinates. Determine the globally hyperbolic spacetime region in the analytic extension of $\mathcal{M}_{\text{Krusk}}$ which has H as a Cauchy hypersurface and draw this region in a Penrose diagram.]

[Exercise: Construct the analytic extension of extremal Reissner–Nordström exterior spacetimes $(\mathcal{M}_{\text{ext}}, g_{M,e})$ across $\{r = r_+\}$. Construct a Penrose diagram of this extended spacetime and discuss the singularity properties at the boundary of the Penrose diagram.]

[Exercise: Construct a Penrose diagram of super-extremal Reissner–Nordström exterior spacetime $(\mathcal{M}_{\text{ext}}, g_{M,e})$. Describe the properties of possible analytic extensions of these spacetimes.]

5 Spherically symmetric spacetimes

In this section, we will focus on spherically symmetric spacetimes. We will see that the vacuum Einstein equations have a strong *rigidity* property in spherical symmetry, i.e. the only possible spherically symmetric vacuum spacetimes are isometric to a subset of the Schwarzschild spacetime family or the Minkowski spacetime.

Via the inclusion of suitable matter, however, we can nevertheless see many of the interesting phenomena characteristic to dynamical black holes spacetimes.

We will give a precise definition of what we mean by a "spherically symmetric" spacetime. First, we need to introduce some notation.

Consider the manifold $\mathbb{R}^2 \times \mathbb{S}^2$ and the associated canonical projection maps:

$$\begin{aligned} \pi : \mathbb{R}^2 \times \mathbb{S}^2 &\to \mathbb{R}^2, \\ \pi(x, y) &= x, \\ \pi_{\mathbb{S}^2} : \mathbb{R}^2 \times \mathbb{S}^2 &\to \mathbb{S}^2, \\ \pi_{\mathbb{S}^2}(x, y) &= y. \end{aligned}$$

Now consider the manifold $\mathbb{R}_t \times \mathbb{R}^3$. Then we define $\pi : \mathbb{R}_t \times \mathbb{R}^3 \to \mathbb{R} \times [0, \infty)$ as the map that sends all points in the spheres $S^2_{t',r}$ of area radius $r = \sqrt{x^2 + y^2 + z^2}$ in $\{t = t'\}$, to the point $(t', r) \in \mathbb{R} \times (0, \infty)$ and $\pi(t', 0) = (t', 0)$. We also define $\pi_{\mathbb{S}^2} : \mathbb{R}_t \times (\mathbb{R}^3 \setminus \{0\}) \to \mathbb{S}^2$ as the projection map to the unit round sphere $S^2_{t,1}$.

Definition 5.1. *Let $(\mathcal{M}, g)$ be a Lorentzian manifold and assume that there exists a smooth diffeomorphism:*

A) *$\psi : \mathcal{M} \to \mathbb{R}^2 \times \mathbb{S}^2$, or*

B) *$\psi : \mathcal{M} \to \mathbb{R} \times \mathbb{R}^3$.*

*In case **B)** we denote*

$$\Gamma = \psi^{-1}(\mathbb{R} \times \{0\}).$$

We will refer to Γ as the centre of spherical symmetry.

We say $(\mathcal{M}, g)$ is a spherically symmetric Lorentzian manifold *if ψ and g satisfy the following properties:*

1. *The metric g on $\mathcal{M} \setminus \Gamma$ can be decomposed as follows:*

$$g = (\pi \circ \psi)^* \overline{g} + (r^2 \circ \pi \circ \psi) \cdot (\pi_{\mathbb{S}^2} \circ \psi)^* g_{\mathbb{S}^2}, \tag{5.1}$$

   *where $g_{\mathbb{S}^2}$ is the unit round 2-sphere metric, $\overline{g}$ is a smooth Lorentzian metric on **A)** $\mathbb{R}^2$ or **B)** $\mathbb{R} \times [0, \infty)$ and where **A)** $r : \mathbb{R}^2 \to (0, \infty)$ is a smooth function, or **B)** $r : \mathbb{R} \times [0, \infty) \to [0, \infty)$ is a smooth function with $r(t, 0) = 0$.*[31]

2. *Γ is a timelike curve in $(\mathcal{M}, g)$.*

*We will denote by $\mathcal{Q}$ the 2-dimensional manifolds(-with-boundary): **A)** $\mathbb{R}^2$ and **B)** $\mathbb{R} \times [0, \infty)$. In case **B)**, $\partial\mathcal{Q} = \mathbb{R} \times \{0\} \subset \mathbb{R}^2$, with respect to $\mathbb{R}^2$.*

In slight abuse of notation we will also denote by Γ the curve $\partial\mathcal{Q}$ in case **B**). To simplify the notation, we write the requirement (5.1) as follows:

$$g = \overline{g} + r^2 \mathring{g}.$$

[31] A metric of the above form is called a *warped product metric*. We write **A**) $\mathcal{M} \cong \mathbb{R}^2 \times_r \mathbb{S}^2$ or **B**) $\mathcal{M} \setminus \Gamma \cong (\mathbb{R} \times (0, \infty)) \times_r \mathbb{S}^2$. The regularity requirements on $\overline{g}$ and r can be relaxed, but we will assume smoothness for the sake of convenience.

Lemma 5.1. *There exists global coordinates $u, v : \mathcal{Q} \to \mathbb{R}$ and a C^2 function $\Omega^2 : \mathcal{Q} \to (0, \infty)$, such that*

$$\overline{g} = -\Omega^2 du dv.$$

Proof. (*non-examinable*) Let $p \in \mathcal{Q}$ be arbitrary. Let $\underline{L}_p$ and L_p be null tangent vectors at p, such that $\overline{g}_p(\underline{L}_p, L_p) = -2$ and consider corresponding affinely-parametrized null geodesics $\gamma_{\underline{L}}$ ("ingoing"') and γ_L ("outgoing") emanating from p, where the respective affine parameters u and v are fixed by taking: $\dot{\gamma}_{\underline{L}}(p) = \underline{L}_p$ and $\dot{\gamma}_L(p) = L_p$ and setting $(u, v) = 0$ at p. We will first construct double null coordinates (u, v) in a sufficiently small open neighbourhood U_p of p.

Denote by $\overline{\nabla}$ the covariant derivative associated to $\overline{g}$. By taking U_p, sufficiently small, we can extend $\underline{L}_p$ to $\gamma_{\underline{L}} \cap U_p$, by defining $\underline{L}_{\gamma(s)} := \dot{\gamma}_{\underline{L}}(s)$, or $\overline{\nabla}_{\underline{L}} \underline{L} = 0$ along $\gamma_{\underline{L}} \cap U_p$. Similarly, we extend L_p to $\gamma_L \cap U_p$ by demanding $\overline{\nabla}_L L = 0$. We also uniquely extend L to $\gamma_{\underline{L}} \cap U_p$ by demanding $\overline{\nabla}_{\underline{L}} L = 0$ along $\gamma_{\underline{L}} \cap U_p$, and we extend $\underline{L}$ to $\gamma_L \cap U_p$ by demanding $\overline{\nabla}_L \underline{L} = 0$ along $\gamma_L \cap U_p$. This guarantees that $\overline{g}(L, \underline{L}) = -1$ along $(\gamma_{\underline{L}} \cup \gamma_L) \cap U_p$.

Now consider the points $\gamma_{\underline{L}}(u) \cap U_p$. Then we can consider affinely parametrized null geodesics $\tilde{\gamma}_u$ with initial tangent vector $L_{\gamma(u)}$ emanating from the point $\gamma_{\underline{L}}(u)$. This provides a further extension of L to U_p that we will denote by L', with $L' = \dot{\tilde{\gamma}}_u$ along $\tilde{\gamma}_u$, provided we take U_p suitably small. We can carry out an analogous procedure for affinely parametrized null geodesics emanating from $\gamma_L(v)$ with initial tangent vector $\underline{L}$ to obtain a vector field $\underline{L}'$ in U_p.

Denote $2\Omega^{-2} := -\overline{g}(L', \underline{L}')$. Note that $\Omega^2 > 0$, since L' and $\underline{L}'$ cannot be proportional anywhere. In order to define double null coordinates u and v we rescale the vector fields L' and $\underline{L}'$ to obtain the following alternative extensions of L_p and $\underline{L}_p$ to U_p:

$$L = \frac{1}{2}\Omega^2 L' \quad \text{and} \quad \underline{L} = \frac{1}{2}\Omega^2 \underline{L}'.$$

[Exercise: Show that $\overline{g}([L, \underline{L}], L) = \overline{g}([L, \underline{L}], \underline{L}) = 0$ and hence $[L, \underline{L}] = 0$. Hint: Show first that $\overline{g}([L', \underline{L}'], L') = 2\Omega^{-4} L'(\Omega^2)$.]

We now extend u and v away from p as functions defined on U_p, by requiring $L(u) = 0$ and $\underline{L}(v) = 0$. Since $[L, \underline{L}] = 0$, it follows that (u, v) must be well-defined coordinates in U_p and that $L = \partial_v$ and $\underline{L} = \partial_u$.[32] Furthermore, since $\overline{g}(\underline{L}, L) = -\frac{1}{2}\Omega^2$, we can write in U_p:

$$\overline{g} = -\Omega^2(u, v)\, du dv.$$

In dimensions greater than 2, two ingoing (or two outgoing) null geodesic could in principle intersect, which would prevent us from extending the domain of double null coordinates to the full $\mathcal{Q}$. This happens, for example, at the vertex of a lightcone in Minkowski, so the corresponding double null coordinates would be ill-defined past the intersection point. The reason that this cannot happen in two dimensions is that each point $x \in \mathcal{Q}$ the subspace of $T_x\mathcal{Q}$ spanned by either ingoing or outgoing null geodesics is 1-dimensional. We will exploit this fact in the argument below.

We provide below a precise argument for extending the domain of the coordinates (u, v) constructed above in U_p to the full $\mathcal{Q}$. Consider the subset

$$\mathcal{D} = \{q \in \mathcal{Q} \,|\, \text{there exists a neighbourhood of } q \text{ covered by the } (u, v) \text{ coordinates}\}.$$

- Since $p \in \mathcal{D}$, $\mathcal{D} \neq \emptyset$.

[32] On a general n-manifold $\mathcal{M}$, if $X_1, \ldots, X_n$ are n linearly independent smooth vector fields in a neighbourhood of $p \in \mathcal{M}$, such that $[X_i, X_j] = 0$ for all $i, j \in \{1, \ldots, n\}$, then there exists a coordinate chart $(y^1, \ldots, y^n)$ in a neighbourhood of p such that $X_i = \frac{\partial}{\partial y^i}$. The idea of the proof is to use that the local flows corresponding to these vector fields commute.

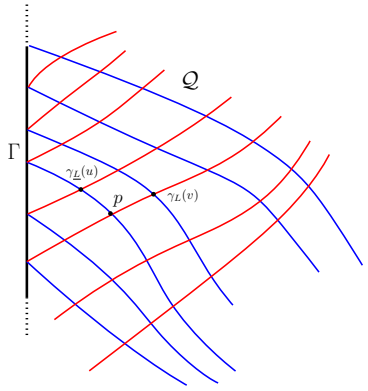


Figure 18: The set $\mathcal{Q}$ (with possible centre Γ) and an illustration of the global foliation by ingoing and outgoing null geodesics.

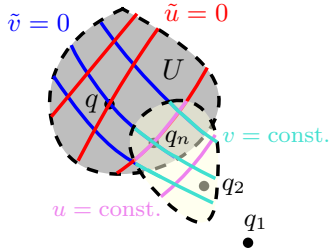


Figure 19: Showing that $\mathcal{D}$ is closed.

- Furthermore, if $q \in \mathcal{A}$, then we automatically get the existence of an open neighbourhood of q that is also contained in $\mathcal{A}$ by the way we defined $\mathcal{A}$, so $\mathcal{A}$ is open.
- Let $q_n \in \mathcal{A}$, such that $q_n \to q$ (convergence with respect to the $\mathbb{R}^2$ topology). We can repeat the above local construction of double null coordinates around q to obtain an open neighbourhood U_q of q covered by possible different double null coordinates $(\tilde{u}, \tilde{v})$, such that $q_n \in U_q$ for n suitably large. Since the tangent spaces $T_x\mathcal{Q}$ are 2-dimensional, the ingoing null geodesics in U with respect to $(\tilde{u}, \tilde{v})$ must agree with the null geodesics in a neighbourhood of q_n associated to (u, v) coordinates up to a reparametrization, and similarly, the outgoing null geodesics must agree up to a reparametrization. So $(\tilde{u}, \tilde{v})$ and (u, v) are related by a rescaling in u and v and a shift of the origin $(\tilde{u}, \tilde{v}) = 0$. Therefore the coordinates (u, v) can be extended to also cover U_q, so $q \in \mathcal{D}$. The set $\mathcal{D}$ is therefore also closed in $\mathcal{Q}$.

Since $\mathcal{D}$ is a non-empty, open and closed subset of the connected set $\mathcal{Q}$, it must be the whole $\mathcal{Q}$. □

The u-level sets and v-level sets in a spherically symmetric spacetime constitute a *global double null foliation* of the spacetime.

Remark 5.1. *The existence of global coordinates (u, v) such that $\overline{g} = -\Omega^2 dudv$ implies that the Lorentzian spacetime $(\mathcal{Q}, \overline{g})$ is conformally isometric to a subset of the 1+1-dimensional Minkowski spacetime.*[33] *This is a Lorentzian analogue of the Uniformization Theorem from Riemannian geometry, which can be stated as follows: let $(\mathcal{Q}, \overline{g})$ be an orientable compact 2-dimensional Riemannian*

[33] A conformal isometry between $(\mathcal{M}, g)$ and $(\mathcal{N}, \tilde{g})$ is a diffeomorphism $\psi : \mathcal{M} \to \mathcal{N}$ such that $\psi^*\tilde{g} = \Omega^2 g$ for some (smooth) function $\Omega^2 : \mathcal{M} \to (0, \infty)$.

manifold. Then $(\mathcal{Q}, \overline{g})$ is conformally isometric to an open subset of a surface of constant Gauss curvature with values 1, 0 or -1, i.e. a quotient of the unit round sphere $\mathbb{S}^2$, the Euclidean plane $\mathbb{E}^2$ or the hyperbolic plane $\mathbb{H}^2$.

- **We now fix the time orientation on $(\mathcal{M}, g)$ by declaring the vector field $\partial_u + \partial_v$ to be future-directed.** We will also assume that $\Phi_*(\partial_u + \partial_v)$ is future directed. This makes $(\mathcal{M}, g, [\partial_u + \partial_v])$ a *spherically symmetric spacetime.*
- We will also refer to ∂_v (and its integral curves) as "outgoing" and ∂_u (and its integral curves) as "ingoing".

5.1 Penrose diagrams

We now use the existence of global double null coordinates on spherically symmetric spacetimes to represent the spacetimes as bounded subsets of $\mathbb{R}^2$ via Penrose diagrams. As we will later see, Penrose diagrams remain an invaluable (schematic) tool even when considering more general spacetimes that are not spherically symmetric. The special property of spherically symmetric spacetimes is that they *always* admits global double null coordinates and it is this property that underlies the ability to construct Penrose diagrams, as we already saw in the examples of Minkowski, Schwarzschild and Reissner–Nordström.

Definition 5.2. *A* (Carter–)Penrose diagram *of a spherically symmetric spacetime $(\mathcal{M}, g)$ is a pair $(\Phi, \mathcal{PD})$, with $\mathcal{PD} \subset \mathbb{R}^2$ a bounded subset that is a manifold (with boundary, if $\partial\mathcal{Q} \neq \emptyset$, i.e. $\Gamma \neq \emptyset$), such that:*

1. $$\Phi : \mathcal{Q} \to \mathcal{PD}$$
   is a smooth diffeomorphism.
2. *Timelike curves in $(\mathcal{Q}, \overline{g})$ are mapped to timelike curves in $(\mathcal{PD}, -dudv)$ and null curves in $(\mathcal{Q}, \overline{g})$ are mapped to null curves in $(\mathcal{PD}, -dudv)$ with $-dudv$ the 1+1-dimensional Minkowski metric in standard double null coordinates.*

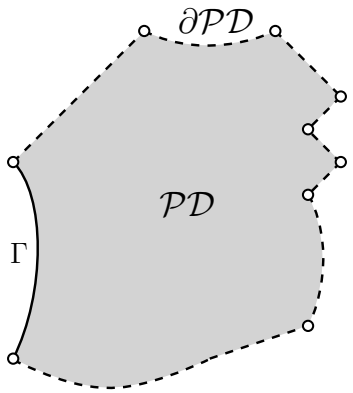


Figure 20: Example of a Penrose diagram of a spherically symmetric spacetime.

There is a slight ambiguity with what we mean by the boundary of the Penrose diagram $\partial\mathcal{PD}$. We will use $\partial\mathcal{PD}$ to denote the boundary of $\mathcal{PD}$ as subset of $\mathbb{R}^2$ and <u>not</u> as a manifold with boundary. For example, when $\Gamma = \partial\mathcal{Q} \neq \emptyset$, we have that $\Phi(\partial\mathcal{Q}) \subset \partial\mathcal{PD}$, but $\Phi(\Gamma) \neq \partial\mathcal{PD}$. And

when $\partial\mathcal{Q} = \emptyset$ we still have that $\partial\mathcal{PD} \neq \emptyset$. For example, when $(\mathcal{M}, g)$ is the maximally-extended Schwarzschild spacetime, we have that $\partial\mathcal{Q} = \emptyset$, but

$$\partial\mathcal{PD} = \mathcal{I}_L^+ \cup \mathcal{I}_R^+ \cup \mathcal{I}_L^- \cup \mathcal{I}_R^- \cup \mathcal{S}^+ \cup \mathcal{S}^- \cup i_L^+ \cup i_R^+ \cup i_L^- \cup i_R^- \cup i_L^0 \cup i_R^0 \neq \emptyset.$$

We will often refer to $\mathcal{PD}$ or $\overline{\mathcal{PD}}$ as a Penrose diagram and omit the map Φ.

Corollary 5.2. *Let $\phi : \mathcal{Q} \to \phi(\mathcal{Q}) \subset \mathbb{R}^2$ be a global double null coordinate chart. Define $(\Phi \circ \phi^{-1})(u, v) = (\arctan u, \arctan v)$. Then $(\Phi, \Phi(\mathcal{Q}))$ is a Penrose diagram of $(\mathcal{M}, g)$.*

A main advantage of Penrose diagrams is that causal curves in $(\mathcal{M}, g)$ are depicted by causal curves in $(\mathcal{PD}, -dudv)$.

Proposition 5.3.

1. *Timelike curves in $(\mathcal{M}, g)$ are represented by timelike curves in $(\mathcal{PD}, -dudv)$.*
2. *Γ is represented by a timelike curve in $(\mathcal{PD}, -dudv)$.*
3. *Null curves in $(\mathcal{M}, g)$ are represented by causal curves in $(\mathcal{PD}, -dudv)$. Radial null curves in $(\mathcal{M}, g)$ are represented by null curves in $\mathcal{PD}$.*
4. *A null curve in $(\mathcal{M}, g)$ that intersects with Γ at a single point $p \in \Gamma$ is represented by two causal curve segments in $(\mathcal{PD}, -dudv)$, such that one segment is tangent to a line of constant v at $\Phi(\pi(\psi(p)))$ and the other segment is tangent to a line of constant u at $\Phi(\pi(\psi(p)))$ (see Figure 21).*

Proof. Let $\gamma : I \to \mathcal{M}$ be a timelike curve. Then we can express with respect to the coordinate chart $\phi = (u, v, \theta, \varphi)$:

$$(\phi \circ \gamma)(s) = \begin{pmatrix} \gamma^u \\ \gamma^v \\ \gamma^\theta \\ \gamma^\varphi \end{pmatrix}(s)$$

and by the timelike property, we have that

$$0 > g_{\gamma(s)}(\dot\gamma(s), \dot\gamma(s)) = -\Omega^2(\gamma(s))\dot\gamma^u(s)\dot\gamma^v(s) + r^2(\gamma(s))((\dot\gamma^\theta)^2(s) + \sin^2\theta(\gamma(s))(\dot\gamma^\varphi)^2(s)).$$

Hence, the associated curve in $\mathcal{Q}$:

$$\overline{\gamma}(s) = \begin{pmatrix} \gamma^u \\ \gamma^v \end{pmatrix}(s)$$

must also satisfy $-\Omega^2\dot{\overline{\gamma}}^u\dot{\overline{\gamma}}^v < 0$, so $-\dot{\overline{\gamma}}^u\dot{\overline{\gamma}}^v < 0$, which means that $\overline{\gamma}$ is timelike with respect to $(\mathcal{Q}, -dudv)$ so $\Phi \circ \overline{\gamma}$ is also timelike with respect to $(\mathcal{PD}, -dudv)$ (after appropriately rescaling u and v).

Similarly, if γ is null, then by the above argument $-\Omega^2\gamma^u\gamma^v \leq 0$, so we can only say that $\overline{\gamma}$ is causal with respect to $(\mathbb{R}^2, -dudv)$, unless $\dot\gamma^\theta = \dot\gamma^\varphi = 0$, in which case $\overline{\gamma}$ must be null.

[Exercise: Show that Γ is represented by a timelike curve in $(\mathcal{PD}, -dudv)$. *Hint:* Suppose that $\partial\mathcal{Q}$ has a spacelike or null segment S and use that Γ timelike implies that there exist $p, q \in \Gamma$ and a timelike curve $\gamma : [0, 1] \to \mathcal{M}$, such that $\gamma(0) = p$, $\gamma(1) = q$ and $\gamma(s) \in \mathcal{M} \setminus \Gamma$ for $s \in (0, 1)$ to reach a contradiction.]

Now let $\gamma : (-1, 1) \to \mathcal{M}$ be a future-directed null curve in $\mathcal{M}$, such that $p = \gamma(0) \in \Gamma$. Then there exists a $\epsilon > 0$ such that $\gamma(s) \in \mathcal{M} \setminus \Gamma$ for $s \in (-\epsilon, 0) \cup (0, \epsilon)$. We can reparametrize the curve so that without loss of generality we may assume that $\gamma(s) \in \mathcal{M} \setminus \Gamma$ for $s \in (-1, 0) \cup (0, 1)$. Let $\overline{\gamma} : (-1, 0) \to \mathcal{PD}$ be defined as $\overline{\gamma}(s) = (\gamma^u(s), \gamma^v(s))$ and $\tilde\gamma : (0, 1) \to \mathcal{PD}$ and $\tilde\gamma$ be defined as $\tilde\gamma(s) = (\gamma^u(s), \gamma^v(s))$. By 3., $\overline{\gamma}$ and $\tilde\gamma$ must be causal.

By the null property of γ, we have that $g_p(\dot\gamma(0),\dot\gamma(0)) = 0$. Furthermore, without loss of generality, we can rescale our null coordinates, so that at $\Omega^2(u_p,v_p) = 1$, with (u_p,v_p) the limiting value of the (u,v) coordinates at p. Then, using moreover that $r|_{\partial\mathcal{Q}} \equiv 0$,

$$-\dot{\overline{\gamma}}^u(0)\dot{\overline{\gamma}}^v(0) = g_p(\dot\gamma(0),\dot\gamma(0)) = -\dot{\tilde\gamma}^u(0)\dot{\tilde\gamma}^v(0).$$

Hence, $\dot{\overline{\gamma}}^u(0) = 0$ or $\dot{\overline{\gamma}}^v(0) = 0$, and $\dot{\tilde\gamma}^u(0) = 0$ or $\dot{\tilde\gamma}^v(0) = 0$.

Without loss of generality, assume that γ is future-directed and that $(u_p,v_p) = (0,0)$. By 2., we can assume WLOG that $\Phi(\partial\mathcal{Q})$ bounds $\mathcal{PD}$ to the left. Suppose that $\dot{\overline{\gamma}}^u(0) = 0$. Then $\dot{\overline{\gamma}}^v(0) = \lambda > 0$, which means that $\overline{\gamma}^v(s) = \lambda s + O(s^2)$ and $\overline{\gamma}^u(s) = O(s^2)$ so $\overline{\gamma}^v(s) < \overline{\gamma}^u(s)$ for s close to 0 (since $s < 0$), which is in contradiction with the fact that $\Phi(\partial\mathcal{Q})$ is a timelike left-boundary. Hence, $\dot{\overline{\gamma}}^v(0) = 0$. Similarly, it follows that $\dot{\tilde\gamma}^u(0) = 0$. This concludes that $\overline{\gamma}$ must be ingoing null at (u_p,v_p). We can repeat this argument to conclude that $\tilde\gamma$ must be outgoing null at (u_p,v_p). □

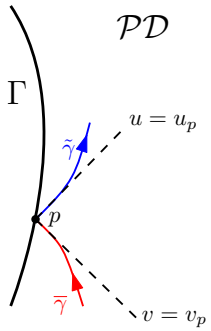


Figure 21: Representation of future-directed null curves intersecting Γ at $p \in \Gamma$.

[Exercise: Show that spacelike curves in $(\mathcal{M},g)$ need not be represented by spacelike curves in $(\mathcal{PD},-dudv)$.]

An important role will be played by the boundary $\partial\mathcal{PD}$ with respect to the ambient space $\mathbb{R}^2$. We can also investigate their spacelike, timelike or null nature, provided that they are sufficiently regular curves in $\mathbb{R}^2$. For example, $\mathcal{S}^\pm$ in the maximally-extended Schwarzschild spacetime are spacelike curves in $(\mathbb{R}^2,-dudv)$ and $\mathcal{I}^\pm$ are null curves in $(\mathbb{R}^2,-dudv)$.

An important type of Penrose diagram boundary segment is "null infinity", $\mathcal{I}$ (also called "scri"), which we already encountered in Minkowski, Schwarzschild and Reissner–Nordström.

Definition 5.3. *Let $p \in \partial\mathcal{PD}$. Choose coordinates (u,v) on $\mathcal{Q}$ such that $p = (0,0)$. Assume that there exists an $\epsilon > 0$, such that $(0,v) \in \mathcal{Q}$ for all $v \in (-\epsilon,0)$ or $v \in (0,\epsilon)$, or $(u,0) \in \mathcal{Q}$ for all $u \in (-\epsilon,0)$ or $u \in (0,\epsilon)$. We then define $\mathcal{I} \subseteq \partial\mathcal{PD}$ as follows: $p \in \mathcal{I}$ if and only if:*

$$\limsup_{v\to 0} r(0,v) := \lim_{V\to 0}\left[\sup_{|v|\le V} r(0,v)\right] = \infty \quad \text{or} \quad \limsup_{u\to 0} r(u,0) := \lim_{U\to 0}\left[\sup_{|u|\le U} r(u,0)\right] = \infty.$$

We refer to the subset $\mathcal{I} \subseteq \partial\mathcal{PD}$ as null infinity. *Suppose that $\mathcal{I}$ is achronal. Then we define $\mathcal{I}^+$,* future null infinity *("scri-plus"), and $\mathcal{I}^-$,* past null infinity *("scri-minus") as follows:*

- *$p \in \mathcal{I}^+ \subseteq \mathcal{I}$ if and only if*

$$\limsup_{v\uparrow 0} r(0,v) = \infty \quad \text{or} \quad \limsup_{u\uparrow 0} r(u,0) = \infty.$$

- $p \in \mathcal{I}^- \subseteq \mathcal{I}$ *if and only if*

$$\limsup_{v\downarrow 0} r(0,v) = \infty \quad \text{or} \quad \limsup_{u\downarrow 0} r(u,0) = \infty.$$

Penrose diagrams are very convenient tools for representing causal properties of hypersurfaces and open subsets of the manifold. Let $S \subset \mathcal{PD}$. Then S represents the set $\mathcal{M} \supset \Sigma := (\Phi \circ \pi \circ \psi)^{-1}(S)$. If S is a finite union of smooth curves (with boundaries), then Σ is a smooth hypersurface(-with-boundary) of $\mathcal{M}$.

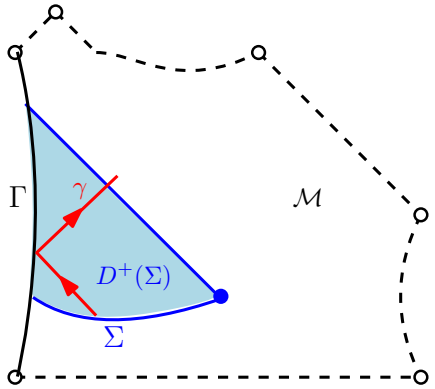


Figure 22: The representation of the domain of dependence of a hypersurface-with-boundary Σ in a spherically symmetric spacetime with $\Gamma \neq \emptyset$. The line γ represents a radial null curve passing through the centre of spherical symmetry Γ in $\mathcal{M}$.

For such sets we can characterize the causal/chronological future/past and the future/past domain of dependence on the Penrose diagram $\mathcal{PD}$:

Corollary 5.4.
- $J^{\pm}(\Sigma)$ *is represented by* $J^{\pm}(S)$ *with respect to* $(\mathcal{PD}, -dudv)$.
- $I^{\pm}(\Sigma)$ *is represented by* $I^{\pm}(S)$ *with respect to* $(\mathcal{PD}, -dudv)$.

Furthermore, since radial null geodesics in $(\mathcal{M}, g)$ *are represented by lines in* $\mathcal{Q}$ *that are reflected at* Γ*, we have to be a little more careful when discussing the domain of dependence. We have that:*

- *If* $\Sigma \cap \Gamma = \emptyset$*, then* $D^{\pm}(\Sigma)$ *is simply represented by* $D^{\pm}(S)$ *with respect to* $(\mathcal{PD}, -dudv)$.
- *If* $\Sigma \cap \Gamma \neq \emptyset$*, then* $D^{\pm}(\Sigma)$ *is represented by* $D^{\pm}(S \cup \Gamma)$ *with respect to* $(\mathcal{PD}, -dudv)$.

Proof. Suppose that $q \in J^+(\Sigma)$, then there exist $p \in \Sigma$ and a future-directed causal curve $\gamma : (0,1) \to \mathcal{M}$ such that $\gamma(0) = p$ and $\gamma(1) = q$. Let $\overline{\gamma} = \Phi \circ \pi \circ \psi \circ \gamma : (0,1) \to \mathcal{PD}$. By construction, $\overline{\gamma}$ is future-directed, since $\partial_u + \partial_v$ is future directed in both $\mathcal{M}$ and $\mathcal{Q}$. By Proposition 5.3 1.–3., $\overline{\gamma}$ is also causal, so $\Phi(\pi(\psi(q))) \in J^+(\Sigma)$. Now suppose that $q' \in J^+(S)$. Then there there exists $p' \in S$ and a future-directed causal curve $\overline{\gamma} : (0,1) \to \mathcal{PD}$ such that $\gamma(0) = p'$ and $\gamma(1) = q'$. Let $\gamma : (0,1) \to \mathcal{M}$ be the radial curve satisfying $\overline{\gamma} = \Phi \circ \pi \circ \psi \circ \gamma$, i.e. $\dot{\gamma}^{\theta} = \dot{\gamma}^{\varphi} \equiv 0$. Then γ must also be future-directed and causal and connects Σ to elements in $(\Phi \circ \pi \circ \psi)^{-1}(J^+(S))$. The above arguments can be repeated when considering $J^-(\Sigma)$ and $I^{\pm}(\Sigma)$.

Without loss of generality, consider D^+. Suppose that $q \in D^+(\Sigma)$. Then for all past-inextendible, future-directed causal curves $\gamma : I \to \mathcal{M}$ with $q = \gamma(s_1)$ for some $s_1 \in I$, there exists an $s_0 \in I$ such that $p = \gamma(s_0) \in \Sigma$. Consider $q' = \Phi(\pi(\psi(q)))$ and a past-inextendible, future-directed causal curve $\overline{\gamma} : I \to \mathcal{PD}$. Suppose that $\overline{\gamma} \cap \Gamma = \emptyset$. Let $\gamma : I \to \mathcal{M}$ be the radial curve satisfying $\overline{\gamma} = \Phi \circ \pi \circ \psi \circ \gamma$. We must then have that there exists an $s_0 \in I$ such that

$p = \gamma(s_0) \in \Sigma$, so $\overline{\gamma}(s_0) \in S$. Hence, $q' \in D^+(S)$. Suppose now that $\overline{\gamma}(s_0) \in \Gamma$ for some $s_0 \in I$. Then $q' \in D^+(\Gamma)$. We conclde that $q' \in D^{\pm}(S \cup \Gamma)$.

EXERCISE: Show that $(\Phi \circ \pi \circ \psi)^{-1}(D^+(S \cup \Gamma)) \subseteq D^+(\Sigma)$. □

In slight abuse of notation, we will use the notations $J^{\pm}(\Sigma)$ and $J^{\pm}(S)$, or $D^{\pm}(\Sigma)$ and $D^{\pm}(S)$ interchangeably.

The Penrose diagram allows us to consider the sets $J^{\pm}(S)$, even when $S \cap (\mathbb{R}^2 \setminus \mathcal{PD}) \neq \emptyset$. In that case, we will denote with $J^{\pm}(S)$ the intersection of the causal past/future of S with $\mathcal{PD}$. For example, we can consider $J^-(\mathcal{I}^+) \subset \mathcal{PD}$ to define the notion of a "black hole" region in spherical symmetry.

Definition. *Let $(\mathcal{M}, g)$ be a spherically symmetric spacetime with $\mathcal{I}^+ \neq \emptyset$.*

- *We define the* black hole region $\mathcal{BH}$ *of $(\mathcal{M}, g)$ to be the subset of $\mathcal{M}$ represented by the set:*

$$\mathcal{PD} \setminus J^-(\mathcal{I}^+)$$

  in a Penrose diagram. We denote $\mathcal{BH} =: \mathcal{M} \setminus J^-(\mathcal{I}^+)$.

- *We denote with $J^-(\mathcal{I}^+)$ both the subset of $\mathcal{PD}$ and the points in $\mathcal{M}$ that it represents and call these the* domain of outer communications.

- *The boundary $\mathcal{H}^+ = \partial J^-(\mathcal{I}^+)$ in $\mathcal{PD}$ and the points in $\mathcal{M}$ that it represents are called the* future event horizon.

[EXERCISE: Convince yourself that the above subsets of $\mathcal{M}$ do not depend on the precise choice of Penrose diagram $\mathcal{PD}$.]

We can think of $\mathcal{I}^+$ as an idealization of the location of far-away observers. It is possible to make sense of "energy/mass radiated to far-away observers" (in spherical symmetry) by studying properties along $\mathcal{I}^+$.

5.2 Einstein equations in spherical symmetry

Consider the Christoffel symbols:

$$\Gamma^{\rho}_{\mu\nu}[g] = \frac{1}{2}(g^{-1})^{\rho\sigma}\left(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}\right).$$

Denote by:

- $\overline{\Gamma}^{c}_{ab}$, $a, b, c \in \{0, 1\}$ the Christoffel symbols of $(\mathcal{Q}, \overline{g})$,
- $\mathring{\Gamma}^{C}_{AB}$, $A, B \in \{2, 3\}$, the Christoffel symbols of $(\mathbb{S}^2, \mathring{g}_{\mathbb{S}^2})$.

We will use the small Latin indices $a, b, c, \ldots$ to denote elements in $\{0, 1\}$ and the capital Latin indices $A, B, \ldots$ to denote components in $\{2, 3\}$. In view of the fact that $g = \overline{g} + r^2\mathring{g}$, the remaining Christoffel symbols take the following form:

$$\begin{aligned}
\Gamma^{a}_{bC} &= 0, \\
\Gamma^{a}_{BC} &= -\frac{1}{2}(g^{-1})^{ab}(\partial_b r^2)\mathring{g}_{BC}, \\
\Gamma^{A}_{bc} &= 0, \\
\Gamma^{A}_{Bc} &= \frac{1}{2}r^{-2}\partial_c(r^2)\delta^A_B = (\partial_c \log r)\delta^A_B.
\end{aligned}$$

We denote with $\overline{R}_{abcd}$ the components of the Riemann tensor associated to $\overline{g}$. Since $\mathcal{Q}$ is 2-dimensional, the symmetries of the Riemann tensor imply that the space of all Riemann tensors on $\mathcal{Q}$ is 1-dimensional (see Problem Sheet 3), and we can therefore write:

$$\overline{R}_{abcd} = K(\overline{g}_{ac}\overline{g}_{bd} - \overline{g}_{ad}\overline{g}_{bc}),$$

with K the Gaussian curvature of $(\mathcal{Q}, \overline{g})$. Similarly, let $\mathring{R}_{ABCD}$ be the components of the Riemann tensor associated to $\mathring{g}$. Then:

$$\mathring{R}_{ABCD} = \mathring{g}_{AC}\mathring{g}_{BD} - \mathring{g}_{AD}\mathring{g}_{BC},$$

since the Gaussian curvature of the unit round sphere is equal to 1.

The corresponding Ricci tensors take the following form:

$$\overline{R}_{ac} = (\overline{g}^{-1})^{bd}\overline{R}_{abcd} = K\overline{g}_{ac},$$
$$\mathring{R}_{AC} = (\mathring{g}^{-1})^{BD}\mathring{R}_{ABCD} = \mathring{g}_{AC}.$$

Now we will relate the Ricci tensor components R_{ab} to $\overline{R}_{ab}$. Recall first that

$$R_{\mu\nu\rho}{}^{\sigma} = \partial_\nu\Gamma^\sigma_{\rho\mu} - \partial_\rho\Gamma^\sigma_{\nu\mu} + \Gamma^\alpha_{\rho\mu}\Gamma^\sigma_{\nu\alpha} - \Gamma^\alpha_{\nu\mu}\Gamma^\sigma_{\rho\alpha}.$$

Hence,

$$\begin{aligned}
R_{ac} = R_{a\nu c}{}^{\nu} &= \partial_\nu\Gamma^\nu_{ca} - \partial_c\Gamma^\nu_{\nu a} + \Gamma^\alpha_{ca}\Gamma^\nu_{\nu\alpha} - \Gamma^\alpha_{\nu a}\Gamma^\nu_{c\alpha} \\
&= \overline{R}_{ac} + \partial_A\Gamma^A_{ac} - \partial_c\Gamma^A_{Aa} + \Gamma^b_{ca}\Gamma^A_{Ab} - \Gamma^b_{Aa}\Gamma^A_{cb} + \Gamma^B_{ca}\Gamma^A_{AB} - \Gamma^A_{Ba}\Gamma^B_{cA} + \Gamma^A_{ca}\Gamma^b_{bA} - \Gamma^A_{ba}\Gamma^b_{cA} \\
&= K\overline{g}_{ac} - \partial_c\Gamma^A_{Aa} + \Gamma^b_{ca}\Gamma^A_{Ab} - \Gamma^A_{Ba}\Gamma^B_{cA} \\
&= K\overline{g}_{ac} - 2(\partial_c\partial_a\log r) + 2\Gamma^b_{ca}(\partial_b\log r) - 2(\partial_a\log r)(\partial_c\log r) \\
&= K\overline{g}_{ac} - 2\partial_c(r^{-1}\partial_a r) + 2\Gamma^b_{ca}(\partial_b\log r) - 2(\partial_a\log r)(\partial_c\log r) \\
&= K\overline{g}_{ac} - 2r^{-1}\partial_c\partial_a r + 2\Gamma^b_{ca}(\partial_b\log r) \\
&= K\overline{g}_{ac} - 2r^{-1}\overline{\nabla}_c\overline{\nabla}_a r,
\end{aligned}$$

with $\overline{\nabla}$ the Levi–Civita connection with respect to $\overline{g}$.

Similarly,

$$\begin{aligned}
R_{AC} = R_{A\nu C}{}^{\nu} &= \partial_\nu\Gamma^\nu_{CA} - \partial_C\Gamma^\nu_{\nu A} + \Gamma^\alpha_{CA}\Gamma^\nu_{\nu\alpha} - \Gamma^\alpha_{\nu A}\Gamma^\nu_{C\alpha} \\
&= \mathring{R}_{AC} + \partial_a\Gamma^a_{AC} - \partial_C\Gamma^a_{aA} + \Gamma^B_{CA}\Gamma^a_{aB} - \Gamma^a_{BA}\Gamma^B_{Ca} + \Gamma^b_{CA}\Gamma^a_{ab} - \Gamma^a_{bA}\Gamma^b_{Ca} + \Gamma^a_{CA}\Gamma^B_{Ba} - \Gamma^a_{BA}\Gamma^B_{Ca} \\
&= \mathring{g}_{AC} + \partial_a\Gamma^a_{AC} - \Gamma^a_{BA}\Gamma^B_{Ca} + \Gamma^b_{CA}\Gamma^a_{ab} + \Gamma^a_{CA}\Gamma^B_{Ba} - \Gamma^a_{BA}\Gamma^B_{Ca} \\
&= \mathring{g}_{AC} - \partial_a((g^{-1})^{ab}r\partial_b r)\mathring{g}_{AC} + (g^{-1})^{ab}(\partial_b r)(\partial_a r)\mathring{g}_{AC} + \Gamma^a_{ab}(g^{-1})^{cb}r(\partial_c r)\mathring{g}_{AC} \\
&\quad - 2(g^{-1})^{ab}(\partial_b r)(\partial_a r)\mathring{g}_{AC} + (g^{-1})^{ab}(\partial_b r)(\partial_a r)\mathring{g}_{AC} \\
&= (1 - \partial_a((g^{-1})^{ab}r(\partial_b r) + \Gamma^a_{ab}(g^{-1})^{cb}r(\partial_c r))\mathring{g}_{AC} \\
&= \left(1 - (\overline{g}^{-1})^{ab}\overline{\nabla}_a(r\overline{\nabla}_b r)\right)\mathring{g}_{AC}.
\end{aligned}$$

[Exercise: Show that $R_{aA} = 0$.]

From the Einstein equations:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi\mathbb{T}_{\mu\nu},$$

it follows that $-R = 8\pi \operatorname{tr} \mathbb{T}$ and hence

$$R_{\mu\nu} = 8\pi \left(\mathbb{T}_{\mu\nu} - \frac{1}{2} \operatorname{tr} \mathbb{T} g_{\mu\nu} \right).$$

Note also that $\operatorname{tr} \mathbb{T} = (\overline{g}^{-1})^{ab} \mathbb{T}_{ab} + r^{-2} (\mathring{\not{g}}^{-1})^{AB} \mathbb{T}_{AB}$. It therefore follows immediately that

$$8\pi \mathbb{T}_{Ab} = R_{Ab} - \frac{1}{2} R g_{Ab} = 0, \tag{5.2}$$

$$\left(1 - (\overline{g}^{-1})^{ab} \overline{\nabla}_a (r \overline{\nabla}_b r)\right) \mathring{\not{g}}_{AB} = R_{AB} = 8\pi (\mathbb{T}_{AB} - \frac{1}{2} \operatorname{tr} \mathbb{T} r^2 \mathring{\not{g}}_{AB}) =: r^2 S \mathring{\not{g}}_{AB}, \tag{5.3}$$

$$K \overline{g}_{ab} - 2 r^{-1} \overline{\nabla}_a \overline{\nabla}_b r = R_{ab} = 8\pi (\mathbb{T}_{ab} - \frac{1}{2} \operatorname{tr} \mathbb{T} \overline{g}_{ab}) \tag{5.4}$$

Equation (5.3) is equivalent to:

$$r^{-1} \Box_{\overline{g}} r = r^{-2} - S - (\overline{g}^{-1})^{ab} r^{-2} \partial_a r \partial_b r$$

and, taking the trace of the right-hand side of (5.3) with respect to $\mathring{\not{g}}$, we obtain:

$$S = -4\pi (\overline{g}^{-1})^{ab} \mathbb{T}_{ab}.$$

Now we can take the trace of (5.4) with respect to $\overline{g}$ to obtain:

$$K = r^{-1} \Box_{\overline{g}} r - S - 4\pi \operatorname{tr} \mathbb{T} = r^{-2} - r^{-2} (\overline{g}^{-1})^{ab} \partial_a r \partial_b r + 8\pi (\overline{g}^{-1})^{ab} \mathbb{T}_{ab} - 4\pi \operatorname{tr} \mathbb{T}. \tag{5.5}$$

Then we can rearrange (5.4) to obtain:

$$r^{-1} \overline{\nabla}_a \overline{\nabla}_b r = \frac{1}{2r^2} \left[1 - (\overline{g}^{-1})^{cd} \partial_c r \partial_d r\right] \overline{g}_{ab} - 4\pi (\mathbb{T}_{ab} - (\overline{g}^{-1})^{cd} \mathbb{T}_{cd} \overline{g}_{ab}). \tag{5.6}$$

Now consider double null coordinates (u, v), so that $\overline{g}_{uv} = -\frac{1}{2}\Omega^2$ and $\overline{g}_{uu} = \overline{g}_{vv} = 0$. Furthermore, $(\overline{g}^{-1})^{uv} = -2\Omega^{-2}$ and $(\overline{g}^{-1})^{uu} = (\overline{g}^{-1})^{vv} = 0$. Then

$$\begin{aligned}
\overline{\Gamma}^u_{uu} &= \frac{1}{2} (\overline{g}^{-1})^{uv} (\partial_u \overline{g}_{uv} + \partial_u \overline{g}_{uv} - \partial_v \overline{g}_{uu}) = \Omega^{-2} \partial_u \Omega^2, \\
\overline{\Gamma}^v_{vv} &= \Omega^{-2} \partial_v \Omega^2, \\
\overline{\Gamma}^u_{vv} &= \Gamma^u_{uv} = \Gamma^v_{uu} = \Gamma^v_{uv} = 0.
\end{aligned}$$

Hence,

$$-\frac{1}{2} K \Omega^2 = \overline{R}_{uv} = \partial_b \overline{\Gamma}^b_{vu} - \partial_v \overline{\Gamma}^b_{bu} + \overline{\Gamma}^a_{vu} \overline{\Gamma}^b_{ba} - \overline{\Gamma}^a_{bu} \overline{\Gamma}^b_{va} = -\partial_v (\Omega^{-2} \partial_u \Omega^2) = -\partial_v \partial_u \log \Omega^2.$$

From (5.5), it therefore follows that

$$\partial_v \partial_u \log \Omega^2 = \frac{\Omega^2}{2r^2} \left[1 + 4\Omega^{-2} (\partial_u r)(\partial_v r) - 4\pi r^2 \operatorname{tr} \mathbb{T} - 32\pi r^2 \Omega^{-2} \mathbb{T}_{uv}\right]. \tag{5.7}$$

From (5.6) we conclude the following equations:

$$\partial_u \partial_v r = -\frac{\Omega^2}{4r} \left[1 + 4\Omega^{-2} (\partial_u r)(\partial_v r)\right] + 4\pi r \mathbb{T}_{uv}, \tag{5.8}$$

$$\Omega^2 \partial_u (\Omega^{-2} \partial_u r) = -4\pi r \mathbb{T}_{uu}, \tag{5.9}$$

$$\Omega^2\partial_v(\Omega^{-2}\partial_v r) = -4\pi r\mathbb{T}_{vv}. \tag{5.10}$$

We refer to (5.9) and (5.10) as the *Raychaudhuri equations*.

We define the *Hawking mass* $m : \mathcal{Q} \to \mathbb{R}$ as follows:

$$m = \frac{r}{2}(1 + 4\Omega^{-2}(\partial_u r)(\partial_v r)) = \frac{r}{2}(1 - (\overline{g}^{-1})^{ab}(\partial_a r)(\partial_b r)) = \frac{r}{2}(1 - \overline{g}^{-1}(dr, dr)).$$

We will later see, when discussing self-gravitating fluids, why the quantity m is compatible with a notion of mass typically attributed to matter.

Note that in the Schwarzschild case $g = g_M$, $m = M$ is constant. In general,

$$\partial_u m = -8\pi r^2\Omega^{-2}(\partial_v r\mathbb{T}_{uu} - \partial_u r\mathbb{T}_{uv}), \tag{5.11}$$

$$\partial_v m = 8\pi r^2\Omega^{-2}(-\partial_u r\mathbb{T}_{vv} + \partial_v r\mathbb{T}_{uv}). \tag{5.12}$$

[Exercise: Derive (5.11) and (5.12).]

Note that these equations imply immediately that in the case of vacuum ($\mathbb{T} \equiv 0$), the Hawking mass is constant on $\mathcal{Q}$. Note also that the metric tensor g can be reconstructed from the pairs (r, Ω^2) or (r, m). The Einstein equations are equivalent to (5.7)–(5.10).

Lemma 5.5. *If the stress-energy tensor $\mathbb{T}$ satisfies the null energy condition, then*

$$\mathbb{T}_{uu}, \mathbb{T}_{vv} \geq 0.$$

If the stress-energy tensor $\mathbb{T}$ satisfies the dominant energy condition, then additionally:

$$\mathbb{T}_{uv} \geq 0.$$

Proof. Exercise □

[Exercise: Show that the Schwarzschild family of metrics are solutions to the vacuum Einstein equations. Hint: Show that g_M corresponds to a solution to (5.7)–(5.10).]

5.3 The Einstein-scalar field system

In the case of the Einstein-scalar field system with $\mathfrak{m} = 0$, we recall that

$$\mathbb{T}_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}(g^{-1})^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi$$

We can express the wave equation $\Box_g\phi = 0$ for a spherically symmetric ϕ as follows:

$$0 = \Box_g\phi = \frac{1}{\sqrt{-\det g}}\partial_\alpha(\sqrt{-\det g}(g^{-1})^{\alpha\beta}\partial_\beta\phi) = -2r^{-2}\Omega^{-2}[\partial_u(r^2\partial_v\phi) + \partial_v(r^2\partial_u\phi)].$$

Hence,

$$\partial_u\partial_v\phi = -r^{-1}\partial_u r\partial_v\phi - r^{-1}\partial_v r\partial_u\phi.$$

Furthermore, $\mathbb{T}_{uu} = (\partial_u\phi)^2$, $\mathbb{T}_{vv} = (\partial_v\phi)^2$, $\mathbb{T}_{uv} = 0$ and

$$\operatorname{tr}\mathbb{T} = (g^{-1})^{\alpha\beta}\mathbb{T}_{\alpha\beta} = -(g^{-1})^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi = 4\Omega^{-2}\partial_u\phi\partial_v\phi.$$

The Einstein–scalar field system of equations is therefore equivalent to the following coupled system of equations for $r, \Omega^2 : \mathcal{Q} \to \mathbb{R}_+$ and $\phi : \mathcal{Q} \to \mathbb{R}$:

$$\partial_v\partial_u \log\Omega^2 = \frac{1}{2}\Omega^2 r^{-2} + 2r^{-2}(\partial_u r)(\partial_v r) - 8\pi\partial_u\phi\partial_v\phi, \tag{5.13}$$

$$\partial_u\partial_v r = -\frac{\Omega^2}{4r}\left[1+4\Omega^{-2}(\partial_u r)(\partial_v r)\right], \tag{5.14}$$

$$\Omega^2\partial_u(\Omega^{-2}\partial_u r) = -4\pi r(\partial_u\phi)^2, \tag{5.15}$$

$$\Omega^2\partial_v(\Omega^{-2}\partial_v r) = -4\pi r(\partial_v\phi)^2, \tag{5.16}$$

$$\partial_u\partial_v\phi = -r^{-1}\partial_u r\partial_v\phi - r^{-1}\partial_v r\partial_u\phi. \tag{5.17}$$

Remark 5.2. *The equations* (5.13), (5.14) *and* (5.17) *are called* propagation equations *and they form a system of* 1+1-dimensional semilinear[34] wave equations. *Note that the products of two derivative terms never feature two u or two v derivatives. This is called the* null structure of the Einstein-scalar field equations *and it is a very important structural property that also holds outside of spherical symmetry and is important for addressing global dynamics of small perturbations of spacetimes as well as the nature of singularities.*

This kind of structure does not apply, for example, to the relativistic Euler fluid equations of fluid dynamics, where "singular" behaviour, in the form of shocks, can occur even if one considers initial data corresponding to a small perturbation of the constant state.

Remark 5.3. *The equations* (5.15) *and* (5.16) *are* constraint equations.

We moreover have that

$$\partial_u m = -8\pi r^2\Omega^{-2}\partial_v r(\partial_u\phi)^2, \tag{5.18}$$

$$\partial_v m = 8\pi r^2\Omega^{-2}(-\partial_u r)(\partial_v\phi)^2. \tag{5.19}$$

5.4 Local existence and uniqueness of the characteristic initial value problem for the Einstein-scalar field system

We will now show that we can sensibly study *dynamics* for the spherically symmetric Einstein–scalar field problem. First we will state a local existence and uniqueness statement for general systems of 1+1-dimensional wave equations.

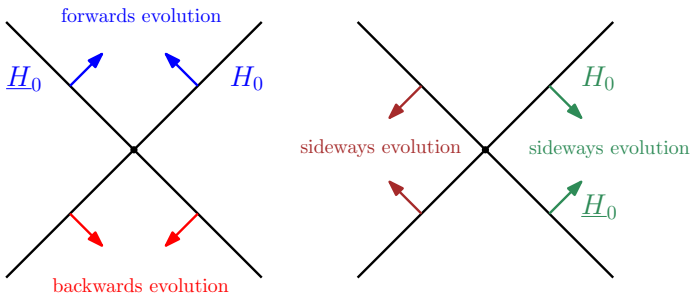


Figure 23: The evolution problem for systems of 1+1-dimensional wave equations spherical symmetry.

Theorem 5.6 (Local existence and uniqueness characteristic initial value problem for 1+1-dimensional semilinear wave equations (with null structure))**.** *Consider the following system of semilinear 1+1-dimensional wave equations: let $M \in \mathbb{N}$ and $\Psi : \mathbb{R}^2 \supseteq U \to \mathbb{R}^M$, with $0 \in U$ and*

$$\partial_u\partial_v\Psi^I = \sum_{J,K=1}^{M} N^I_{JK}(\Psi)\partial_u\Psi^J\partial_v\Psi^K + \sum_{J=1}^{M}\left[L^I_J(\Psi)\partial_u\Psi^J + R^I_J(\Psi)\partial_v\Psi^J\right] + f^I(\Psi), \tag{5.20}$$

[34] Semilinear PDE are nonlinear PDE that are linear in the highest derivative terms (second-order in this case).

with $N^I_{JK}, L^I_J, R^I_J, f^I : \mathbb{R}^K \to \mathbb{R}$ *smooth functions for all* $I, J, K \in \{1, \dots, M\}$.

Consider the union of line segments H_0 *and* $\underline{H}_0$ *in* $\mathbb{R}^2$*, with:*

$$H_0 := \{0\}_u \times [0, \epsilon)_v, \quad \underline{H}_0 := [0, \epsilon)_u \times \{0\}_v$$

and prescribe the following characteristic initial data*:*

$$\begin{aligned} \Psi_0 &\in \mathbb{R}^M \quad \text{on } H_0 \cap \underline{H}_0, \\ \underline{\Psi}'_0 : \underline{H}_0 &\to \mathbb{R}^M \quad \text{on } \underline{H}_0, \\ \Psi'_0 : H_0 &\to \mathbb{R}^M \quad \text{on } H_0. \end{aligned}$$

(i) *(Forwards and backwards evolution) Then, for* $\epsilon > 0$ *suitably small depending on* $\Psi_0, \underline{\Psi}'_0, \Psi'_0$*, there exists a unique smooth solution* $\Psi : [0, \epsilon) \times [0, \epsilon) \to \mathbb{R}^M$ *to* (5.20)*, such that*

$$\begin{aligned} \Psi|_{H_0 \cap \underline{H}_0} &= \Psi_0, \\ \partial_u \Psi|_{\underline{H}_0} &= \underline{\Psi}'_0, \\ \partial_v \Psi|_{H_0} &= \Psi'_0. \end{aligned}$$

The same statement holds with ϵ *replaced by* $-\epsilon$ *in the definitions of* H_0 *and* $\underline{H}_0$*.*

(ii) *(Sideways evolution) The above statement also holds for* $\Psi : (-\epsilon, 0] \times [0, \epsilon) \to \mathbb{R}^M$ *with the following choices for* H_0 *and* $\underline{H}_0$*:*

$$H_0 := \{0\}_u \times [0, \epsilon)_v, \quad \underline{H}_0 := (-\epsilon, 0]_u \times \{0\}_v.$$

and also with ϵ *replace by* $-\epsilon$*.*

We can also compare two global solutions in the following way:

Theorem 5.7.

(i) *(*Cauchy stability*) Let* $U_0, V_0 \in (0, \infty)$ *and let* $\Psi : [0, U_0) \times [0, V_0) \to \mathbb{R}^M$ *be a smooth solution to* (5.20)*. Then, for any* $0 < \epsilon < 1$ *and* $n > 0$*, there exists a* $\delta > 0$ *such that for any initial data*

$$\begin{aligned} \widetilde{\Psi}_0 &\in \mathbb{R}^M \quad \text{on } H_0 \cap \underline{H}_0, \\ \widetilde{\underline{\Psi}}'_0 : \underline{H}_0 &\to \mathbb{R}^M \quad \text{on } \underline{H}_0, \\ \widetilde{\Psi}'_0 : H_0 &\to \mathbb{R}^M \quad \text{on } H_0. \end{aligned}$$

with $|\widetilde{\Psi}_0 - \Psi|_{H_0 \cap \underline{H}_0}| < \delta$ *and for all* $k \leq n$*,* $\sup_{u \in [0, U_0)} |\partial_u^k (\widetilde{\underline{\Psi}}'_0 - \partial_u \Psi|_{\underline{H}_0})| < \delta$*,* $\sup_{v \in [0, V_0)} |\partial_v^k (\widetilde{\Psi}'_0 - \partial_v \Psi|_{H_0})| < \delta$*, we have that the corresponding solution* $\widetilde{\Psi} : [0, (1-\epsilon)U_0) \times [0, (1-\epsilon)V_0) \to \mathbb{R}^M$ *satisfies* $\widetilde{\Psi} \in C^{n+1}([0, (1-\epsilon)U_0) \times [0, (1-\epsilon)V_0); \mathbb{R}^M)$ *and*

$$\sum_{k_1 + k_2 \leq n+1} \sup_{(u,v) \in [0, (1-\epsilon)U_0) \times [0, (1-\epsilon)V_0)} |\partial_u^{k_1} \partial_v^{k_2} (\widetilde{\Psi} - \Psi)| < \epsilon.$$

(ii) *(*Global uniqueness*) Solutions corresponding to the same initial data agree.*

Remark 5.4. *In higher dimensions, systems of nonlinear wave equations do not satisfy local well-posedness for "sideways evolution", only for "forwards" and "backwards" evolution!*

We can apply the above lemma to obtain local well-posedness (existence, uniqueness, Cauchy stability) for the spherically symmetric Einstein–scalar field equations.

Theorem 5.8 (Local well-posedness of the characteristic initial value problem for the spherically symmetric Einstein–scalar field system)**.** *Let $\epsilon > 0$. Consider $H_0 \cup \underline{H}_0 \subset \mathbb{R}^2$, with*

$$H_0 := \{0\}_u \times [0, \epsilon)_v, \quad \underline{H}_0 := [0, \epsilon)_u \times \{0\}_v.$$

Prescribe the following numbers and smooth functions:

$$\begin{aligned}
(r_0, (r^{-1}\Omega^2)_0, \phi_0, \underline{r}'_0, r'_0) \in \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}^3 \quad & \text{on } H_0 \cap \underline{H}_0, \\
\underline{\phi}'_0 : \underline{H}_0 \to \mathbb{R} \quad & \text{on } \underline{H}_0, \\
\phi'_0 : H_0 \to \mathbb{R} \quad & \text{on } H_0, \\
(r^{-1}\underline{\Omega}^2)'_0 : \underline{H}_0 \to \mathbb{R} \quad & \text{on } \underline{H}_0, \\
(r^{-1}\Omega^2)'_0 : H_0 \to \mathbb{R} \quad & \text{on } H_0.
\end{aligned}$$

Then, for $\epsilon > 0$ suitably small, there exists a unique solution $(r, \Omega^2, \phi) : [0, \epsilon) \times [0, \epsilon) \to \mathbb{R}^2_+ \times \mathbb{R}$ to (5.13)–(5.17), such that

$$\begin{aligned}
(r, r^{-1}\Omega^2, \phi, \partial_u r, \partial_v r)(0,0) &= (r_0, (r^{-1}\Omega^2)_0, \phi_0, \underline{r}'_0, r'_0), \\
\partial_u \phi|_{\underline{H}_0} &= \underline{\phi}'_0, \\
\partial_v \phi|_{H_0} &= \phi'_0, \\
\partial_u (r^{-1}\Omega^2)|_{\underline{H}_0} &= (r^{-1}\underline{\Omega}^2)'_0, \\
\partial_u (r^{-1}\Omega^2)|_{H_0} &= (r^{-1}\Omega^2)'_0.
\end{aligned}$$

The above statement also holds with any of the following alternative choices for H_0 and $\underline{H}_0$ (with appropriately modified domain of (r, Ω^2, ϕ):

$$H_0 := \{0\}_u \times (-\epsilon, 0]_v, \quad \underline{H}_0 := (-\epsilon, 0]_u \times \{0\}_v.$$

Proof. **Strategy**: We will show that the prescribed initial data determines the values of r, Ω^2, ϕ on $H_0 \cup \underline{H}_0$ by imposing the Raychaudhuri equations along the initial null lines. The statement of the theorem then follows from a direct application of (5.20) together with a verification that the Raychaudhuri equations also remain valid away from the initial null lines.

Note that if $\phi(0,0) = \phi_0$, then we need

$$\begin{aligned}
\phi(0, v) &= \phi_0 + \int_0^v \phi'_0(v')\, dv', \\
\phi(u, 0) &= \phi_0 + \int_0^u \underline{\phi}'_0(u')\, du'.
\end{aligned}$$

Similarly, if $r^{-1}\Omega^2(0,0) = (r^{-1}\Omega^2)_0$, then we need

$$\begin{aligned}
r^{-1}\Omega^2(0, v) &= (r^{-1}\Omega^2)_0 + \int_0^v (r^{-1}\Omega^2)'_0(v')\, dv', \\
r^{-1}\Omega^2(u, 0) &= (r^{-1}\Omega^2)_0 + \int_0^u (r^{-1}\underline{\Omega}^2)'_0(u')\, du'.
\end{aligned}$$

Note that for $\epsilon > 0$ suitably small, depending on the choice $(r^{-1}\Omega^2)'_0$ and $(r^{-1}\underline{\Omega}^2)'_0$, we have that $r^{-1}\Omega^2(0, v) > 0$ and $r^{-1}\Omega^2(u, 0) > 0$.

Let $r(0,0) = r_0$, $\partial_u r(0,0) = \underline{r}'_0$ and $\partial_v r(0,0) = r'_0$. We can rewrite (5.15) and (5.16) as follows:

$$\partial_u \left(\frac{r}{\Omega^2} r^{-1} \partial_u r \right) = -4\pi \frac{r}{\Omega^2} (\partial_u \phi)^2,$$

$$\partial_v\left(\frac{r}{\Omega^2}r^{-1}\partial_v r\right) = -4\pi\frac{r}{\Omega^2}(\partial_v\phi)^2,$$

The left-hand sides we can further rewrite as $\partial_u(\frac{r}{\Omega^2}\partial_u \log r)$ and $\partial_v(\frac{r}{\Omega^2}\partial_v \log r)$.

Then integrating (5.15) and (5.16) gives for $\epsilon > 0$ suitably small:

$$\begin{aligned}\frac{r}{\Omega^2}(u,0)(\partial_u \log r)(u,0) =& \frac{1}{(r^{-1}\Omega^2)_0}r_0^{-1}\underline{r}'_0 - 4\pi\int_0^u \frac{1}{r^{-1}\Omega^2(u',0)}(\underline{\phi}'_0)^2(u')\,du',\\ \frac{r}{\Omega^2}(0,v)(\partial_v \log r)(0,v) =& \frac{1}{(r^{-1}\Omega^2)_0}r_0^{-1}r'_0 - 4\pi\int_0^v \frac{1}{r^{-1}\Omega^2(0,v')}(\phi'_0)^2(v')\,dv'\end{aligned}$$

and determines $\partial_u \log r(u,0)$ and $\partial_v \log r(0,v)$. Integrating these then gives:

$$\begin{aligned}(\log r)(0,v) =& \log r_0 + \int_0^v \partial_v \log r(0,v')\,dv',\\ (\log r)(u,0) =& \log r_0 + \int_0^u \partial_u \log r(u',0)\,du'.\end{aligned}$$

We take $r(u,0)$ and $r(0,v)$ to be the corresponding exponentials.

[Exercise: Show that, the solution (r,Ω^2,ϕ) to (5.13), (5.14) and (5.17) corresponding to the prescribed initial data satisfies also (5.15) and (5.16).] □

Corollary 5.9. *Two solutions (r_1,Ω_1^2,ϕ_1) and (r_2,Ω_2^2,ϕ_2) to (5.13)–(5.17) corresponding to the same initial data on H_0 and $\underline{H}_0$ must agree on the intersection of their domains.*

Proof. This follows from Theorem 5.8 together with the global uniqueness part of Theorem 5.7. □

Even if two solutions (r_1,Ω_1^2,ϕ_1) and (r_2,Ω_2^2,ϕ_2) agree, the corresponding metrics g_1 and g_2 might satisfy the relation $g_2 = \psi^* g_1$ for a suitable diffeomorphism $\psi : \mathcal{M} \to \mathcal{M}$. Equivalently, *different* solutions (r,Ω^2,ϕ) can nevertheless correspond to the *same* metric g, but described in *different* double null coordinates.

Corollary 5.10. *Consider solutions (r_1,Ω_1^2,ϕ_1) and (r_2,Ω_2^2,ϕ_2) to (5.13)–(5.17) arising from initial data on H_0 and $\underline{H}_0$.*

(i) *There exist double null coordinates on the corresponding spacetimes, with respect to which $\Omega_i^2(u,0) = \Omega_i^2(0,v) = 1$ for $i \in \{1,2\}$. With respect to those coordinates, (r_1,Ω_1^2,ϕ_1) and (r_2,Ω_2^2,ϕ_2) agree if:*

$$\begin{aligned}(r_1,\phi_1,\partial_u r_1,\partial_v r_1)(0,0) &= (r_2,\phi_2,\partial_u r_2,\partial_v r_2)(0,0),\\ \partial_u\phi_1(u,0) &= \partial_u\phi_2(u,0),\\ \partial_v\phi_1(0,v) &= \partial_v\phi_2(0,v),\end{aligned}$$

(ii) *Suppose that $\partial_u r(0,0) \neq 0$ or $\partial_v r(0,0) \neq 0$. Then there exist double null coordinates on the corresponding spacetimes, with respect to which $\Omega_i^2(u,0) = \Omega_i^2(0,v) = 1$ and, respectively, $\partial_u r_i(0,0) = 1$ or $\partial_v r_i(0,0) = 1$ for $i \in \{1,2\}$. In both those cases, (r_1,Ω_1^2,ϕ_1) and (r_2,Ω_2^2,ϕ_2) agree if:*

$$\begin{aligned}(r_1,m_1,\phi_1)(0,0) &= (r_2,m_2,\phi_2)(0,0),\\ \partial_u\phi_1(u,0) &= \partial_u\phi_2(u,0),\\ \partial_v\phi_1(0,v) &= \partial_v\phi_2(0,v).\end{aligned}$$

In particular, in the special case that $\partial_u r_i(0,0) = 0$ or $\partial_v r_i(0,0) = 0$ for $i \in \{1,2\}$, we have that $r_i(0,0) = 2m_i(0,0)$, so $m_2(0,0) = m_1(0,0)$ is equivalent to $r_2(0,0) = r_1(0,0)$.

Proof. We will show that we can rescale any double null coordinates to obtain double null coordinates $(\tilde{u}, \tilde{v})$, with respect to which $g = -\widetilde{\Omega}^2 d\tilde{u}d\tilde{v} + r^2 \mathring{\not{g}}$ with $\widetilde{\Omega}^2(\tilde{u}, 0) = 1$ and $\widetilde{\Omega}^2(0, \tilde{v}) = 1$.

We have that

$$\widetilde{\Omega}^2(\tilde{u}, \tilde{v}) = \Omega^2(u(\tilde{u}), v(\tilde{v})) \frac{du}{d\tilde{u}}(\tilde{u}) \frac{dv}{d\tilde{v}}(\tilde{v}).$$

Let $(\tilde{u}, \tilde{v})(0, 0) = (0, 0)$ and

$$\frac{du}{d\tilde{u}}(\tilde{u}) = \frac{1}{\Omega^2(u(\tilde{u}), 0)},$$
$$\frac{dv}{d\tilde{v}}(\tilde{v}) = \frac{\Omega^2(0, 0)}{\Omega^2(0, v(\tilde{v}))}.$$

Then

$$\widetilde{\Omega}^2(\tilde{u}, 0) = \Omega^2(u(\tilde{u}), 0) \frac{1}{\Omega^2(u(\tilde{u}), 0)} \frac{\Omega^2(0, 0)}{\Omega^2(0, 0)} = 1,$$
$$\widetilde{\Omega}^2(0, \tilde{v}) = \Omega^2(0, v(\tilde{v})) \frac{1}{\Omega^2(0, 0)} \frac{\Omega^2(0, 0)}{\Omega^2(0, v(\tilde{v}))} = 1.$$

Hence, no matter what the restriction of Ω_i^2 is to $H_0 \cup \underline{H}_0$, we can always rescale our double coordinates, so that $\Omega_2^2|_{H_0 \cup \underline{H}_0} = \Omega_2^2|_{H_0 \cup \underline{H}_0}$, so the initial data for Ω^2 automatically agree.

We can further rescale:

$$\hat{u}(\tilde{u}) = \lambda \tilde{u},$$
$$\hat{v}(\tilde{v}) = \lambda^{-1} \tilde{v},$$

with $\lambda \in \mathbb{R} \setminus \{0\}$ without changing the conclusions in the argument above. Suppose that $\partial_u r(0, 0) \neq 0$. Then take $\lambda = \partial_{\tilde{u}} r(0, 0)$. By the chain rule, we must therefore have that

$$(\partial_{\hat{u}} r)(0, 0) = \frac{d\tilde{u}}{d\hat{u}} (\partial_{\tilde{u}} r)(0, 0) = \frac{(\partial_{\tilde{u}} r)(0, 0)}{\lambda} = 1,$$

so $(\partial_{\hat{u}} r)(0, 0)$ is fixed. Furthermore,

$$\frac{2m(0, 0)}{r(0, 0)} = 1 + 4 \frac{\partial_{\tilde{u}} r \partial_{\tilde{v}} r}{\Omega^2}(0, 0) = 1 + 4(\partial_{\hat{v}} r)(0, 0),$$

so $(\partial_{\hat{v}} r)(0, 0)$ can be determined from $m(0, 0)$ and $r(0, 0)$. Suppose that $\partial_v r(0, 0) \neq 0$, then we can repeat the above argument with $\lambda = \frac{1}{\partial_{\tilde{v}} r(0,0)}$ to determine $(\partial_{\tilde{u}} r)(0, 0)$.

Furthermore, by the above expression for $m(0, 0)$, $\partial_u r(0, 0) = 0$ or $\partial_v r(0, 0) = 0$ guarantees that $\frac{2m(0,0)}{r(0,0)} = 1$. □

Making a particular choice of double null coordinates is also called "fixed the gauge", in analogy with fixing a choice of gauge in the context of gauge field theories.

5.5 Birkhoff's theorem

Using the uniqueness property from Corollary 5.9, we can show that all spherically symmetric spacetimes in vacuum must be isometric to a subset of a Schwarzschild or Minkowski spacetime. This result is called Birkhoff's theorem and is attributed to Jebsen 1921, Birkhoff 1923.

Theorem 5.11 (Birkhoff's theorem)**.** *A spherically symmetric spacetime solution $(\mathcal{M}, g)$ to $\mathrm{Ric}[g] = 0$ is isometric to an open subset of a maximally-extended Schwarzschild spacetime with $M \in \mathbb{R}$ or to an open subset of the Minkowski spacetime.*

Remark 5.5. *Note that the region $\mathbb{R}^{3+1} \setminus (\mathbb{R} \times \{0\})$ of the Minkowski spacetime is isometric to the $M = 0$ Schwarzschild spacetime.*

Proof. Let $p \in \mathcal{M} \setminus \Gamma$. Without loss of generality, we can assume p is represented by $(u, v) = (0, 0)$. By Corollary 5.10 (with $\phi \equiv 0$), we can always change double null coordinates so that the metric g is uniquely determined by $r(0, 0)$ and $m(0, 0)$ in the region represented by the smallest rectangle in the Penrose diagram containing the null lines H_0 and $\underline{H}_0$ that pass through p. Denote $M := m(0, 0)$ and $r_0 = r(0, 0)$ and consider the Schwarzschild spacetime with mass parameter M. For any $r_0 > 0$, there exists a sphere in Schwarzschild with mass M such that the Schwarzschild radius $r_{\text{Schw.}}$ agrees with r_0. Hence, the metric g must be isometric to a Schwarzschild metric if $M \neq 0$ or $\Gamma = \emptyset$, or the Minkowski metric if $M = 0$ and $\Gamma \neq \emptyset$.

We can cover the full Penrose diagram by a finite number of open rectangles corresponding containing intersecting ingoing and outgoing null lines, by compactness of the diagram in $\mathbb{R}^2$, and none of the rectangles is disjoint, by connectedness of $\mathcal{Q}$, the full spacetime must be isometric to an open subset in Schwarzschild with mass M or to an open subset in Minkowski. □

5.6 Global properties of general spherically symmetric spacetimes

It turns out that even with minimal information about the matter model under consideration, it is possible to make statements about global features of the spacetime.

Definition 5.4. *Let $(\mathcal{M}, g)$ be a spherically symmetric spacetime.*

1. *A sphere at $p \in \mathcal{Q} \setminus \Gamma$ is* trapped *if $(\partial_u r)(p), (\partial_v r)(p) < 0$.*
2. *A sphere at $p \in \mathcal{Q} \setminus \Gamma$ is* marginally trapped *if $(\partial_u r)(p) < 0$ and $(\partial_v r)(p) = 0$.*
3. *A sphere at $p \in \mathcal{Q}$ is* anti-trapped *if $(\partial_u r)(p), (\partial_v r)(p) > 0$ and* marginally anti-trapped *if $(\partial_u r)(p) = 0$ and $(\partial_v r)(p) > 0$.*

[Exercise: Determine which spheres in the maximally-extended Schwarzschild spacetime (expressed in Kruskal coordinates) are trapped, marginally trapped, anti-trapped and marginally anti-trapped.]

The existence of a trapped sphere is a local property in the spacetime, but it has global consequences.

Definition 5.5.

- *A null geodesic in a spacetime $(\mathcal{M}, g)$ is* future/past complete *if it is geodesically inextendible (or maximal)*[35] *and its affine parameters are unbounded from above/below. A geodesically inextendible null geodesic that is not future/past complete is said to be* future/past incomplete.
- *If all inextendible null geodesics in $(\mathcal{M}, g)$ are future/past complete null geodesics, we say that $(\mathcal{M}, g)$ is* future/past null-geodesically complete. *If there exists a null geodesic that is future/past incomplete, we say that $(\mathcal{M}, g)$ is* future/past null-geodesically incomplete.

One can similarly define "timelike-geodesic future/past incompleteness" by replacing the word "null" in the above definitions with "timelike". One can analogously define "spacelike-geodesic completeness" (dropping additionally the words "future/past").

[35] A null geodesic $\gamma : I \to \mathcal{M}$ is geodesically extendible if there exists a null geodesic $\gamma' : J \to \mathcal{M}$ with $I \subset J$, such that $\gamma'(s) = \gamma(s)$ for all $s \in I$. If a null geodesic is not geodesically extendible, we say it is geodesically inextendible or maximal.

Proposition 5.12 (Spherically symmetric Penrose incompleteness theorem)**.** *Let $(\mathcal{M}, g)$ be a spherically symmetric spacetime, such that* $\mathrm{Ric}[g](X, X) \geq 0$ *for any null vector field X. Assume that $(\mathcal{M}, g)$ has a trapped sphere. Then $(\mathcal{M}, g)$ is future-null-geodesically incomplete.*

Remark 5.6. *If we define the stress energy tensor corresponding to $(\mathcal{M}, g)$ via $8\pi\mathbb{T} = \mathrm{Ric}[g] - \frac{1}{2}R[g]g$, then we require $\mathbb{T}$ to satisfy the null energy condition. Since this theorem does not involve the equations satisfied by matter coupled to the Einstein equations, it is really a result in Lorentzian geometry. We do not need to appeal to the Einstein equations!*

Remark 5.7. *The Penrose incompleteness theorem is sometimes also called the Penrose singularity theorem. This name is a bit misleading, as it suggests that something singular is happening when we reach the end points of future/past-incomplete null geodesics, as in the case of maximally-extended Schwarzschild. However, there need not be any singularity. Consider, for example, the region $\{r < r_-\}$ in Reissner–Nordström spacetimes. We will come back to this observation when we discuss the strong cosmic censorship conjecture.*

Remark 5.8. *The assumption of spherical symmetry of $(\mathcal{M}, g)$ is actually not necessary. The general statement of the theorem is as follows:*

Theorem 5.13 (The general Penrose incompleteness theorem)**.** *Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime with a non-compact Cauchy hypersurface and a* closed trapped surface.[36] *Then $(\mathcal{M}, g)$ is future-null-geodesically incomplete.*

Proof of Theorem 5.12. There exists a $p \in \mathcal{Q}$ such that $\partial_v r(p) < 0$ and $\partial_u r(p) < 0$. Let $p = (0, 0)$ and consider the outgoing null line $\ell = \{(0, v) \in \mathcal{Q}\,,\, v \geq 0\}$. By rescaling v so that $\Omega^2|_\ell \equiv 1$, we have that

$$\nabla_{\partial_v}\partial_v|_\ell = \Gamma^v_{vv}\partial_v|_\ell = \Omega^{-2}\partial_v\Omega^2\partial_v|_\ell = 0,$$

so ℓ is the imagine of an inextendible affinely parametrized null geodesic γ in $(\mathcal{M}, g)$ with affine parameter v. We will show that v is bounded along ℓ, which implies that γ is future-null-geodesically incomplete or reaches Γ.

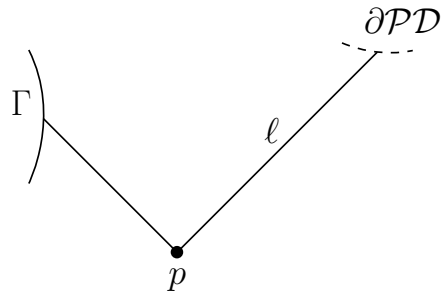


Figure 24: Null lines emanating from a trapped sphere at $p \in \mathcal{Q}$.

The key equation in this argument is the Raychaudhuri equation (5.10), which implies that

$$\partial_v^2 r(0, v) = -4\pi r\mathbb{T}_{vv}(0, v),$$

where $\mathbb{T}_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$. Hence, $\mathbb{T}_{vv} = R_{vv} = \mathrm{Ric}[g](\partial_v, \partial_v) \geq 0$, by assumption. Hence $\partial_v^2 r(0, v) \leq 0$, so $\partial_v r(0, v) \leq \partial_v r(0, 0) < 0$ for all $v \geq 0$.

[36] A compact 2-dimensional submanifold without boundary S with induced metric $\not{g}$, such that $\sqrt{\det \not{g}}$ decreases in the null directions normal to S.

We can now integrate $\partial_v r(0,v)$ to obtain:

$$r(0,v) = r(0,0) + \int_0^v \partial_v r(0,v')\, dv' \leq r(0,0) - v(-\partial_v r)(0,0).$$

Since $r(0,v) > 0$, this means that

$$v < \frac{r(0,0)}{(-\partial_v r)(0,0)}.$$

We can repeat the same argument with v replaced by u to show that the future-directed ingoing line emanating from $(0,0)$ must also either each $\partial\mathcal{PD} \setminus \Gamma$ in finite affine time or reach Γ. Since the future-directed ingoing and outgoing lines cannot both reach Γ, we conclude that the spacetime admits a future-null-geodesically incomplete □

[Exercise: Explain why in the case of spherically symmetric spacetimes satisfying the null energy condition, which contain an anti-trapped sphere, there are null geodesics did not exist for infinite affine time in the past.]

We already encountered future-directed null geodesics which did not exist for ever in the Schwarzschild spacetime, where such geodesics were hidden behind a black hole event horizon. We will now show that, this remains true in very general spherical symmetric spacetimes that have a non-empty future null infinity $\mathcal{I}^+$.

We will show that the existence of a trapped sphere implies the existence of a black hole region.[37]

Definition 5.6. *We define the* regular set $\mathcal{R} \subseteq \mathcal{Q}$*, the* trapped set $\mathcal{T} \subseteq \mathcal{Q}$ *and the* apparent horizon $\mathcal{A} \subset \mathcal{Q}$ *as spacetime subsets represented by the following points in* $\mathcal{Q}$*:*

$$\begin{aligned}
\mathcal{R} &:= \{(u,v) \in \mathcal{Q} \,|\, \partial_u r(u,v) < 0,\ \partial_v r(u,v) > 0\}, \\
\mathcal{T} &:= \{(u,v) \in \mathcal{Q} \,|\, \partial_u r(u,v) < 0,\ \partial_v r(u,v) < 0\}, \\
\mathcal{A} &:= \{(u,v) \in \mathcal{Q} \,|\, \partial_u r(u,v) < 0,\ \partial_v r(u,v) = 0\}.
\end{aligned}$$

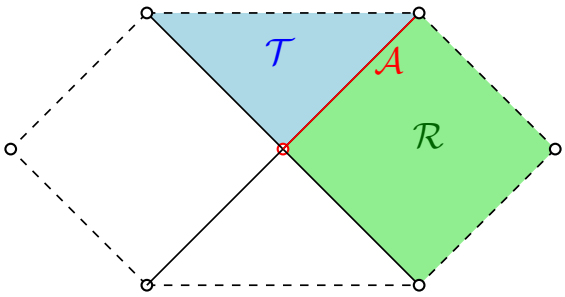


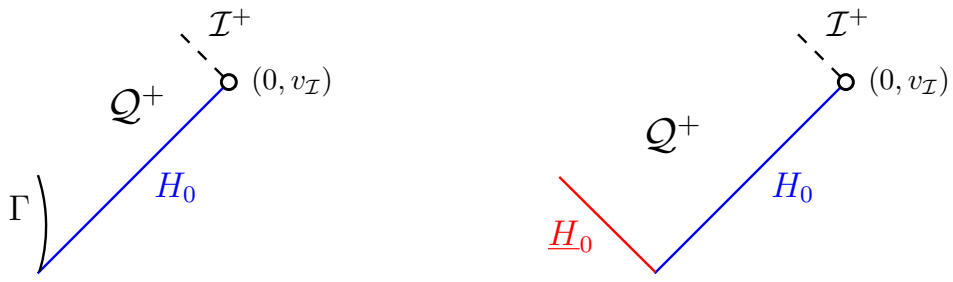
Figure 25: The regions $\mathcal{R}$, $\mathcal{T}$ and $\mathcal{A}$ in a maximally-extended Schwarzschild spacetime.

We consider the following to set-ups:

Set-up 1: Suppose $\Gamma \neq \emptyset$. Let $\Sigma = H_0$ be an outgoing line emanating from Γ. Assume that the future limit point on H_0 is an element of null infinity $\mathcal{I}$ and that $\partial_u r < 0$ on H_0. Denote $\Sigma = H_0$.

[37] This is a sufficient condition, but not a necessary one. Exercise: Show this by considering the extremal Reissner–Nordström black hole spacetimes.

Set-up 2: Suppose $\Gamma = \emptyset$. Consider two intersecting null lines H_0 (outgoing) and $\underline{H}_0$ (ingoing) in $\mathcal{Q}$. Let $\Sigma = H_0 \cup \underline{H}_0$. Assume that the future limit point on H_0 is an element of null infinity $\mathcal{I}$ and that $\partial_u r < 0$ on H_0.



(a) Set-up 1 (b) Set-up 2

Figure 26: Penrose diagrams illustrating $\mathcal{Q}^+$ for the two different set-ups with the additional assumption that $\mathcal{I}$ consists of more than one point.

In both cases, we define $\mathcal{Q}^+ = D^+(\Sigma \cup \Gamma)$ (with respect to $(\mathcal{PD}, -dudv)$). We will choose (u, v) coordinates such that $H_0 \subset \{u = 0\}$ and $\underline{H}_0 \subset \{v = 0\}$. We will take v to have bounded range so that the future limit point of H_0 is given by $(0, v_{\mathcal{I}})$.

Lemma 5.14. *Assume that $\mathbb{T} = \mathrm{Ric}[g] - \frac{1}{2}R[g]g$ satisfies the null energy condition. Then*

$$\mathcal{Q}^+ = (\mathcal{R} \cup \mathcal{T} \cup \mathcal{A}) \cap \mathcal{Q}^+.$$

Proof. By construction, $H_0 \subset (\mathcal{R} \cup \mathcal{T} \cup \mathcal{A}) \cap \mathcal{Q}^+$. Since any point in $\mathcal{Q}^+$ can be connected to H_0 via a past-directed ingoing null line and by the Raychaudhuri equation (5.9) $\partial_u(\Omega^{-2}\partial_u r) \leq 0$ along that null line. Hence $\partial_u r < 0$ everywhere in $\mathcal{Q}^+$. □

Proposition 5.15. *Null infinity $\mathcal{I}$ of $\mathcal{Q}^+$ is either a single point: $\mathcal{I} = (0, v_{\mathcal{I}})$, or it can be represented by an ingoing null segment $\mathcal{I} = \{(u, v_{\mathcal{I}}) \in \mathbb{R}^2 \,|\, 0 \leq u < u_{\mathcal{I}}\}$, for some $u_{\mathcal{I}} > 0$. In the latter case, $\mathcal{I} = \mathcal{I}^+$.*

Proof. Consider the line segments $\ell_v = \{(u, v) \in \mathbb{R}^2 \,|\, u \geq 0\}$ with $v \leq v_{\mathcal{I}}$ that emanate from H_0. Since $\partial_u r < 0$ in $\mathcal{Q}^+$, we have that $r|_{\ell_v \cap \mathcal{Q}^+} \leq r(0, v)$. Hence, the lines ℓ_v do not intersect $\mathcal{I}$ for $v < v_{\mathcal{I}}$ and we conclude that $\mathcal{I} \subset \ell_{v_{\mathcal{I}}}$. Suppose that there exists a $u_0 > 0$ such that $(u_0, v_{\mathcal{I}}) \in \mathcal{I}$. Then, by $\partial_u r < 0$, we must have that $(u, v_{\mathcal{I}}) \in \mathcal{I}$) for all $0 \leq u \leq u_0$. Let $u_{\mathcal{I}}$ denote the supremum of such u_0. Since $\mathcal{I}$ is an ingoing future boundary of $\mathcal{Q}^+$, it follows immediately that $\mathcal{I} = \mathcal{I}^+$.

If there exist no such u_0, then $\mathcal{I}$ consists of the single point $(0, v_{\mathcal{I}})$. □

We will now additionally assume that $\mathcal{I}$ consists of more than one point, so by the above proposition, $\mathcal{I} = \mathcal{I}^+ = \{(u, v_{\mathcal{I}}) \in \mathbb{R}^2 \,|\, u < u_{\mathcal{I}}\}$. We denote $i^+ := (u_{\mathcal{I}}, v_{\mathcal{I}})$. Then i^+ is the future limit point of $\mathcal{I}^+$ in $\mathbb{R}^2$.

Recall that the black hole region $\mathcal{BH} \subset \mathcal{M}$ is represented by the following points in the Penrose diagram $\mathcal{PD}$ associated to $\mathcal{Q}^+$:

$$\mathcal{PD} \setminus J^-(\mathcal{I}^+)$$

and $\mathcal{BH}$ can be empty. If $\mathcal{BH} \neq \emptyset$, then the future event horizon $\mathcal{H}^+$ is represented by future boundary of $J^-(\mathcal{I}^+)$ in $\mathcal{PD}$. We will also use the notation $\mathcal{BH}$ to indicate the points in $\mathcal{PD}$ and $\mathcal{Q}^+$ representing points in the black hole region of $\mathcal{M}$. Similarly, we also use $\mathcal{H}^+$ to denote the null line segments in $\mathcal{PD}$ and $\mathcal{Q}^+$ representing points on the future event horizon of $\mathcal{M}$.

We will now show that the presence of a (marginally) trapped surface implies the non-emptyness of the black hole region $\mathcal{BH}$.

Proposition 5.16. *Assume that $8\pi\mathbb{T} = \mathrm{Ric}[g] - \frac{1}{2}R[g]g$ satisfies the null energy condition. Suppose that $\mathcal{T} \cup \mathcal{A} \neq \emptyset$. Then:*

(i) *$\mathcal{T} \cup \mathcal{A} \subseteq \mathcal{BH}$. In particular, the black hole region of $\mathcal{Q}^+$ is non-empty.*

(ii) *$\mathcal{H}^+$ is an outgoing null line with future limit point i^+ and $\mathcal{H}^+ \subset \mathcal{R} \cup \mathcal{A}$.*

(iii) *Let $(u_*, v_*) \in \mathcal{T} \cup \mathcal{A}$. Then $(u_*, v) \in \mathcal{T} \cup \mathcal{A}$ for all $v \geq v_*$.*

(iv) *On $\mathcal{A}$, we have that $1 - \frac{2m}{r} \equiv 0$.*

Proof. "(iii)" Let $p = (u_*, v_*) \in \mathcal{Q}^+$, such that $\partial_v r(p) \leq 0$. Since the null energy condition is satisfied, we have by Raychaudhuri that $\partial_v^2 r(u_*, v) \leq 0$ after rescaling v, so $\partial_v r(u_*, v) \leq 0$.

"(i)": Suppose additionally that $p \in J^-(\mathcal{I}^+)$. Then the line $\{(u_*, v) \in \mathbb{R}^2\ ,\ v \geq v_*\}$ must intersect $\mathcal{I}^+$. By definition of $\mathcal{I}^+$, this means that $\limsup_{v \geq v_*} r(u_*, v) = \infty$. From the above, it also follows that $\limsup_{v \geq v_*} r(u_*, v) \leq r(u_*, v_*) < \infty$, which is a contradiction. We conclude that $p \notin J^-(\mathcal{I}^+)$, so $\mathcal{Q}^+ \setminus J^-(\mathcal{I}^+) \neq \emptyset$ and $p \in \mathcal{BH}$.

"(ii)": By definition, $\mathcal{H}^+$ is the future boundary of $J^-(\mathcal{I}^+)$, which is an outgoing null line with limit point i^+. Suppose $\mathcal{H}^+ \cap \mathcal{T} \neq \emptyset$. Then there would exist a $p \in \mathcal{H}^+ \cap \mathcal{T}$ and a q in neighbourhood of p such that $q \in J^-(\mathcal{I}^+)$ and $\partial_v r(q) < 0$. But this is a contradiction by (i). Hence $\mathcal{H}^+ \subset \mathcal{R} \cup \mathcal{A}$.

"(iv)": This is immediate from the definition of m and the fact that $\partial_v r = 0$ on $\mathcal{A}$. □

[Exercise: Suppose $\mathcal{H}^+ \cap \mathcal{A} \neq \emptyset$. Show that then $\mathbb{T}_{vv} \equiv 0$ on $\mathcal{H}^+ \cap \mathcal{A}$.]

We immediately obtain the following corollary.

Corollary 5.17 (Hawking's area theorem in spherical symmetry)**.** *$\partial_v(4\pi r^2) \geq 0$ on $\mathcal{H}^+$.*

This statement is known as the *second law of black hole mechanics*.

By imposing the dominant energy condition, we can obtain the following *monotonicity properties* for the Hawking mass.

Proposition 5.18. *Assume that $8\pi\mathbb{T} = \mathrm{Ric}[g] - \frac{1}{2}R[g]g$ satisfies the dominant energy condition. Then:*

(i) *In $(\mathcal{R} \cup \mathcal{A}) \cap \mathcal{Q}^+$:*

$$\partial_v m \geq 0,$$
$$\partial_u m \leq 0.$$

(ii) *Assume that m is bounded from above along H_0, i.e. there exists an $M_0 \in \mathbb{R}$, such that $m(0, v) \leq M_0$ for all $0 \leq v < v_{\mathcal{I}}$. Then m can be continuously extended to $\mathcal{I}^+$, i.e.*

$$M_{\mathrm{Bondi}}(u) := \lim_{\substack{v \to v_{\mathcal{I}} \\ u < u_{\mathcal{I}}}} m(u, v)$$

is well-defined. Furthermore,

$$M_{\mathrm{Bondi}}(u_2) \leq M_{\mathrm{Bondi}}(u_1).$$

for $0 \leq u_1 \leq u_2 < u_{\mathcal{I}}$.

(iii) *If $\Gamma = \emptyset$, assume that m is bounded from below on $\underline{H}_0 \cap \{u \leq u_{\mathcal{I}}\}$, i.e. there exists an $M_1 \in \mathbb{R}$, such that $m|_{\underline{H}_0}(u) \geq M_1$ for $u \leq u_{\mathcal{I}}$. Then the following limit is well-defined $M_f := \lim_{u \uparrow u_{\mathcal{I}}} M_{\text{Bondi}}(u)$.*

(iv) *If $\Gamma \neq \emptyset$, then $m|_{\Gamma} \equiv 0$ and $m \geq 0$ in $\mathcal{R} \cup \mathcal{A}$.*

Proof. "(i)": The monotonicity properties follow directly from (5.11) and (5.12), using that $\mathbb{T}_{uu}, \mathbb{T}_{vv}, \mathbb{T}_{uv} \geq 0$ by Lemma 5.5. Since $\partial_u m \leq 0$ and m is bounded by M_0 along H_0, we must have that m stays bounded by M_0 in $\mathcal{R} \cup \mathcal{A}$.

"(ii)": Let $(u_*, v_*) \in J^-(\mathcal{I}^+)$. Then $(u, v) \in \mathcal{R}$ for all $v \geq v_*$ and $u \leq u_*$ by Proposition 5.16. Since $\partial_u m \leq 0$ in $\mathcal{R}$, this implies that $m(u_*, v) \leq m(0, v)$, which is bounded from by M_0 by assumption. Since moreover $\partial_v m(u_*, v) \geq 0$, $m(u, v)$ is monotonically non-decreasing and must attain a finite limit as $v \uparrow v_{\mathcal{I}}$.[38]. Hence, $M_{\text{Bondi}}(u)$ is well-defined and using that the limit respects inequalities:

$$0 \geq \lim_{v \to v_{\mathcal{I}}} [m(u_2, v) - m(u_1, v)] = M_{\text{Bondi}}(u_2) - M_{\text{Bondi}}(u_1)$$

and we conclude (ii).

"(iv)": Consider $p \in \Gamma$. We can write:

$$m(p) = \frac{r}{2}(1 - \overline{g}^{-1}(dr, dr))(p).$$

Since r is certainly a C^1 function on $\mathbb{R} \times [0, \infty)$ and $\overline{g}$ is certainly continuous on $\mathbb{R} \times [0, \infty)$, the expression $\overline{g}^{-1}(dr, dr)(p)$ must be finite. Since $r(p) = 0$, we must therefore have that $m(p) = 0$.

Now let $q = (u_*, v_*) \in \mathcal{R} \cup \mathcal{A}$. Then $q \in J^+(\Gamma)$. By (iii) of Proposition 5.16, $(u_*, v) \in \mathcal{R} \cup \mathcal{A}$ for all $v \leq v_*$. Since $\partial_v m \geq 0$ in $\mathcal{R} \cup \mathcal{A}$, we have that $m \geq 0$ at q.

"(iii)": To conclude (iii) we use that $M_{\text{Bondi}}(u)$ non-increasing and, if we can show that $M_{\text{Bondi}}(u) \geq M_1$ for all u, we can conclude that it must attain a limit as $u \uparrow u_{\mathcal{I}}$ via the argument in the proof of (ii). When $\Gamma \neq \emptyset$, we have that $m \geq 0$ by (iv), so $M_{\text{Bondi}}(u) \geq 0$ which is a uniform bound from below. When $\Gamma = \emptyset$, we use that assumption that $m|_{\underline{H}_0 \cap \{u \leq u_{\mathcal{I}}\}} \geq M_1$ together with $\partial_v m \geq 0$ in $\mathcal{R} \cup \mathcal{A}$ to conclude that $M_{\text{Bondi}}(u) \geq \inf m|_{\underline{H}_0} \geq M_1$.

□

We refer to M_f as the *final Bondi mass* of the spacetime and $M_i := M_{\text{Bondi}}(0)$ as the *initial Bondi mass*.

Definition 5.7. *The outermost apparent horizon $\mathcal{A}'$ is the following subset of $\mathcal{A}$:*

$$\mathcal{A}' := \{(u, v) \in \mathcal{A} \,|\, (u', v) \in \mathcal{R} \text{ for all } u' < u\}.$$

Proposition 5.19 (Penrose inequality in spherical symmetry)**.** *$r \leq 2M_f$ along $\mathcal{A}'$.*

Proof. Exercise. Hint: Use that $r(u, v) = 2m(u, v)$ for $(u, v) \in \mathcal{A}'$ and connect (u, v) to $\mathcal{I}^+$ via ingoing and outgoing null segments. □

Proposition 5.20 (Positive mass theorem in spherical symmetry)**.** *Assume that $8\pi\mathbb{T} = \text{Ric}[g] - \frac{1}{2}R[g]g$ satisfies the dominant energy condition and that $\Gamma \neq \emptyset$. Then $M_i \geq 0$, with $M_i = 0$ if and only if $(\mathcal{M}, g)$ is isometric to a region of the Minkowski spacetime.*

Proof. Let $\Gamma \neq \emptyset$, then it follows from Proposition 5.18 (iv) that $m \geq 0$ in $\mathcal{R} \cup \mathcal{A}$, so in particular $M_i \geq 0$.

Now suppose that $M_i = 0$. Since $H_0 \subset \mathcal{R}$, we have that $\partial_v m|_{H_0} \geq 0$, so $m|_{H_0} \leq 0$. By the fact that also $m|_{H_0} \geq 0$, we must therefore have that $m|_{H_0} = 0$. Let $p \in \mathcal{R}$. Then there exists a

[38] This follows from the following fact from real analysis: bounded monotone sequences converge.

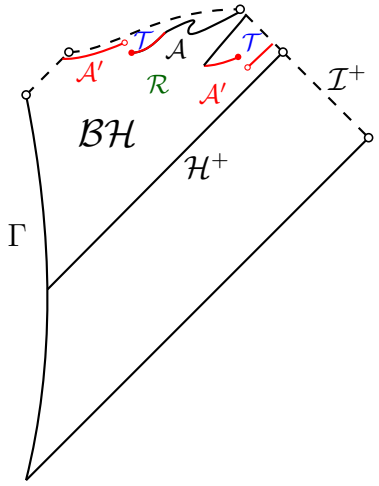


Figure 27: Example of a spherically symmetric spacetime with a complicated apparent horizon.

past-directed ingoing null line from p to H_0, since $p \in D^+(H_0 \cup \Gamma)$ and past-directed ingoing null lines cannot intersect Γ, as it is timelike and forms a left boundary of the spacetime in the Penrose diagram. If $\mathcal{R} = \mathcal{Q}^+$, we moreover have that $\partial_u m \leq 0$ along this ingoing null line, so together with $m \geq 0$ in $\mathcal{R}$, this implies that $m \equiv 0$ everywhere in $\mathcal{Q}^+$.

We will now show that $\mathcal{Q}^+ = \mathcal{R}$. Suppose that $\mathcal{T} \cup \mathcal{A} \neq \emptyset$. Then we must have that $\mathcal{A}' \neq \emptyset$ (using continuity of $\partial_v r$ and the fact that there exist null lines along which $\partial_v r > 0$ and $\partial_v r < 0$ at different points). Let $q \in \mathcal{A}'$, then $\frac{2m}{r}(q) = 1$, so $m(q) \geq \frac{r}{q} > 0$. But q can also be connected to H_0 via a past-directed null line, which is moreover contained in $\mathcal{R} \cup \mathcal{A}$, by the defining property of $\mathcal{A}'$, so $\partial_u m \leq 0$ along the null line and we must have that $m(q) \leq 0$, which is a contradiction.

By (5.11) and (5.12), together with the fact that $\partial_u r < 0$, $\partial_v r \geq 0$ in $\mathcal{R}$ and $\mathbb{T}_{uu}, \mathbb{T}_{uv}, \mathbb{T}_{vv} \geq 0$, we must have that $\mathbb{T}_{uu} = \mathbb{T}_{uv} = \mathbb{T}_{vv} = 0$. Now recall the equations for r (equations (5.8)–(5.10)):

$$\begin{aligned} \partial_u \partial_v r &= -\frac{\Omega^2 m}{2r^2} + 4\pi r \mathbb{T}_{uv}, \\ \Omega^2 \partial_u(\Omega^{-2} \partial_u r) &= -4\pi r \mathbb{T}_{uu}, \\ \Omega^2 \partial_v(\Omega^{-2} \partial_v r) &= -4\pi r \mathbb{T}_{vv}. \end{aligned}$$

Using that $\mathbb{T}_{uu} = \mathbb{T}_{uv} = \mathbb{T}_{vv} = 0$ and $m = 0$, they simplify as follows:

$$\begin{aligned} \partial_u \partial_v r &= 0, \\ \partial_u(\Omega^{-2} \partial_u r) &= 0, \\ \partial_v(\Omega^{-2} \partial_v r) &= 0. \end{aligned}$$

We can now apply Theorem 5.11. We provide here an alternative, more explicit argument. Without loss of generality, we can rescale the v coordinate so that $\partial_v r = \frac{1}{2}$ along H_0. By integrating $\partial_u \partial_v r = 0$ in u, we must therefore have that $\partial_v r = \frac{1}{2}$ everywhere in $\mathcal{Q}^+$. Furthermore, by $m = 0$, we have that $-4\Omega^{-2} \partial_u r \partial_v r = 1$, so $-\Omega^{-2} \partial_u r = \frac{1}{2}$. By integrating $\partial_v \partial_u r = 0$ in v, we moreover have that $\partial_u r(u, v) = \partial_u r|_\Gamma(u)$. By rescaling the u coordinate, we can assume without loss of generality that $\partial_u r|_\Gamma(u) = -\frac{1}{2}$, so that $\partial_u r = -\frac{1}{2}$ everywhere and hence, $\Omega^2 = 1$. We conclude that $r = \frac{1}{2}(v - u)$ and $g = -dudv + \frac{(v-u)^2}{4} \mathring{\not{g}}$, which is the Minkowski metric in standard double null coordinates. □

[Exercise: 1) Explain why in the case $\Gamma \neq \emptyset$, $\mathcal{Q}^+$ must be future-null-geodesically incomplete. 2) Drop the assumption that $\partial_u r < 0$ along H_0. Assume that $\Gamma = \emptyset$ and $m|_{H_0} < 0$ and *derive* $\partial_u r|_{H_0} < 0$.]

Note finally that the apparent horizon $\mathcal{A}$ can have a very complicated structure (even in spherical symmetry!). Unlike the future event horizon $\mathcal{H}^+$, it does not need to be connected, for example.

5.7 Cold static stars

We consider spherically symmetric spacetimes, with metrics that are solutions to the Einstein–Euler equations with the perfect fluid stress-energy tensor:

$$\mathbb{T}_{\mu\nu} := (\rho + p)U_\mu U_\nu + p(g^{-1})^{\mu\nu}.$$

Assume that $U \in \mathcal{T}(\mathcal{M})$ is spherically symmetric, which we take to mean that U is independent of θ, φ and $U^A \equiv 0$, where for $A \in \{\theta, \varphi\}$.[39]

Therefore,

$$-1 = g(U, U) = -\Omega^2 U^u U^v.$$

Furthermore $U_u = -\frac{1}{2}\Omega^2 U^v$ and $U_v = -\frac{1}{2}\Omega^2 U^u$.

We therefore obtain:

$$\begin{aligned}
\mathbb{T}_{uu} &= (\rho + p)(U_u)^2 = \frac{1}{4}(\rho + p)\Omega^4 (U^v)^2, \\
\mathbb{T}_{vv} &= (\rho + p)(U_v)^2 = \frac{1}{4}(\rho + p)\Omega^4 (U^u)^2 \\
\mathbb{T}_{uv} &= (\rho + p)U_u U_v - \frac{1}{2}p\Omega^2 = \frac{1}{4}(\rho - p)\Omega^2.
\end{aligned}$$

We will first investigate static solutions describing "stars" of some fixed radius $r = R > 0$ at zero temperature. That is to say. We make the following assumptions on our spherically symmetric spacetime solutions to the Einstein–Euler equations:

- $(\mathcal{M}, g)$ is spherically symmetric with $\Gamma \neq \emptyset$ and ρ and p are independent of θ, φ, U is independent of θ, φ and $U^A \equiv 0$,
- $\mathbb{T}$ satisfies the dominant energy condition, i.e. $\rho \geq |p|$.
- In the region $\{r > R\}$, $p = \rho \equiv 0$. By Birkhoff's theorem, the region $\{r > R\}$ must then be isometric to the region $\{r > R\}$ in a Schwarzschild spacetime with mass $M \in \mathbb{R}$. Since $\Omega^{-2}\partial_u r < 0$ everywhere in Schwarzschild, we can integrate (5.9) and use that $\mathbb{T}_{uu} \geq 0$ by the null energy condition to conclude that $\Omega^{-2}\partial_u r < 0$ along all ingoing null lines emanating from $\{r > R\}$ and in particular, $dr_p \neq 0$ for all p in the corresponding spacetime region.
- $(\mathcal{M}, g)$ is static. More precisely, we will assume that there exists a complete timelike Killing vector field K,[40] such that we can decompose:

  $$g = -e^{\Phi} dt^2 + \underline{g},$$

  with $K = \frac{\partial}{\partial t}$, $\underline{g}$ is the induced Riemannian metric on $\Sigma := \{t = 0\}$, with $\Sigma \cong \mathbb{R}^3$ and $\Phi : \Sigma \to \mathbb{R}$.

[39] This is equivalent to: $\mathcal{L}_{O_i} U \equiv 0$, where O_i, $i \in \{1, 2, 3\}$ are the angular momentum vector fields.

[40] This means that the domain of the integral curves of K is $\mathbb{R}$.

Together with the spherical symmetry assumption, $K(r) = 0$ and the fact that $dr \neq 0$ in the region connecting $\{r < R\}$ via ingoing null lines, this means that $dr \neq 0$ everywhere in $\mathcal{M}$ and we can write $\underline{g} = e^{-\Psi}dr^2 + r^2 \mathring{\not{g}}$, so that:

$$g = -e^{\Phi}dt^2 + e^{-\Psi}dr^2 + r^2 \mathring{\not{g}},$$

with $\Phi, \Psi : [0, \infty)_r \to \mathbb{R}$. We will assume that K is future-directed. We denote $X = \partial_r$ in (t, r, θ, φ) coordinates.

- The fluid is at rest in (t, r, θ, φ) coordinates, so U points in the ∂_t direction and ρ, p do not depend on t. By the condition $g(U, U) = -1$, this means that $U = e^{-\frac{1}{2}\Phi}\partial_t$.
- We will assume that $p \geq 0$. This can be interpreted as fluid trying to resist compression.
- We will assume that p is a function of ρ. This is called a *barotropic equation of state*. For more general equations of state, p can be thought of as a function of ρ and a temperature T. The barotropic assumption may be motivated by the assumption that the temperature of the star is negligible; the star is "cold" and no longer radiating.
- We assume that $\frac{dp}{d\rho} > 0$, a necessary condition for the initial value problem of the Euler equations to be locally well-posed. The speed of sound in the fluid is then given by $\sqrt{\frac{dp}{d\rho}}$. We could additionally assume that $\frac{dp}{d\rho} < 1$, which means that the speed of sound is smaller than the speed of light.

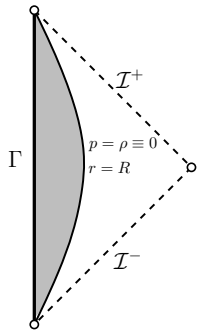


Figure 28: The Penrose diagram of a spherically symmetric static star.

We can write the metric in double null coordinates by introducing $u = t - r_*$ and $v = t + r_*$ ($t = \frac{1}{2}(u + v)$ and $r_* = \frac{1}{2}(v - u)$), with $r_*(0) = 0$ and

$$\frac{dr_*}{dr} = e^{-\frac{1}{2}(\Phi+\Psi)}.$$

Indeed, then

$$g = e^{\Phi}(-dt^2 + dr_*^2) + r^2 \mathring{\not{g}} = -e^{\Phi}dudv + r^2 \mathring{\not{g}}.$$

[Exercise: Construct the corresponding Penrose diagram.]

We have the following identities:

$$\Omega^2 = e^{\Phi},$$
$$\partial_v = \frac{1}{2}(K + e^{\frac{1}{2}(\Phi+\Psi)}X),$$

$$\begin{aligned}\partial_u &= \frac{1}{2}(K - e^{\frac{1}{2}(\Phi+\Psi)}X),\\ U &= e^{-\frac{1}{2}\Phi}(\partial_u + \partial_v),\\ X &= e^{-\frac{1}{2}(\Phi+\Psi)}(\partial_v - \partial_u),\\ \partial_v r &= -\partial_u r = \frac{1}{2}e^{\frac{1}{2}(\Phi+\Psi)}.\end{aligned}$$

By definition of m, we therefore have that

$$m = \frac{r}{2}(1 + 4\Omega^{-2}(\partial_u r)(\partial_v r)) = \frac{r}{2}\left(1 - e^{\Psi}\right).$$

Rearranging the above, we have that

$$e^{\Psi} = 1 - \frac{2m}{r}.$$

This implies in particular that $m(r) < \frac{r}{2}$.

Recall also that the Hawking mass satisfies:

$$\begin{aligned}\partial_u m &= -8\pi r^2\Omega^{-2}(\partial_v r\mathbb{T}_{uu} - \partial_u r\mathbb{T}_{uv})\\ \partial_v m &= 8\pi r^2\Omega^{-2}(-\partial_u r\mathbb{T}_{vv} + \partial_v r\mathbb{T}_{uv}).\end{aligned}$$

Note that therefore

$$\begin{aligned}X(m) &= e^{-\frac{1}{2}(\Phi+\Psi)}(\partial_v m - \partial_u m)\\ &= 8\pi r^2\Omega^{-2}e^{-\frac{1}{2}(\Phi+\Psi)}(-\partial_u r\mathbb{T}_{vv} + \partial_v r\mathbb{T}_{uu} + (\partial_v r - \partial_u r)\mathbb{T}_{uv})\\ &= 4\pi r^2\Omega^{-2}e^{-\frac{1}{2}(\Phi+\Psi)}\frac{dr}{dr_*}(\mathbb{T}_{vv} + \mathbb{T}_{uu} + 2\mathbb{T}_{uv})\\ &= \pi r^2\Omega^2\left[(\rho+p)((U^v)^2 + (U^u)^2) + 2(\rho-p)\Omega^{-2}\right]\\ &= \pi r^2\Omega^2\left[2(\rho+p)e^{-\Phi} + 2(\rho-p)e^{-\Phi}\right]\\ &= 4\pi\rho r^2 \qquad (5.21)\end{aligned}$$

By Birkhoff's theorem, the region $r > R$ must be isometric to a region in a Schwarzschild spacetime with mass M, so $\Phi(r) = \Psi(r) = \log\left(1 - \frac{2M}{r}\right)$ for $r > R$. Furthermore, by (5.21) and the dominant energy condition, we obtain $\rho \geq 0$ and

$$M = m(R) = 4\pi\int_0^R \rho(r)\, r^2 dr \geq 0.$$

Since $m(r) < \frac{r}{2}$, this means that $R > 2M$. There can therefore exist no static stars with radius $R \leq 2M$. We will see below that this lower bound can be sharpened.

[Exercise: We define E, the total energy of the fluid by integrating the matter energy density $\mathbb{T}_{00}$, i.e. $E = \int_{\{t=0\}}\mathbb{T}_{00}\sqrt{-\det \underline{g}}d\theta d\varphi dr$, with $\underline{g} = e^{-\Psi}dr^2 + r^2\mathring{\not{g}}$ the induced metric on $\{t=0\}$. Show that $M < E$. Hence, the total energy of the star, M, is smaller than the energy of the matter, E. One can interpret the difference $E - M$ as the gravitational binding energy of the star.]

Proposition 5.21. *Under the above assumptions, (m, p, Φ) satisfy the following equations:*

$$\frac{dm}{dr} = 4\pi r^2\rho, \qquad (5.22)$$

$$\frac{d\Phi}{dr} = 2\frac{m + 4\pi r^3 p}{r(r-2m)}, \qquad (5.23)$$

$$\frac{dp}{dr} = -(\rho + p)\frac{m + 4\pi r^3 p}{r(r-2m)}. \tag{5.24}$$

The above equations are called the Tolman–Oppenheimer–Volkoff *(TOV) equations of hydrostatic equilibrium.*

Proof. First of all, note that the equation for m follows directly from (5.21). The Raychaudhuri equations (5.9) and (5.10), together with (5.21) then give (Exercise):

$$\left(1 - \frac{2m}{r}\right)^{\frac{1}{2}} e^{\frac{\Phi}{2}} \frac{d}{dr}\left(\left(1 - \frac{2m}{r}\right)^{\frac{1}{2}} e^{-\frac{\Phi}{2}}\right) = -4\pi r(\rho + p).$$

We apply the Leibniz rule for differentiation to obtain:

$$-\frac{d}{dr}\left(\frac{m}{r}\right) - \frac{1}{2}\left(1 - \frac{2m}{r}\right)\frac{d\Phi}{dr} = -4\pi r(\rho + p).$$

Using that $\frac{d(r^{-1}m)}{dr} = 4\pi\rho r - r^{-2}m$, we obtain:

$$2mr^{-2} - \left(1 - \frac{2m}{r}\right)\frac{d\Phi}{dr} = -8\pi r p.$$

so

$$\frac{d\Phi}{dr} = 2\frac{m + 4\pi r^3 p}{r(r-2m)}.$$

The Euler equations give:

$$(\rho + p)U^\alpha \nabla_\alpha U^\mu = -((g^{-1})^{\mu\nu} + U^\mu U^\nu)\nabla_\nu p.$$

Using that $U = e^{-\frac{1}{2}\Phi}K$, with K a Killing vector field, $g(X, U) = 0$, and $(\mathcal{L}_K g)_{\alpha\mu} = \nabla_\alpha K_\mu + \nabla_\mu K_\alpha = 0$, so

$$\begin{aligned}
\frac{dp}{dr} = X(p) = (g^{-1})^{\mu\nu} X_\mu \nabla_\nu p &= ((g^{-1})^{\mu\nu} + U^\mu U^\nu) X_\mu \nabla_\nu p \\
&= -(\rho + p)U^\alpha X^\mu \nabla_\alpha U_\mu = -(\rho + p)e^{-\frac{\Phi}{2}} K^\alpha X^\mu \nabla_\alpha (e^{-\frac{1}{2}\Phi} K)_\mu \\
&= -(\rho + p)e^{-\Phi} K^\alpha X^\mu \nabla_\alpha K_\mu = (\rho + p)e^{-\Phi} K^\alpha X^\mu \nabla_\mu K_\alpha \\
&= \frac{1}{2}(\rho + p)e^{-\Phi} X(g(K, K)) = -\frac{1}{2}(\rho + p)e^{-\Phi}\frac{d}{dr}(e^\Phi) \\
&= -\frac{1}{2}(\rho + p)\frac{d\Phi}{dr} = -(\rho + p)\frac{m + 4\pi r^3 p}{r(r-2m)}.
\end{aligned}$$

□

Lemma 5.22. *The following identity holds:*

$$\frac{d}{dr}\left(r^{-1} e^{\frac{\Psi}{2}} \frac{de^{\frac{\Phi}{2}}}{dr}\right) = e^{\frac{\Phi - \Psi}{2}} \frac{d}{dr}\left(\frac{m}{r^3}\right). \tag{5.25}$$

Proof. We apply the Leibniz rule to obtain:

$$\frac{d}{dr}\left(r^{-1} e^{\frac{\Psi}{2}} \frac{de^{\frac{\Phi}{2}}}{dr}\right) = r^{-1}\frac{d}{dr}\left(e^{\frac{\Psi}{2}} \frac{de^{\frac{\Phi}{2}}}{dr}\right) - \frac{1}{2} r^{-2} e^{\frac{\Psi + \Phi}{2}} \frac{d\Phi}{dr}.$$

By (5.23) together with $r(r-2M) = r^2 e^{\Psi}$, we therefore have:

$$\frac{d}{dr}\left(r^{-1}e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) = r^{-1}\frac{d}{dr}\left(e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) - r^{-4}e^{\frac{\Phi-\Psi}{2}}(m+4\pi r^3 p).$$

Using that $X = e^{-\frac{\Phi+\Psi}{2}}(\partial_v - \partial_u)$, we obtain:

$$r^{-1}\frac{d}{dr}\left(e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) = r^{-1}e^{-\frac{\Phi+\Psi}{2}}(\partial_v-\partial_u)\left(e^{-\frac{\Phi}{2}}(\partial_v-\partial_u)(e^{\frac{\Phi}{2}})\right) = \frac{1}{2r}e^{-\frac{\Phi+\Psi}{2}}(\partial_v-\partial_u)^2\Phi.$$

Now we use that $e^{\Phi} = \Omega^2$ and $(\partial_u+\partial_v)\Phi = 0$ to obtain $(\partial_v-\partial_u)^2\Phi = -4\partial_u\partial_v \log\Omega^2$ and hence:

$$r^{-1}\frac{d}{dr}\left(e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) = -\frac{2}{r}e^{-\frac{\Phi+\Psi}{2}}\partial_u\partial_v\log\Omega^2.$$

We then use that $\frac{2m}{r} = 1 + 4\Omega^{-2}\partial_u r\partial_v r$, $\operatorname{tr}\mathbb{T} = -\rho+3p$ and $\Omega^{-2}\mathbb{T}_{uv} = \frac{1}{4}(\rho-p)$ to put (5.7) in the following form:

$$\partial_v\partial_u\log\Omega^2 = \frac{\Omega^2}{2}\left[-2mr^{-3} - 4\pi(-\rho+3p) - 8\pi r^2(\rho-p)\right] = -\frac{\Omega^2}{2}\left(-2mr^{-3}+4\pi(\rho+p)\right).$$

Hence,

$$r^{-1}\frac{d}{dr}\left(e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) = e^{\frac{\Phi-\Psi}{2}}(-2mr^{-4}+4\pi(\rho+p)r^{-1}).$$

Combining the above and using (5.22), we obtain:

$$\frac{d}{dr}\left(r^{-1}e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) = e^{\frac{\Phi-\Psi}{2}}(-3mr^{-4}+4\pi\rho r^{-1}) = e^{\frac{\Phi-\Psi}{2}}\frac{d}{dr}(mr^{-3}).$$ □

Proposition 5.23 (Buchdahl inequality)**.** *There exists a solution (m,p,Φ) to the TOV equations if and only if $R > \frac{9}{4}M$.*

Spherically symmetric, cold static stars must therefore have a radius bounded below by $\frac{9}{4}M$.

Remark 5.9. *There is no lower bound on the radius of a self-gravitating spherical fluid ball in Newtonian theory!*

Proof. We will derive the inequality by combining (5.25) with $e^{\frac{1}{2}\Phi(0)} > 0$, since Φ is well defined at $r = 0$ by assumption.

Consider the average ρ, measured with respect to the volume form $4\pi r^2 dr$:

$$\overline{\rho}(r) := \frac{1}{\frac{4}{3}\pi r^3}\int_0^r 4\pi\rho(r')r'^2\,dr = \frac{m(r)}{\frac{4}{3}\pi r^3}.$$

Then we can express (5.25) as follows:

$$\frac{d}{dr}\left(r^{-1}e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) = e^{\frac{\Phi-\Psi}{2}}\frac{4}{3}\pi r^3\frac{d\overline{\rho}}{dr}(r).$$

By assumption, $\frac{dp}{d\rho} > 0$ and by (5.24), $\frac{dp}{dr} \leq 0$, so $\frac{d\rho}{dr} \leq 0$, which implies that for all $0 \leq r_1 < r_2$:

$$\overline{\rho}(r_1) \geq \overline{\rho}(r_2).$$

In particular, this implies that

$$m(r) = \frac{4}{3}\pi r^3 \overline{\rho}(r) \geq \frac{4}{3}\pi r^3 \overline{\rho}(R) = MR^{-3}r^3. \tag{5.26}$$

and $\frac{d\overline{\rho}}{dr}(r) \leq 0$, so

$$\frac{d}{dr}\left(r^{-1}e^{\frac{\Psi}{2}}\frac{de^{\frac{\Phi}{2}}}{dr}\right) \leq 0. \tag{5.27}$$

It will now be convenient to introduce the auxiliary variable s, defined as follows:

$$s(r) = \int_0^r r'(1-\frac{2m(r')}{r'})^{-\frac{1}{2}}\,dr'.$$

Note that $\frac{ds}{dr}(r) = r(1-\frac{2m(r)}{r})^{-\frac{1}{2}} = re^{-\frac{\Psi}{2}} \geq 0$. Then we can write (5.27) as follows:

$$\frac{d^2}{ds^2}(e^{\frac{\Phi}{2}}) \leq 0$$

Since $\frac{d\Phi}{dr} >\geq 0$, we also have that $\frac{d}{ds}(e^{\frac{\Phi}{2}}) \geq 0$. In particular, $e^{\frac{\Phi}{2}}$ is concave as a function of s. Write $s_R = s(R)$. Then we apply (5.23) to obtain:

$$\frac{d}{ds}(e^{\frac{\Phi}{2}})(s_R) = \left(\frac{ds}{dr}\right)^{-1}(R)\frac{d}{dr}(e^{\frac{\Phi}{2}})(R) = (1-\frac{2M}{R})^{-\frac{1}{2}}R^{-3}e^{\frac{\Phi}{2}}(s_R)(M+4\pi R^3 p(R)) = (1-\frac{2M}{R})^{-\frac{1}{2}}MR^{-3}e^{\frac{\Phi}{2}}(s_R).$$

Since $\frac{d}{ds}(e^{\frac{\Phi}{2}})(s) \geq \frac{d}{ds}(e^{\frac{\Phi}{2}})(s_R)$, for all $0 \leq s \leq s_R$ by $\frac{d^2}{ds^2}(e^{\frac{\Phi}{2}}) \leq 0$, we can write:

$$e^{\frac{\Phi}{2}}(0) = e^{\frac{\Phi}{2}}(s_R) - \int_0^{s_R}\frac{d}{ds}(e^{\frac{\Phi}{2}})(s)\,ds \leq e^{\frac{\Phi}{2}}(s_R) - \frac{d}{ds}(e^{\frac{\Phi}{2}})(s_R)s_R = e^{\frac{\Phi}{2}}(s_R)\left[1-(1-2MR^{-1})^{-\frac{1}{2}}MR^{-3}s_R\right].$$

Now we apply (5.26) to estimate $2m(r)r^{-1} \geq 2MR^{-3}r^2$ and hence:

$$s_R = \int_0^R r'(1-\frac{2m(r')}{r'})^{-\frac{1}{2}}\,dr' \geq \int_0^R r'\left(1-\frac{2Mr^2}{R^3}\right)^{-\frac{1}{2}}\,dr' = \frac{R^3}{2M}\left(1-\left(1-\frac{2M}{R}\right)^{\frac{1}{2}}\right),$$

so that:

$$e^{\frac{\Phi}{2}}(0) \leq e^{\frac{\Phi}{2}}(s_R)\left[1-\frac{1}{2}\left((1-2MR^{-1})^{-\frac{1}{2}}-1\right)\right] = e^{\frac{\Phi}{2}}(s_R)\left[\frac{3}{2}-\frac{1}{2}(1-2MR^{-1})^{-\frac{1}{2}}\right].$$

Since $e^{\frac{\Phi}{2}}(0) > 0$, this implies: $(1-2MR^{-1}) > \frac{1}{9}$, which is equivalent to $R > \frac{9}{4}M$. □

5.7.1 Collapsing dust clouds

Now consider the spherically symmetric Einstein-dust model by taking $p \equiv 0$ in Euler equations and setting $\rho \geq 0$. Then the Euler equations reduce to:

$$\begin{aligned}\mathrm{div}(\rho U) &= 0,\\ \nabla_U U &= 0\end{aligned}$$

Assume $\Gamma \neq \emptyset$, so $\mathcal{M} \cong \mathbb{R}\times\mathbb{R}^3$.

Assume that there exists a function $F : \mathcal{Q} \to \mathbb{R}$, such that $\rho \equiv 0$ in $\{F > 0\}$, so $(\{F > 0\}, g)$ must be isometric to an open subset in Schwarzschild with mass $M > 0$, by Birkhoff's theorem and the presence of a centre of symmetry.

Assume that there exists a spherically symmetric spacelike hypersurface Σ along which $\rho(r) \equiv \rho_0$, for $r \leq R$, with $\rho_0 > 0$ constant, and $\rho(r) = 0$ for $r > R$. The discontinuity of ρ along Σ implies that g cannot be a C^2 solution to the Einstein equations. We will nevertheless assume it is C^1 in global double null coordinates.

Assume that U extends continuously to $\{F = 0\}$ and $U(F)|_{\{F=0\}} = 0$ (with slight abuse of notation), so U is tangential to the curve $\{F = 0\}$. Since $\nabla_U U = 0$ and the metric is C^1 along $\{F = 0\}$, we must have that $\{F = 0\}$ represents the image of a radial timelike geodesic $\gamma : I \to \mathcal{M}$ in Schwarzschild and $\dot{\gamma}(\tau) = U_{\gamma(\tau)}$, with τ the proper time that we can set to 0 at Σ. We will assume that $\dot{r_\gamma}(0) = 0$. Then must have that $r_\gamma(\tau) \to 0$ in finite proper time τ.[41] That is to say, there is no pressure to hold off gravitational collapse of the dust cloud. Hence, the spacetime contains a subset of the Schwarzschild black hole interior and therefore the black hole region $\mathcal{BH} = \mathcal{Q} \setminus J^-(\mathcal{I}^+) \neq \emptyset$ is non-empty.

One class of spacetimes that fit the above assumptions are called the *Oppenheimer–Snyder spacetimes* (Oppenheimer–Snyder 1939). For these spacetimes, the metric in the interior of the

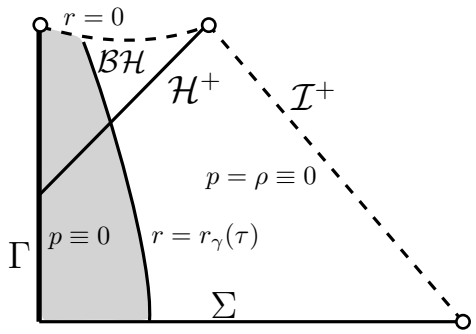


Figure 29: The Penrose diagram of an Oppenheimer–Snyder spacetime with initial hypersurface Σ.

dust cloud takes the form:

$$g = -d\tau^2 + a^2(\tau)\left[d\chi^2 + \sin^2\chi \mathring{g}\right],$$

with $U = \partial_\tau$, $r(\tau, \chi) = a(\tau)\sin\chi$, where $0 \leq \chi \leq \chi_b < \frac{\pi}{2}$, $R = a(0)\sin\chi_b$ and $r_\gamma(\tau) = a(\tau)\sin\chi_b$. The interior of the cloud is therefore spatially homogeneous.

Within the spherically symmetric dust model, the behaviour of spherically symmetric, homogeneous collapsing dust clouds is special. By considering instead a more general class of initial data including initially *inhomogeneous* spherically symmetric dust (of compact support), Christodoulou proved in 1984 that within this model, for an *open* subset of initial data (within spherical symmetry), $\mathcal{T} \cup \mathcal{A} = \emptyset$ and $\mathcal{BH} = \emptyset$. Nevertheless, the boundary $\partial\mathcal{PD}$ has a singular point $\mathcal{O}$ that can be reached in finite affine time, where moreover $\rho \to \infty$. Furthermore, observers moving along any curve $\{r = r_0 > 0\}$ inside the support of the fluid exist for only finite time and outgoing null geodesics emanating from the centre Γ get infinitely red-shifted when they are intercepted by the observer at their final time of existence. The singularity can be "seen" from far-away, in contrast with the singularity inside a black hole region, which is hidden behind an event horizon. These types of spacetimes are therefore called "naked singularities".

It is fortuitous that the naked singularities of Christodoulou were only discovered decades after the discovery of the Oppenheimer–Snyder spacetimes, since the later played a fundamental role in developing intuition regarding black hole formation and trapped surfaces.

[41] This follows from the timelike geodesic equation in Schwarzschild and the consideration of the potential $V_1(r)$ with total angular momentum $L = 0$.

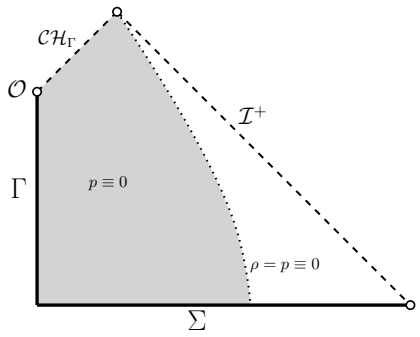


Figure 30: The Christodoulou Einstein-dust naked singularities. The radius r has a non-zero limit on $\mathcal{CH}_\Gamma$.

The existence of naked singularities in the spherically symmetric dust model can be attributed to the highly idealized nature of the model and the singular nature of the Einstein–Euler equations without pressure. Indeed, since $\rho \to \infty$, the assumption of $p = 0$ becomes questionable. One might therefore *hope* that these kind of spacetimes will be entirely excluded by considering less singular matter models.

In 2021, however, Guo–Hadžić–Jang established the existence of naked singularity solutions to the spherically symmetric Einstein–Euler equations with pressure, following earlier numerical work by Ori–Piran in 1987–1990.

In this next section, we will see that naked singularities can even arise in much simpler matter models.

5.8 The cosmic censorship conjectures

Christodolou's naked singularities can be characterized from future null infinity as follows: consider a sequence of inextendible, ingoing, future-directed, affinely parametrized radial null geodesics $\gamma_n : I_n \to \mathcal{M}$, such that

$$\gamma_n(0) = (u_0, v_n),$$
$$\dot{\gamma}_n(0) = (\partial_u)_{(u_0, v_n)},$$

with $v_n \to \infty$ as $n \to \infty$. Let $s_n^{\sup} = \sup I_n$. In the naked singularity spacetimes, there exists an $S < \infty$ such that $s_n^{\sup} \leq S$ for all n. You can interpreted this as the statement that "$\mathcal{I}^+$ is future-null-geodesically incomplete".

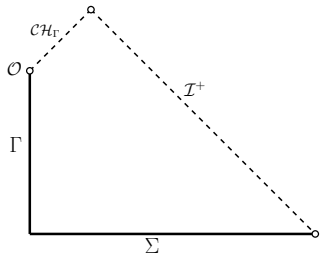
If we think of the corresponding spacetimes as arising from initial data on some Cauchy hypersurface Σ, there is no *global existence in time* of solutions to the Einstein–Euler equations. From the perspective $\mathcal{I}^+$, we cannot use the equations to predict what happens "for all times".

In 1994, Christodoulou proved that naked singularities also exist for the spherically symmetric Einstein–scalar field system, arising from initial data that are not smooth, but still sufficiently regular to have a well-posed initial value problem for the equations.[42] The singular behaviour at $\mathcal{O}$, on the other hand, is too strong to extend the metric across it. In 2019, it was shown by Shlapentokh-Rothman and Rodnianski that (non-spherically symmetric) naked singularities exist even in vacuum.

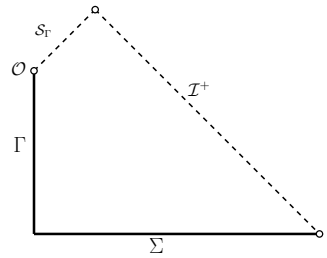
Sometimes, the $M < 0$ Schwarzschild spacetimes or the $M < e$ super-extremal Reissner–Nordström spacetimes are called "naked singularities". In these cases, however, the singularity

[42] The existence of naked singularties arising from smooth initial data for the Einstein–scalar field system remains an open problem and the time of writing of these notes.

is already present at the level of initial data and does not *form* in the evolution of more regular initial data.



(a) A naked singularity solution to the spherically symmetric Einstein-scalar field system with a null boundary $\mathcal{CH}_\Gamma$ emanating from a singular point $\mathcal{O}$ across which the metric can be extended in a suitably regular manner.



(b) A naked singularity solution to the spherically symmetric Einstein-scalar field system with a null boundary $\mathcal{S}_\Gamma$ emanating from a singular point $\mathcal{O}$ at which $r = 0$ and the metric cannot be extended in a suitably regular manner.

Figure 31: Penrose diagrams of naked singularity solutions to the spherically symmetric Einstein-scalar field system.

The *weak cosmic censorship conjecture* asserts that naked singularities are nevertheless "special".

Conjecture 5.24 (The weak cosmic censorship conjecture)**.** *Spacetimes arising from "generic", asymptotically flat, geodesically complete initial data to the Einstein equations coupled to a "reasonable" matter model have a future-null-geodesically complete future null infinity (i.e. there are no naked singularities).*

[Exercise: Find spacetimes arising from geodesically incomplete (characteristic) initial data with a null infinity that is not future-null-geodesically complete. Hint: consider spacetime regions in Minkowski arising from appropriate characteristic initial data to the spherically symmetric Einstein-scalar field equations with a vanishing scalar field.]

We already encountered another failure in the predictability of the Einstein equations, namely the lack of "global uniqueness" in the case of Reissner–Nordström spacetimes, which could be thought of as arising from geodesically complete initial data, but nevertheless had Cauchy horizons across which the spacetime could be extended smoothly in a *highly non-unique* manner. The *strong cosmic censorship conjecture* asserts that there should generically be *global uniqueness* of solutions:

Conjecture 5.25 (The strong cosmic censorship conjecture)**.** *Spacetimes arising from "generic", geodesically complete initial data to the Einstein equations coupled to a "reasonable" matter model are inextendible as "suitably regular" solutions to the equations.*

Despite what the naming suggests, the strong cosmic censorship conjecture does not imply the weak cosmic censorship conjecture. Indeed, this is the case in the Einstein–scalar field singularities of Christodoulou; see the Penrose diagrams in Figure 31.

6 Kerr black hole spacetimes

We will now consider our first non-spherically symmetric black hole spacetimes: the *Kerr black holes spacetimes* (Kerr 1963). Just like in the Schwarzschild case, we start by defining the *exterior* of the spacetimes, before extending into the black hole interior. In contrast with Schwarzschild, we will not construct a global double null foliation (this is considerably harder on Kerr spacetimes!). Since the existence of a global double null foliation is directly connected to the construction of a Penrose diagram, we will, strictly speaking, not be able to draw the corresponding Penrose diagram. Nevertheless, we will use "Penrose-like" diagrams to draw boundaries of different spacetime regions schematically.

6.1 Kerr exterior

The Kerr exterior is the spacetime $(\mathcal{M}_{\text{ext}}, g_{M,a})$ with $M \in \mathbb{R}$ and $a \in \mathbb{R}$. Here, $\mathcal{M}_{\text{ext}} = \mathbb{R} \times (r_+, \infty) \times \mathbb{S}^2$. When $a^2 \leq M^2$, $0 < r_- < r_+$ are the roots of of the following polynomial:

$$\Delta(r) := r^2 - 2Mr + a^2,$$

More explicitly,

$$r_\pm = M \pm \sqrt{M^2 - a^2}.$$

When $a^2 > M^2$, we can take $r_+ := 0$.

We interpret the parameter M as the mass or energy of the spacetime and $M|a|$ as the magnitude of the "total angular momentum of the spacetime".[43] We will assume for now that $|a| \leq M$. Note that when $a = 0$, we recover the Schwarzschild metric.

The Kerr metric $g_{M,a}$ is defined as follows:[44]

$$g_{M,a} = -\frac{\Delta}{\rho^2}(dt - a\sin^2\theta d\varphi)^2 + \frac{\rho^2}{\Delta}dr^2 + \rho^2 d\theta^2 + \frac{\sin^2\theta}{\rho^2}(adt - (r^2+a^2)d\varphi)^2,$$
$$\rho^2(r,\theta) := r^2 + a^2\cos^2\theta.$$

One may check that the above metric is a solution to the vacuum Einstein equations.

In the above form, the Kerr metric does not immediately fit the definition of a spherically symmetric or static spacetime when $a \neq 0$. However, that alone does not rule out the possibility that there exist coordinates in which the Kerr metric is manifestly spherically symmetric or static. For example, when $M = 0$, $(\mathcal{M}_{\text{ext}}, g_{M,a})$ can be shown to be isometric to a region in Minkowski, which is definitely spherically symmetric and static. We can use the non-vanishing of the total angular momentum vector of the spacetime in the case $Ma \neq 0$, which is a coordinate-invariant statement by the upcoming discussion in §7.1, to conclude that the spacetime cannot be static when $a \neq 0$ and $M \neq 0$. By Birkhoff's theorem, spherical symmetry would require the spacetime to be isometric to a region in the Schwarzschild exterior, which is static, so we conclude that $(\mathcal{M}_{\text{ext}}, g_{M,a})$ can also not be spherically symmetric.

We say a Kerr spacetime is *extremal* if $|a| = M$, *sub-extremal* if $|a| < M$ and *super-extremal* if $|a| > M$. Note that for $|a| = M$, $r_+ = r_- = M$.

Since $(g_{M,a})_{\mu\nu}$ are independent of t and φ, the vector fields $T := \partial_t$ and $\Phi := \partial_\varphi$ are Killing vector fields. The vector field Φ can be interpreted as the angular momentum vector field in the

[43] The notion of mass and angular momentum of a spacetime can be defined precisely in a very general setting at the level of asymptotically flat initial data; see §7.1.

[44] Technically, we also have to complement (θ, φ) with another spherical coordinate chart to extend the metric to the full sphere $\mathbb{S}^2$. It is a nice exercise to convince yourself that this is indeed the case by considering the induced metric on $\{t = \text{const.}\} \cap \{r = \text{const.}\}$.

z-direction. The Kerr metric is therefore said to be *axisymmetric*. The coordinates (t, r, θ, φ) are called *Boyer–Lindquist coordinates*.

Consider T and note that

$$g_{M,a}(T,T) = (g_{M,a})_{tt} = -\frac{\Delta}{\rho^2} + \frac{a^2 \sin^2\theta}{\rho^2} = -\frac{r^2 - 2Mr + a^2 - a^2\sin^2\theta}{\rho^2} = -\frac{r^2 - 2Mr + a^2\cos^2\theta}{\rho^2}.$$

The above expression is non-negative when

$$r \leq M + \sqrt{M^2 - a^2\cos^2\theta}.$$

The region $\{r \leq M + \sqrt{M^2 - a^2\cos^2\theta}\}$ is called the *ergoregion*. Outside the ergoregion T is timelike. The Kerr spacetime is said to be *stationary* since it admits a Killing vector field that is timelike outside a bounded region in r. We will fix the time orientation by demanding T outside the ergoregion to be timelike.

[Exercise: 1) Show that the vector field $dt^\sharp$ is timelike everywhere and hence the level sets $\{t = \text{const.}\}$ are spacelike. 2) Show that Φ is everywhere spacelike . 3) Show that the Killing vector fields T and Φ evaluated at any $p \in \mathcal{M}_{\text{ext}}$ span a timelike hyperplane in $T_p\mathcal{M}_{\text{ext}}$ and explain why around every point $p \in \mathcal{M}_{\text{ext}}$ there exists a Killing vector field that is timelike in a neighbourhood of p.]

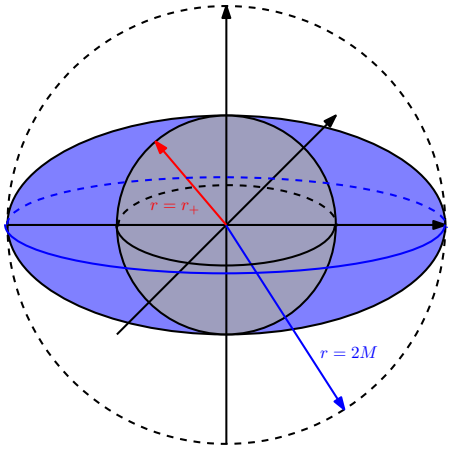


Figure 32: The ergoregion in a Kerr exterior spacetime restricted to a t-level set.

6.2 Kerr black hole interior

In this section, we will restrict to $|a| \leq M$.

Recall that in Schwarzschild, we could extend the spacetime across $\{r = r_+ = 2M\}$, into the black hole exterior. One way doing this, was to consider ingoing (or outgoing) Eddington–Finkelstein coordinates (v, r, θ, φ). We then obtained:

$$g_{M,0} = -\frac{\Delta}{r^2}dv^2 + 2dvdr + r^2 \mathring{\not{g}}.$$

Note that $Y = \partial_r$ is null so the integral curve to $-Y$ is a null vector field that points in the direction of decreasing r.[45]

[45] And the curve is in fact an affinely parametrized null geodesic with affine parameter $-r$, which reaches $r = r_+$ at finite affine time. [Exercise: Revisit the Schwarzschild case and convince yourself of this.]

[Exercise: Show that with respect to (t, r) coordinates on Schwarzschild, we can express:

$$Y = -\frac{r^2}{\Delta}\partial_t + \partial_r.]$$

We would like to consider similar coordinates to ingoing Eddington–Finkelstein coordinates (v, r) in Kerr with $a \neq 0$. The above choice of Y will no longer be null. However, we can consider the following modification (which agrees with the Schwarzschild choice if $a = 0$):

$$Y = -\frac{r^2 + a^2}{\Delta}\partial_t + \partial_r - \frac{a}{\Delta}\partial_\varphi.$$

Then, writing $g = g_{M,a}$:

$$\begin{aligned}
\rho^2\Delta^2 g(Y,Y) &= \rho^2[(r^2+a^2)^2 g_{tt} + 2a(r^2+a^2)g_{t\varphi} + a^2 g_{\varphi\varphi}] + \rho^2\Delta^2 g_{rr} \\
&= (r^2+a^2)^2(-\Delta+\cancel{a^2\sin^2\theta})+2a(r^2+a^2)(a\sin^2\theta\Delta\cancel{-a\sin^2\theta(r^2+a^2)})+a^2(-\Delta a^2\sin^4\theta+\cancel{(r^2+a^2)^2\sin^2\theta}) \\
&\qquad + (r^2+a^2\cos^2\theta)^2\Delta \\
&= -\Delta[(r^2+a^2)^2 - 2a^2\sin^2\theta(r^2+a^2) + a^4\sin^4\theta - (r^2+a^2\cos^2\theta)^2] \\
&= -\Delta[(r^2+a^2)(r^2+a^2(1-\sin^2\theta)) - a^2\sin^2\theta(r^2+a^2(1-\sin^2\theta)) - (r^2+a^2\cos^2\theta)^2] \\
&= -\Delta\rho^2[r^2+a^2-a^2\sin^2\theta-(r^2+a^2\cos^2\theta)] = 0.
\end{aligned}$$

Since Y has a φ-component, we need to change (t, r, φ) to (v, r, φ_*) for an appropriate choice of v and φ_*. To ensure $Y = \partial_r$ in these new coordinates, we moreover need $Y(v) = Y(\varphi_*) = 0$.

Define the tortoise coordinate r_* as follows: $r_* : (r_+, \infty) \to \mathbb{R}$, with $\frac{dr_*}{dr} = \frac{r^2+a^2}{\Delta}$ and define $v = t + r_*$.[46] Then clearly $Y(v) = 0$.

Similarly, let

$$\varphi_* = \varphi - \int_r^\infty \frac{a}{\Delta(r')}\,dr' \mod 2\pi.$$

Then also $Y(\varphi_*) = 0$. We refer to the coordinates $(v, r, \theta, \varphi_*)$ as *Kerr-star coordinates* or *ingoing Kerr coordinates*. With respect to these coordinates, the Kerr metric takes the following form (Exercise):

$$\begin{aligned}
g_{M,a} = -\rho^{-2}\left(\Delta - a^2\sin^2\theta\right)dv^2 + 2dvdr - 4Mar\rho^{-2}\sin^2\theta dvd\varphi_* - 2a\sin^2\theta drd\varphi_* + \rho^2 d\theta^2 \\
+ \rho^{-2}((r^2+a^2)^2 - a^2\Delta\sin^2\theta)\sin^2\theta d\varphi_*^2.
\end{aligned}$$

[Exercise: Show that the inverse metric is given by:

$$\begin{aligned}
g_{M,a}^{-1} &= a^2\rho^{-2}\sin^2\theta\partial_v\otimes\partial_v + \rho^{-2}(r^2+a^2)[\partial_v\otimes\partial_r + \partial_r\otimes\partial_v] + \Delta\rho^{-2}\partial_r\otimes\partial_r \\
&\quad + a\rho^{-2}[(\partial_v+\partial_r)\otimes\partial_{\varphi_*} + \partial_{\varphi_*}\otimes(\partial_v+\partial_r)] + \rho^{-2}[\partial_\theta\otimes\partial_\theta + (\sin^2\theta)^{-1}\partial_{\varphi_*}\otimes\partial_{\varphi_*}].
\end{aligned}$$

Hint: Put the matrix corresponding to $g_{M,a}$ in (t, r, θ, φ) coordinates in block-diagonal form by grouping (r, θ) and (t, φ) together.]

[46] It can be shown that r_* takes the following form:

$$\begin{aligned}
r_* &= r - \frac{M^2}{\sqrt{M^2-a^2}}\log\left(\frac{r-r_-}{r-r_+}\right) + M\log\Delta + c_0 \quad |a| < M, \\
r_* &= r - \frac{2M^2}{r-M} + M\log\Delta + c_0 \quad |a| = M
\end{aligned}$$

with $c_0 \in \mathbb{R}$ a constant.

From the above expression, it follows in particular that the level sets $\{v = \text{const.}\}$ are not null hypersurfaces when $a \neq 0$.

Since the metric is not singular when $0 < r \leq r_+$, we can extend the manifold $\mathcal{M}_{\text{ext}} \cong \mathbb{R}_v \times (r_+, \infty)_r \times \mathbb{S}^2_{\theta,\varphi_*}$ to obtain $\mathcal{M} = \mathbb{R}_v \times (0, \infty)_r \times \mathbb{S}^2_{\theta,\varphi_*}$ and extend $g_{M,a}$ analytically to $\mathcal{M}$.

We denote

$$\mathcal{H}^+_R := \{r = r_+\},$$
$$\mathcal{CH}^+_L := \{r = r_-\}.$$

These level sets are null hypersurfaces. We will refer to $\mathcal{H}^+_R$ as the *right future event horizon* and $\mathcal{CH}^+_L$ as the *left future inner horizon*.

In the region $\{r_- < r < r_+\}$, we moreover can revert "back" to Boyer–Lindquist-like coordinates $(t', r, \theta, \varphi')$, by defining $t' = v - r_*$ and $\varphi' = \varphi_* + \int_r^{r_0} \frac{a}{\Delta(r')}\, dr' \mod 2\pi$, for some $r_- < r_0 < r_+$.

Since the Kerr metric is invariant under the reflection map $(t', \varphi') \mapsto (-t', -\varphi')$, we can carry out similar procedure in the region $\{r_- < r < r_+\}$ with the following null vector field:

$$\widetilde{Y} = \frac{r^2 + a^2}{\Delta}\partial_{t'} + \partial_r + \frac{a}{\Delta}\partial_{\varphi'},$$

which motivates the coordinate changes $u = t' - r_*$ and $\tilde{\varphi}_* = \varphi' + \int_r^{r_0} \frac{a}{\Delta(r')}\, dr' \mod 2\pi$. This allows us to extend the manifold $\{r_- < r < r_+\}$ to obtain $\widetilde{\mathcal{M}} = \mathbb{R}_u \times (0, \infty)_r \times \mathbb{S}^2_{\theta,\tilde{\varphi}_*}$. We define $\widetilde{\mathcal{M}}_{\text{ext}} := \widetilde{\mathcal{M}} \cap \{r > r_+\}$.

In analogy with before, the following level sets are null hypersurfaces:

$$\mathcal{H}^+_L = \{r = r_+\},$$
$$\mathcal{CH}^+_R = \{r = r_-\},$$

which we will call the *left future event horizon* and *right inner horizon*, respectively. When $|a| = M$, $r_+ = r_-$, so $\mathcal{H}^+_L$ and $\mathcal{CH}^+_R$ coincide.

Note that we can also introduce $(u, r, \theta, \tilde{\varphi}_*)$ coordinates in $\mathcal{M}_{\text{ext}}$ to define the left/right past event horizons $\mathcal{H}^-_{L,R}$ and the right/left past inner horizons $\mathcal{CH}^-_{R,L}$.

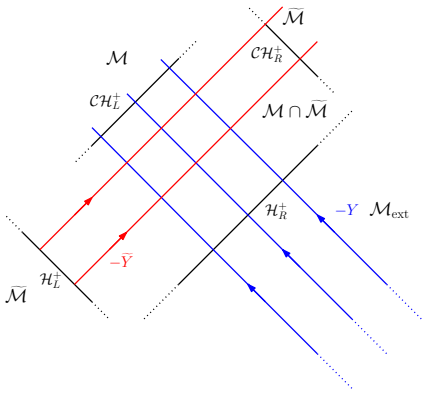


Figure 33: The integral curves of $-Y$ and $-\widetilde{Y}$ and the extensions of $\mathcal{M}_{\text{ext}}$.

We denote

$$\mathcal{BH} = \mathcal{M} \setminus \mathcal{M}_{\text{ext}}.$$

and refer to it as *the black hole region*. Since we do not have a Penrose diagram and we have not defined $\mathcal{I}^+$, it is not immediate that that $\mathcal{BH} \cap J^-(\mathcal{M}_{\text{ext}}) = \emptyset$.

Our choice of time-orientation implies that $-Y$ is future directed. Indeed, this guarantees that $-g(T, Y) = -2 < 0$, which is consistent with T being a future-directed, timelike vector field outside the ergoregion, as we imposed.

Proposition 6.1. *$\mathcal{H}^+_R$ is a null hypersurface, with the null tangent vector field $K = \partial_v + \omega_+ \partial_{\varphi_*}$, where $\omega_+ = \frac{a}{r_+^2 + a^2}$. The vector field K is called the* Hawking vector field *and the constant ω_+ is called the angular velocity of the black hole.*

Proof. First since K does not have an r-component, it is tangent to $\mathcal{H}^+_R$. Note furthermore that

$$g_{M,a}(K, K)|_{\mathcal{H}^+_R} = g_{vv} + 2\omega_+ g_{v\varphi_*} + \omega_+^2 g_{\varphi_*\varphi_*}\Big|_{\mathcal{H}^+_R} = \rho^{-2} \sin^2\theta \left[a^2 - 4Mar_+\omega_+ + (r_+^2 + a^2)^2\omega_+^2\right].$$

So, using that $2Mr_+ = r_+^2 + a^2$, we have that $g(K, K) = 0$ if and only if

$$0 = \omega_+^2 - 2a(r_+^2 + a^2)^{-1}\omega_+ + a^2(r_+^2 + a^2)^{-2} = \left(\omega_+ - a(r_+^2 + a^2)^{-1}\right)^2.$$

Hence $g_{M,a}(K, K) = 0$ if we take $\omega_+ = a(r_+^2 + a^2)^{-1}$.

To conclude that $\mathcal{H}^+_R$ is a null hypersurface, we need to show that $g(K, X)$ for every vector field X tangent to $\mathcal{H}^+_R$. We immediately have that $g_{M,a}(K, \partial_\theta) = 0$. Furthermore,

$$g_{M,a}(K, \partial_{\varphi_*})|_{\mathcal{H}^+_R} = g_{v\varphi_*} + a(r_+^2 + a^2)^{-1} g_{\varphi_*\varphi_*}|_{\mathcal{H}^+_R} = \rho^{-2}\sin^2\theta[-2Mar_+ + a(r_+^2 + a^2)]|_{\mathcal{H}^+_R} = 0.$$

□

[Exercise: Show that $\mathcal{CH}^+_L$ is a null hypersurface and find a null vector field that is tangent to $\mathcal{CH}^+_L$.]

[Exercise: Show that $\varphi_+ := \varphi - \omega_+ t \mod 2\pi$ is constant along the integral curves of K and is well-defined on $\mathcal{H}^+_R$.]

[Exercise: Show that $\nabla_K K|_{\mathcal{H}^+_R} = \kappa_+ K|_{\mathcal{H}^+_R}$, with $\kappa_+ = \frac{r_+ - r_-}{4Mr_+}$]

Proposition 6.2. *$\mathcal{BH} \cap J^-(\mathcal{M}_{\text{ext}}) = \emptyset$.*

Proof. Suppose that $(\mathcal{BH} \setminus \mathcal{H}^+_R) \cap J^-(\mathcal{M}_{\text{ext}}) \neq \emptyset$. Then there exist a future-directed causal curve $\gamma : [0, 2] \to \mathcal{M}$, such that $\gamma(0) \in \mathcal{BH} \setminus \mathcal{H}^+_R$, $\gamma(2) \in \mathcal{M}_{\text{ext}}$ and $p = \gamma(s_*) \in \mathcal{H}^+_R$ for some $s_* \in (0, 2)$ and $\dot{r}_\gamma(s_*) > 0$ (γ is entering $\mathcal{M}_{\text{ext}}$ at time s_*).

Denote $w = \dot{\gamma}(s_*)$. It will be useful to expand:

$$w = w^Y Y_p + w^K K_p + w^\theta \partial_\theta|_p + w^{\varphi_*}\partial_{\varphi_*}|_p.$$

Since γ is future-directed, $0 \leq g_p(Y_p, w)$. By causality of γ we moreover have that:

$$\begin{aligned} 0 \geq g_p(w, w) &= g_p(w^Y Y_p + (w - w^Y Y_p), w^Y Y_p + (w - w^Y Y_p)) \\ &= \cancel{(w^Y)^2 g_p(Y_p, Y_p)} + 2w^Y g_p(Y_p, w) - 2\cancel{(w^Y)^2 g_p(Y_p, Y_p)} + g_p(w - w^Y Y_p, w - w^Y Y_p) \\ &= 2w^Y g_p(Y_p, w) + \cancel{(w^K)^2 g_p(K_p, K_p)} + (w^\theta)^2 g_{\theta\theta}(p) + (w^{\varphi_*})^2 g_{\varphi_*\varphi_*}(p) \geq 2\underbrace{g_p(Y_p, w)}_{\geq 0} w^Y. \end{aligned}$$

Hence, we need $w^Y \leq 0$, which implies that $\dot{r}_\gamma(s_*) \leq 0$. However, since $\dot{r}_\gamma(s_*) > 0$, this is a contradiction.

We conclude that $(\mathcal{BH} \setminus \mathcal{H}^+_R) \cap J^-(\mathcal{M}_{\text{ext}}) = \emptyset$.

It remains to show that $\mathcal{H}^+_R \cap J^-(\mathcal{M}_{\text{ext}}) = \emptyset$. Suppose that γ is a causal curve, such that $\gamma(0), \gamma(2) \in \mathcal{M}_{\text{ext}}$, but there exists an $s_* \in (0,2)$ such that $p = \gamma(s_*) \in \mathcal{H}^+_R$ and $\gamma(s) \in \mathcal{M}_{\text{ext}}$ for $s > s_*$ suitably close to s_* and $w^Y = 0$. If we proceed as above, but we assume that γ is timelike or that $w^\theta \neq 0$ or $w^{\varphi_*} \neq 0$, then we also get a contradiction. It remains to check the case where $w = w^K K_p$. Since γ is causal and future-directed, $w^K > 0$.

We will represent $\gamma(s)$ in the coordinate chart $(v, r, \theta, \varphi_+)$. Note that $\dot{v}_\gamma(s_*) = w^K$. Since γ touches $\mathcal{H}^+_R$, we must have that $r_\gamma(s_*) = r_+$, $\dot{r}_\gamma(s_*) = 0$ and $\ddot{r}_\gamma(s_*) \geq 0$. Let $s \geq s_*$, then $\ddot{r}(s) \geq 0$ for $s_* \leq s \leq s_* + \epsilon$, if ϵ is suitably small. This implies that:

$$0 \leq r_\gamma(s) - r_+ = \int_{s_*}^{s} \dot{r}_\gamma(s')\, ds' \leq \dot{r}_\gamma(s)(s - s_*).$$

Since K is orthogonal to Φ, K, ∂_θ along $\mathcal{H}^+_R$, we can express:

$$\begin{aligned} g_{\gamma(s)}(\dot{\gamma}(s), \dot{\gamma}(s)) &= [2g(K,Y)(\gamma(s_*))\dot{v}(s) + 2g(\Phi, Y)\dot{\varphi}_+(s))\dot{r}(s)] + \dot{v}^2(s) g(K,K) + 2\dot{v}(s)\dot{\varphi}_+(s) g(K,\Phi) \\ &\quad + \dot{\theta}^2 g(\partial_\theta, \partial_\theta) + \dot{\varphi}_+^2 g(\Phi, \Phi) \\ &\geq [2g(K,Y)(\gamma(s_*))\dot{v}_\gamma(s) + 2g(\Phi, Y)\dot{\varphi}_+(s))\dot{r}_\gamma(s)] + O(r_\gamma(s) - r_+) \\ &= 2\left(1 - \frac{a^2}{r_+^2 + a^2}\sin^2\theta\right)(\gamma(s_*)) w^K \ddot{r}_\gamma(s)(1 + O((s - s_*)). \end{aligned}$$

Since $\left(1 - \frac{a^2}{r_+^2+a^2}\sin^2\theta\right)(\gamma(s_*)) \cdot w^K \cdot \ddot{r}_\gamma(s) > 0$ for $s > s_*$ suitably close to s_*, we therefore see that γ must be spacelike for $s > s_*$ suitably close to s_*, so it cannot be a causal curve, which is a contradiction. Hence, $\mathcal{H}^+_R \cap J^-(\mathcal{M}_{\text{ext}}) = \emptyset$. □

An observer in the black hole region $\mathcal{BH}$ can therefore not travel to $\mathcal{M}_{\text{ext}}$, or send signals to $\mathcal{M}_{\text{ext}}$.

Remark 6.1. *Let $R > r_+$ be arbitrarily large. One can also show that $\mathcal{M}_{\text{ext}} \subseteq J^-(\{r \geq R\})$, so we can also write: $\mathcal{BH} = \mathcal{M} \setminus J^-(\{r \geq R\})$. (*Exercise*)*

[Exercise: Show that for $|a| < M$, there exists an $\epsilon > 0$ such that K is timelike in the region $\{r_+ < r \leq (1+\epsilon)r_+\}$. Is this also true for $|a| = M$? Show that integral curves of K in $\{r \leq (1+\epsilon)r_+\}$ rotate at an angular velocity ω_+ with respect to observers represented by integral curves of T outside the ergoregion. Since K is tangent to $\mathcal{H}^+_R$, the boundary of the black hole region, we say that the black hole itself rotates at an angular velocity ω_+.]

6.3 Kruskal-like coordinates and global hyperbolicity

We will show that null hypersurfaces $\mathcal{H}^+_R$ and $\mathcal{H}^+_L$ actually intersect. This will result in a global understanding of the Kerr geometry in terms of Kruskal-like coordinates (U, V). In contrast with Schwarzschild and Reissner–Nordström, U- and V-level sets are not going to be null hypersurfaces.

Recall, $v = t + r_*$ and $u = t - r_*$. We restrict to sub-extremal Kerr ($|a| < M$) and we introduce Kruskal-like rescaled coordinates:

$$\begin{aligned} U &:= -e^{-\kappa_+ u}, \\ V &:= e^{\kappa_+ v}, \end{aligned}$$

with κ_+ the surface gravity of $\mathcal{H}^+_R$, in analogy with the Kruskal-like coordinates on Reissner–Nordström.

We recall moreover the modified azimuthal angular coordinate φ_+:

$$\varphi_+ = \varphi - \omega_+ t \mod 2\pi = \varphi - \frac{a}{r_+^2 + a^2} t \mod 2\pi,$$

which is conserved along the integral curves of the Hawking vector field K.

Proposition 6.3. *Let $|a| < M$. The coordinates $(U, V, \theta, \varphi_+)$ cover*[47] *$(\mathcal{M} \cap \{r > r_-\}) \cup (\widetilde{\mathcal{M}} \cap \{r > r_-\})$ and the components of $g_{M,a}$ are analytic in U, V, θ, φ_+. Furthermore:*

$$\begin{aligned} \mathcal{M}_{\text{ext}} &\cong \{U < 0, V > 0\}, \\ \mathcal{M} \cap \{r > r_-\} &\cong \{U \in \mathbb{R}, V > 0\}, \\ \widetilde{\mathcal{M}}_{\text{ext}} &\cong \{U > 0, V < 0\}, \\ \widetilde{\mathcal{M}} \cap \{r > r_-\} &\cong \{U > 0, V \in \mathbb{R}\}, \end{aligned}$$

We can extend $(\mathcal{M} \cap \{r > r_-\}) \cup (\widetilde{\mathcal{M}} \cap \{r > r_-\}, g_{M,a})$ further to include $\{U \leq 0, V \leq 0\}$, with $\{U = 0,\ V = 0\}$ the bifurcation sphere, and we denote the extended manifold by $\mathcal{M}_{\text{Krus}} = \{(U, V, \theta, \varphi_+) \mid (U, V) \in \mathbb{R}^2,\ (\theta, \varphi) \in \mathbb{S}^2\}$.

Proof. (*Sketch*) The proof relies on two main observations.

1. With respect to $(u, v, \theta, \varphi_+)$:

$$\begin{aligned} (g_{M,a})_{uu} &= (g_{M,a})_{vv} = a^2\Delta^2 h_1(r, \sin^2\theta), \\ (g_{M,a})_{v\varphi_+} &= (g_{M,a})_{u\varphi_+} = a^2\Delta h_2(r, \sin^2\theta), \\ (g_{M,a})_{uv} &= \Delta h_3(r, \sin^2\theta), \end{aligned}$$

   with h_1, h_2, h_3 analytic functions that are well-defined for $r \in (r_-, \infty)$ and h_3 is non-vanishing.

2. We can express:

$$-UV = e^{2\kappa_+(v-u)} = e^{\kappa_+ r_*} = \Delta k(r),$$

   with k a non-vanishing, analytic function on (r_-, ∞).

Then we can analytically extend the function $r(U, V)$ and the components of $g_{M,a}$ from the region $\mathcal{M}_{\text{ext}} = \{U < 0, V > 0\}$ to the extensions of $\mathcal{M}_{\text{ext}}$. Since analytic extensions are unique, $g_{M,a}$ in $\mathcal{M}_{\text{Krus}} \setminus \{U = 0,\ V = 0\}$ must agree with the analytically extended g_M obtained by switching to Kerr-star coordinates.

To extend across $\mathcal{H}^{\pm}_{L,R}$ after changing coordinates, the factor Δ in g_{uv}, $g_{u\varphi_+}$ and $g_{v\varphi_+}$ is important. Similarly, the factor Δ^2 in g_{uu} and g_{vv} is important.

To determine the range of U and V, we observe that we can express in the region $r_- < r < r_+$:

$$U = -e^{-\kappa_+(v - 2r_*)}$$

and $r_* \in \mathbb{R}$, so as we approach $\mathcal{CH}^+_L$, $U \to \infty$. Similarly, as we approach $\mathcal{CH}^+_R$, $V \to \infty$. We similarly have $U, V \to -\infty$ as we approach the time inverses $\mathcal{CH}^-_R$ and $\mathcal{CH}^-_L$, respectively. □

Proposition 6.4. *The hypersurface $\Sigma = \{U + V = 0\}$ is spacelike is a spacelike Cauchy hypersurface for $(\mathcal{M}_{\text{Krus}}, g_{M,a})$.*

Proof. Exercise. Hint: To prove the Cauchy surface property, you could consider a past-directed and past-inextendible causal curve $\gamma(s)$, with s the proper time, and investigate the sign of $\frac{d}{ds}(U + V)(\gamma(s))$. □

[47] At least away from the usual breakdown of spherical coordinates at a meridian connecting the north and south poles of the sphere $\mathbb{S}^2$.

6.4 Dynamics of null geodesics

We will investigate the behaviour of null geodesics on Kerr black hole exteriors $(\mathcal{M}_{\text{ext}}, g_{M,a})$. Let γ be an affinely-parametrized null geodesic. We proceed as in Schwarzschild by writing out the equation $(g_{M,a})_{\gamma(s)}(\dot\gamma(s),\dot\gamma(s)) = 0$. In slight abuse of notation, we will write $g = g_{M,a})_{\gamma(s)}$ and $\dot\gamma = \dot\gamma(s)$:

$$0 = g(\dot\gamma,\dot\gamma) = -\Delta\rho^{-2}(\dot t - a\sin^2\theta\dot\varphi)^2 + \sin^2\theta\rho^{-2}(a\dot t - (r^2+a^2)\dot\varphi)^2 + \frac{\rho^2}{\Delta}\dot r^2 + \rho^2\dot\theta^2. \tag{6.1}$$

Using the Killing property of $T = \partial_t$ and $\Phi = \partial_\varphi$, we identify the following conserved quantities along γ:

$$\begin{aligned} E =& -g(\dot\gamma, T) = -g_{tt}\dot t - g_{t\varphi}\dot\varphi = \rho^{-2}\left[\Delta - a^2\sin^2\theta\right]\dot t + a\sin^2\theta\rho^{-2}\left[r^2+a^2-\Delta\right]\dot\varphi,\\ L =& g(\dot\gamma,\Phi) = g_{\varphi\varphi}\dot\varphi + g_{t\varphi}\dot t = \sin^2\theta\rho^{-2}\left[(r^2+a^2)^2 - a^2\sin^2\theta\Delta\right]\dot\varphi - a\sin^2\theta\rho^{-2}\left[r^2+a^2-\Delta\right]\dot t. \end{aligned}$$

We interpret E as the energy of γ according to integral curves of T and L as the z-component of the angular momentum of γ. Note that E can in principle be non-positive if γ is restricted to the ergoregion (and integral curves of T are not timelike).

We can write these equations as the following single matrix equation:

$$\begin{pmatrix} E \\ L \end{pmatrix} = A \begin{pmatrix} \rho^{-2}\dot t \\ \rho^{-2}\sin^2\theta\dot\varphi, \end{pmatrix}$$

with A a 2×2 matrix that is defined as follows:

$$A = \begin{pmatrix} \Delta - a^2\sin^2\theta & a(r^2+a^2-\Delta) \\ -a\sin^2\theta(r^2+a^2-\Delta) & (r^2+a^2)^2 - a^2\sin^2\theta\Delta \end{pmatrix} = \begin{pmatrix} \rho^2 - 2Mr & 2Mra \\ -2Mar\sin^2\theta & (r^2+a^2)\rho^2 + 2Ma^2r\sin^2\theta \end{pmatrix}.$$

Then

$$\det A = \rho^4(r^2+a^2) + 2Mra^2\sin^2\theta\rho^2 - 2Mr(r^2+a^2)\rho^2 = \rho^4(r^2+a^2-2Mr) = \rho^4\Delta.$$

Therefore

$$A^{-1} = \begin{pmatrix} \rho^{-4}((r^2+a^2)^2\Delta^{-1} - a^2\sin^2\theta) & -\rho^{-4}a((r^2+a^2)\Delta^{-1} - 1) \\ \rho^{-4}a\sin^2\theta((r^2+a^2)\Delta^{-1} - 1) & \rho^{-4}(1 - a^2\sin^2\theta\Delta^{-1}) \end{pmatrix}$$

and

$$\begin{aligned} \rho^2\dot t &= a(L - Ea\sin^2\theta) + \Delta^{-1}(r^2+a^2)(E(r^2+a^2) - aL),\\ \rho^2\sin^2\theta\dot\varphi &= -Ea\sin^2\theta + L + \Delta^{-1}a\sin^2\theta(E(r^2+a^2) - aL). \end{aligned}$$

In contrast with the Schwarzschild case, we cannot simply assume without loss of generality that $\theta \equiv \frac{\pi}{2}$. Instead, we will make use of the existence of an additional, hidden, conserved quantity:

Lemma 6.5. *Let γ be an affinely parametrized causal geodesic, with $g_{\gamma(s)}(\dot\gamma(s),\dot\gamma(s)) = -\sigma$ and define*

$$Q := (\rho^4\circ\gamma)\dot\theta^2 + \frac{L^2}{\sin^2\theta\circ\gamma} - a^2(E^2-\sigma)\cos^2\theta\circ\gamma.$$

Then

$$\frac{d}{ds}Q(s) = 0.$$

Furthermore,

$$Q + a^2E^2 \geq L^2.$$

The quantity Q is called Carter's constant.

The existence of the conserved quantity Q is related to the existence of a Killing tensor $\mathcal{K} \in \mathcal{T}^{(0,2)}(\mathcal{M})$, which is a symmetric tensor field that satisfies the following generalization of the equation satisfied by a Killing vector field: $\nabla_{(\mu}\mathcal{K}_{\nu\rho)} = 0$.

[Exercise: Show that for γ an affinely parametrized geodesic, $\frac{d}{ds}(\mathcal{K}_{\nu\rho}(\gamma(s))\dot{\gamma}^{\nu}(s)\dot{\gamma}^{\rho}(s)) = 0$.]

Using the above lemma with $\sigma = 0$, we obtain the following additional equation:

$$\rho^4\dot{\theta}^2 = Q - \frac{L^2}{\sin^2\theta} + a^2E^2\cos^2\theta = Q + a^2E^2 - 2aEL - \frac{(L - aE\sin^2\theta)^2}{\sin^2\theta}$$

We plug the equations for $\dot{t}$, $\dot{\varphi}$ and $\rho^4\dot{\theta}^2$ in terms of E, L, Q, into (6.1) to obtain:

$$\begin{aligned}
0 = \rho^4\dot{r}^2 - \Delta^2(\dot{t} - a\sin^2\theta\dot{\varphi})^2 + \sin^2\theta\Delta(a\dot{t} - (r^2+a^2)\dot{\varphi})^2 + \rho^4\Delta\dot{\theta}^2 \\
= \rho^4\dot{r}^2 - \rho^{-4}((E(r^2+a^2) - aL)\rho^2)^2 \\
+ \sin^2\theta\Delta\rho^{-4}(a^2(L - Ea\sin^2\theta) - Ea(r^2+a^2) + L(r^2+a^2)(\sin^2\theta)^{-1})^2 \\
+ \Delta\left(Q + a^2E^2 - 2aEL - \frac{(L - aE\sin^2\theta)^2}{\sin^2\theta}\right) \\
= \rho^4\dot{r}^2 - (E(r^2+a^2) - aL)^2 + \Delta\rho^{-4}\sin^2\theta((L - Ea\sin^2\theta)(r^2+a^2)\rho^2(\sin^2\theta)^{-1})^2 \\
+ \Delta\left(Q + a^2E^2 - 2aEL - \frac{(L - aE\sin^2\theta)^2}{\sin^2\theta}\right) \\
= \rho^4\dot{r}^2 - (E(r^2+a^2) - aL)^2 + \Delta\left(Q + a^2E^2 - 2aEL\right) \\
= \rho^4\dot{r}^2 - E^2(r^2+a^2)^2 + 4MaELr - a^2L^2 + \Delta(Q + a^2E^2).
\end{aligned}$$

Hence,

$$\begin{aligned}
\frac{\rho^4}{(r^2+a^2)^2}\dot{r}^2 &= E^2 - V_0(r; E, L, Q), \\
V_0(r; E, L, Q) &:= \frac{1}{(r^2+a^2)^2}\left[4MaELr - a^2L^2 + \Delta(Q + a^2E^2)\right]
\end{aligned}$$

Remark 6.2. *More generally, had we considered causal geodesics satisfying $g(\dot{\gamma}, \dot{\gamma}) = -\sigma$, we would replace V_0 with:*

$$V_\sigma(r; E, L, Q) := \frac{1}{(r^2+a^2)^2}\left[4MaELr - a^2L^2 + \Delta(Q + a^2E^2) + \sigma r^2\Delta\right].$$

Exercise*: Show this.*

Now we can analyze the dynamics of null geodesics by determining the existence and location of maxima/minima of the potential V_0, just like in Schwarzschild. Recall that in Schwarzschild, for $\theta \equiv \frac{\pi}{2}$, we obtain $Q = L^2$ and we can write $V_0(r; L) = r^{-4}L^2\Delta$.

Proposition 6.6. *The potential function $V_0 : [r_+, \infty) \to \mathbb{R}$ either:*

- *is non-increasing,*
- *has a unique critical point $r_+ \leq r_{\max}$ that corresponds to a global maximum, or*
- *has exactly two critical points $r_+ \leq r_{\min} < r_{\max}$ that correspond to a local minimum at $r_{\min}$ and a local maximum at $r_{\max}$.*

Proof. To determine the critical points of V_0, we compute:

$$\begin{aligned}\frac{d}{dr}V_0(r) &= \frac{1}{(r^2+a^2)^3}\left[-4r(4MarEL - a^2L^2 + \Delta(Q+a^2E^2)) + (r^2+a^2)(4MaEL + 2(r-M)(Q+a^2E^2))\right] \\ &= \frac{1}{(r^2+a^2)^3}\left[4MaEL(-3r^2+a^2) + 4ra^2L^2 - 2(Q+a^2E^2)(r^3 - 3Mr^2 + a^2r + Ma^2)\right].\end{aligned}$$

We will investigate first the critical points of $(r^2+a^2)^3\frac{dV_0}{dr}(r)$. We have that:

$$\begin{aligned}\frac{d}{dr}((r^2+a^2)^3\frac{dV_0}{dr})(r) &= -24MaELr + 4a^2L^2 - 2(Q+a^2E^2)(3r^2 - 6Mr + a^2) \\ &= -6(Q+a^2E^2)\left[r^2 - 2M(1-2\xi)r + \frac{a^2}{3}(1-2\mu)\right],\end{aligned}$$

where we introduced the variables $\xi = aEL(Q+a^2E^2)^{-1}$ and $\mu = L^2(Q+a^2E^2)^{-1}$. The critical points of $(r^2+a^2)^3\frac{dV_0}{dr}(r)$ correspond to the roots of the above quadratic polynomial:

$$r_{1,2} = M(1-2\xi) \mp \sqrt{M^2(1-2\xi)^2 - \frac{a^2}{3}(1-2\mu)}$$

We now distinguish two cases:

Case I: $\xi \geq 0$ Since $r_1 \leq M(1-2\xi) \leq M \leq r_+$ (with equality iff $|a| = M$), we have that the only possible critical point in (r_+, ∞) is r_2.

Case II: $\xi < 0$ In this case, we can express:

$$r_1 = M(1-2\xi)\left[1 - \sqrt{1 - \frac{a^2}{3M^2}\frac{1-2\mu}{(1-2\xi)^2}}\right]$$

Note that

$$\frac{a^2}{3M^2}\frac{1-2\mu}{(1-2\xi)^2} < \frac{a^2}{3M^2} \leq \frac{1}{3},$$

where we used that $\xi < 0$, $\mu \geq 0$ and $|a| \leq M$.

Let $0 \leq x \leq \frac{1}{3}$. Then we use that

$$1 - x \geq 1 - x - \frac{x}{3}\left(1 - \frac{4}{3}x\right) = \left(1 - \frac{2}{3}x\right)^2,$$

to obtain $\sqrt{1-x} \geq 1 - \frac{2}{3}x$ and therefore

$$r_1 \leq M(1-2\xi)\frac{2}{3}\frac{a^2}{3M^2}\frac{1-2\mu}{(1-2\xi)^2} = \frac{2a^2}{9M}\frac{1-2\mu}{1-2\xi} \leq \frac{2a^2}{9M} \leq \frac{2}{9}r_+.$$

So in Case II, we also conclude that $(r^2+a^2)^3\frac{dV_0}{dr}(r)$ has at most one critical point, r_2, in (r_+, ∞).

Since $Q + a^2E^2 \geq 0$, we have that

$$\lim_{r\to\infty}(r^2+a^2)^3\frac{dV_0}{dr}(r) = -\infty.$$

This implies the following properties.

- $V_0(r)$ has at most two critical points. Indeed, suppose there are ≥ 3 critical points. Then $(r^2+a^2)^3\frac{dV_0}{dr}(r)$ must have at least 3 zeroes and hence, at least two critical points, which is in contradiction with the existence of at most one critical point, r_2.
- Suppose that $V_0(r)$ has no critical points. Then $(r^2+a^2)^3\frac{dV_0}{dr}(r)$ must have a constant sign. By $\lim_{r\to\infty}(r^2+a^2)^3\frac{dV_0}{dr}(r) = -\infty$, this sign must be negative.
- Suppose that $V_0(r)$ has exactly one critical point. By the limiting property of $(r^2+a^2)^3\frac{dV_0}{dr}(r)$, it must be a maximum.
- Suppose that $V_0(r)$ has exactly two critical points. By the limiting property of $(r^2+a^2)^3\frac{dV_0}{dr}(r)$, the critical point with the largest r-value must be a local maximum. The other critical point cannot be a maximum, since then there would have to be a third critical point in between, which would contradict the existence of exactly two critical points. Similarly, the other critical point cannot be a saddle point, because then it would be a critical point and a zero of $(r^2+a^2)^3\frac{dV_0}{dr}$. However, since $(r^2+a^2)^3\frac{dV_0}{dr}$ must also be zero at the maximum of V_0, it would have another critical point between the maximum and saddle points of V_0, which contradicts the existence of at most one critical point for $(r^2+a^2)^3\frac{dV_0}{dr}$. Hence, V_0 in this case has a local minimum and a local maximum. □

The possible existence of a minimum for $V_0(r)$ might suggest the existence of trapped null geodesics that are stable under small perturbations of their initial data in phase space. The proposition below demonstrates that this is <u>not</u> the case: trapped null geodesics are also unstable in Kerr.

The following conserved quantity will play an important role:

$$E_K := -(g_{M,a})_{\gamma(s)}(\dot\gamma(s), K_{\gamma(s)}) = -(g_{M,a})_{\gamma(s)}\left(\dot\gamma(s), T_{\gamma(s)} + \frac{a}{r_+^2+a^2}\Phi_{\gamma(s)}\right)$$
$$= E - \frac{a}{r_+^2+a^2}L = E - \frac{a}{2Mr_+}L = E - \omega_+ L.$$

At $\mathcal{H}^+$, where K is causal and future-directed, we must have that $E_K \geq 0$ and we can interpret E_K as the energy of γ as measured by observers rotating with the black hole.

Proposition 6.7. *We have that*

$$E^2 \geq V_0(r_{\min})$$

with equality if and only if $E_K = 0$ and $r_{\min} = r_+$, which can only occur when the Kerr black hole is extremal. In particular, if $E_K \neq 0$ or $r_{\min} \neq r_+$, then there are no trapped null geodesics at $r = r_{\min}$.

If $aE_KL < 0$, then $\frac{dV_0}{dr}(r_+) > 0$ and $V_0(r)$ has exactly one critical point: a maximum at $r_{\max}$.

Proof. We first compute, using that $r_+^2 + a^2 - 2Mr_+ = 0$:

$$E^2 - V_0(r_+) = E^2 - \frac{1}{(r_+^2+a^2)^2}(4MaELr_+ - a^2L^2)$$
$$= \frac{E^2(r_+^2+a^2)^2 - 4MaELr_+ + a^2L^2}{(r_+^2+a^2)^2} = \left(\frac{2MEr_+ - aL}{r_+^2+a^2}\right)^2 = E_K^2.$$

Hence, if $E_K \neq 0$, then

$$E^2 > V_0(r_+) \geq V_0(r_{\min}).$$

If $r_{\min} \neq r_+$, then $E^2 \geq V_0(r_+) > V_0(r_{\min})$.

Note moreover that

$$\begin{aligned}(r_+^2+a^2)^3\frac{dV_0}{dr}(r_+) &= -4r_+(r_+^2+a^2)^2V_0(r_+)+(r_+^2+a^2)(4MaEL+\frac{d\Delta}{dr}(Q+a^2E^2))\\ &= -4r_+(4Mar_+EL-a^2L^2)+8MaELr_++2(Q+a^2E^2)(r_+^2+a^2)(r_+-M)\\ &= -8Mr_+^2aE_KL+2(Q+a^2E^2)(r_+^2+a^2)(r_+-M).\end{aligned}$$

Hence, we can only have $r_+ = r_{\min}$ and $E^2 - V_0(r_+) = 0$ if $Q + a^2E^2 = 0$ or $r_+ = M$, the latter which is equivalent to $|a| = M$. Note however that since $Q^2 + a^2E^2 \geq L^2$, the first case implies that $L^2 = Q^2 + a^2E^2 = 0$, so $V_0 \equiv 0$ and there does not exist a $r_{\min}$

The above computation implies moreover that in the case $aE_KL < 0$:

$$\frac{dV_0}{dr}(r_+) > 2(Q+a^2E^2)(r_+^2+a^2)^{-2}(r_+-M) \geq 0.$$

We cannot have the existence of $r_{\min}$, since that would require $\frac{dV_0}{dr}(r_+) \leq 0$. □

Remark 6.3. *The above argument also excludes the existence of trapped null geodesics with non-constant (oscillating) r-values (r bounded below and above).* [Exercise*: Convince yourself of this.*]

[Exercise: Show that any future-directed affinely parametrized null geodesic γ in $\mathcal{M}_{\text{ext}}$ with $E < 0$ must intersect the future event horizon $\mathcal{H}_R^+$. Hint: Use that γ must remain in the ergoregion. Then suppose there exists a future-directed null geodesic with $E < 0$ approaching $r = r_{\max}$ asymptotically. Use that the geodesic can be perturbed by slightly changing (E, L, Q) to obtain $E < 0$ and $E^2 > V_0(r_{\max})$ or $E^2 > V_0(r_{\max})$ and use this to reach a contradiction.]

The presence of an ergoregion allows one to "extract energy" from a black hole. We will first explore this in a heuristic setting. Let γ be a future-directed causal geodesic with energy E and angular momentum L that enters a Kerr black hole with mass M_i and angular momentum (z-component) $J_i = a_iM_i$. We will assume that this dynamical process leads to a dynamical black hole that finally settles down to another Kerr black hole with mass M_f and angular momentum J_f, where $M_f = M_i + E$ and $J_f = J_i + L$. Since E can be negative (if the geodesic did not come from outside the ergoregion), the energy or mass of the black hole can decrease. Extracting this energy difference is known as the *Penrose process*.

We then have:

$$\begin{aligned}J_f - J_i = L = \frac{1}{\omega_+}(E-E_K) = \frac{1}{\omega_+}(M_f-M_i) - \frac{1}{\omega_+}E_K &= \frac{2M_ir_+}{a_i}(M_f-M_i)-\frac{1}{\omega_+}E_K\\ &= \frac{2M_i(M_i^2+\sqrt{M_i^4-J_i^2})}{J_i}(M_f-M_i)-\frac{1}{\omega_+}E_K,\end{aligned}$$

where we plugged in the expression for ω_+ and r_+ with respect to M_i and J_i, assuming that $E \ll M_i$ and $L \ll J_i$. Since γ intersects $\mathcal{H}^+$, we must have that $E_K = E - \omega_+L \geq 0$. We can use the above equality together with the inequality $E_K \geq 0$ to derive properties of the so-called *irreducible mass* M_{irr}, which is defined as follows:

$$M_{\text{irr}}(M,J) := \sqrt{\frac{1}{2}(M^2+\sqrt{M^4-J^2})}.$$

By applying Taylor's theorem, we can obtain

$$(M^2_{\text{irr}})_f - (M^2_{\text{irr}})_i = \frac{\partial M^2_{\text{irr}}}{\partial M}|_{M=M_i,J=J_i}(M_f - M_i) + \frac{\partial M^2_{\text{irr}}}{\partial J}|_{M=M_i,J=J_i}(J_f - J_i) + O(E^2) + O(L^2)$$
$$= \frac{J_i}{2\sqrt{M_i^4 - J_i^4}}\left[\frac{2M_i(M_i^2 + \sqrt{M_i^4 - J_i^2})}{J_i}(M_f - M_i) - (J_f - J_i)\right] + O(E^2) + O(M_i^{-2}L^2)$$
$$= \frac{J_i}{2\sqrt{M_i^4 - J_i^4}\omega_+}E_K + O(E^2) + O(M_i^{-2}L^2) = \frac{M_i^2 r_+}{2\sqrt{M_i^4 - J_i^4}}E_K + O(E^2) + O(M_i^{-2}L^2).$$

Since $EM_i^{-1} \ll 1$, $LM_i^{-2} \ll 1$ and $E_K \geq 0$, the right hand side above is positive if $M_i^{-1}E_K \gg M_i^{-2}E^2 + M_i^{-4}L^2$. In this case, M_{irr} is increasing as consequence of this dynamical process.

The name "irreducible mass" comes from the following inequality:

$$M^2_{\text{irr}} = \frac{1}{2}(M^2 + \sqrt{M^4 - J^2}) \leq M^2,$$

so the mass M must be bounded below by M_{irr}. The Penrose process cannot lead to a decrease in black hole mass below $M_{\text{irr}}(M_i, J_i)$.

The concept of irreducible mass was introduced by Christodoulou in his PhD thesis and it played an important role in the development of black hole thermodynamics, where M^2_{irr} can be interpreted as a non-decreasing *entropy of the Kerr black hole*.

[Exercise: Show that the area A of the spheres of intersection of $\mathcal{H}^+$ and $\{v = \text{const.}\}$ satisfies $A = 16\pi M^2_{\text{irr}}$.]

The phenomenon of *superradiance* is also related to the ability to extract energy from a black hole. Consider solutions ϕ to the wave equation on a (sub)-extremal Kerr background $\Box_{g_{M,a}}\phi = 0$, such that that ϕ vanishes at the past event horizon $\mathcal{H}^-_L$ and has energy equal to 1 at $\mathcal{I}^-_R$. The energy at $\mathcal{I}^+_R$ (that does not escape to the black hole) can then be bigger than 1: there can be an energy amplification caused by the presence of a black hole. If we study this process in frequency space by formally taking a Fourier transform of the solution to the wave equation and decomposing $\Phi = \sum_{m\in\mathbb{Z}}\phi_m$, where $\Phi(\phi_m) = im\phi_m$, we will see that only the following frequency range can lead to an energy increase:

$$0 < am\omega < a\omega_+ m^2.$$

These are called *superradiant frequencies*.

The phenomenon of superradiance is intimately connected to the behaviour of null geodesics. At the level of null geodesics, the superradiance frequency regime corresponds to the condition:

$$0 < aL \cdot E < \frac{a^2L^2}{2Mr_+} = a\omega_+ L^2,$$

or equivalently, $-a\omega_+ L^2 < aLE_K < 0$.

We will refer to null geodesics satisfying this condition as "superradiant null geodesics". For null geodesics with $E > 0$, this amounts to the condition that the geodesic is rotating in the same direction as the black hole and $E_K < 0$. This implies in particular that the geodesic cannot cross $\mathcal{H}^+_R$, where necessarily $E_K \geq 0$, since K and $\dot\gamma$ are causal and future-directed at $\mathcal{H}^+_R$.

The following proposition shows that superadiant null geodesics cannot be (unstably) trapped in sub-extremal black hole exteriors. At the level of the wave equation, this is crucial for establishing decay of waves on Kerr black hole backgrounds and, ultimately, the conjectured (as of the writing of these notes) stability of all sub-extremal Kerr black holes under suitably small initial data perturbations.

Proposition 6.8. *Let γ be a superradiant null geodesic on a sub-extremal Kerr black hole. Then γ cannot be trapped, i.e. $V_0(r_{\max}) - E^2 > 0$.*

Proof. We may assume without loss of generality that $a > 0$, so $a < M$. We have that $aE_K L < 0$ so by the previous proposition, V_0 has exactly one critical point, a maximum $r_{\max}$. Denote $\Lambda := Q + a^2E^2$. We will consider three cases: I) $E_K \leq \epsilon\sqrt{\Lambda}$, II) $E^2 \leq \epsilon\Lambda$ and III) $E_K > \epsilon\sqrt{\Lambda}$ and $E^2 > \epsilon\Lambda$, for some suitably small $\epsilon > 0$.

Case I: $E_K \leq \epsilon\sqrt{\Lambda}$ By the proof of the above proposition

$$E^2 - V_0(r_+) = E_K^2 \leq \epsilon^2\Lambda.$$

We also showed that

$$\frac{dV_0}{dr}(r_+) \geq 2\Lambda(r_+^2 + a^2)^{-2}(r_+ - M) \geq 4r_+^{-1}\Lambda$$

Hence, there exists a $0 < \epsilon^2 < \delta$ such that

$$V_0(r_+(1+\delta)) - E^2 \geq (2\delta - \epsilon^2)\Lambda.$$

so in particular, $V_0(r_{\max}) - E^2 \geq (2\delta - \epsilon^2)\Lambda$.

Case II: $E^2 \leq \epsilon\Lambda$ As $r \to \infty$, we have in this case (using that $L^2 \leq \Lambda$ and $E^2 \leq \epsilon\Lambda$):

$$V_0(r) - E^2 = \Lambda r^{-2} - E^2 + \Lambda O(r^{-3}).$$

Hence, there exists a $R > r_+$ suitably large and an $\epsilon > 0$ suitably small such that

$$V_0(R) - E^2 \geq \frac{1}{2}\Lambda R^{-2} - E^2 \geq (\frac{1}{2} - \epsilon)\Lambda R^{-2}.$$

We conclude also in this case that $V_0(r_{\max}) - E^2 \geq V_0(R) - E^2 \geq \left(\frac{1}{2} - \epsilon\right)\Lambda R^{-2}$.

Case III: $E_K > \epsilon\sqrt{\Lambda}$ **and** $E^2 > \epsilon\Lambda$ Let $r_0 = \frac{aL^2}{2MLE}$. Since $0 < EL \leq \frac{a}{2Mr_+}L^2(1 - \epsilon^2|L|^{-1}\sqrt{\Lambda})$, we can estimate

$$r_0 \geq r_+ \frac{1}{1 - \epsilon^2|L|^{-1}\sqrt{\Lambda}} \geq (1-\epsilon^2)^{-1}r_+ > r_+.$$

Furthermore, it can be shown that (EXERCISE):

$$V_0(r_0) - E^2 = \frac{\Delta(r_0)}{(r_0^2 + a^2)^2}\left(\Lambda - \frac{a^2L^2}{4M^2}(1 + 2Mr_0^{-1} + a^2r_0^{-2})\right) \geq \frac{\Delta(r_0)}{4M(r_0^2+a^2)^2}(M-a)\Lambda$$

where we used that $L^2 \leq \Lambda$ and $1 + 2Mr^{-1} + a^2r^{-2} \leq 1 + 2Ma^{-1} + a^2a^{-2} = 2(1 + \frac{a}{M}) = 4 - 2\frac{M-a}{M}$.

Hence, $V_0(r_{\max}) - E^2 \geq V_0(r_0) - E^2 \geq b\Lambda$, for some $b > 0$. □

EXERCISES:

1. Show that trapped null geodesics that are orthogonal to the integral curves of Φ must lie outside the ergoregion. [Hint: Show that $\frac{dV_0}{dr} > 0$ in the ergoregion.]

2. Show that in the extremal case, these particular null geodesics are located at $r = (1+\sqrt{2})M$. The hypersurface $\{r = (1+\sqrt{2})M\}$ therefore serves as an *effective photon sphere* in this case.

3. ($\star$) Let $b = \frac{|L|}{E}$ be the impact parameter associated to an untrapped null geodesic that does not cross the event horizon. Show that $b_c = \inf_{Q,E} \frac{|L|}{E}$ is larger when $aL < 0$ than when $aL > 0$. [Hint: Show and use that $E^2 < V_0(r_{\max})$, with V_0 attaining its maximum at $r_{\max} \geq r_+$.] Explain why a far-away observer on the equatorial plane ($\theta = \frac{\pi}{2}$) will see a shadow that "bulges outwards" in one direction when looking at a rotating black hole with a uniformly distributed shell of stars surrounding both the observer and black hole, when compared to a round spherical shadow. On which side of the black hole will the shadow bulge outwards? You may use that, as in the Schwarzschild case, b_c can be related to the size of the shadow in the equatorial plane.

We will not discuss in detail the dynamics of timelike geodesics, but it can be shown that, just like in Schwarzschild, there exist stable orbits around Kerr black holes. In contrast with Schwarzschild, these can occur at r-values arbitrarily close to $r = r_+$ for $1 - \frac{|a|}{M}$ suitably small.

7 Asymptotic flatness

Throughout the lectures, we have used the term "asymptotically flat" in a rather vague sense. In particular, we referred to the Kerr and Reissner–Nordström spacetime families as being asymptotically flat. We will now give a precise definition of asymptotic flatness at the level of spacelike hypersurfaces that we can also interpret as "initial data hypersurfaces". We will then use this to state some important propositions and theorems without proof.

First, we need to define the notion of *extrinsic curvature*:

Definition 7.1. *Let $(\mathcal{M}, g)$ be a spacetime with a spacelike hypersurface Σ and let $\iota : \Sigma \to \mathcal{M}$ be the inclusion map. Let $N \in \mathcal{T}(M)$ be (an extension) of the future-directed normal vector field to Σ. Consider the following tensor field $k \in \mathcal{T}^{(0,2)}(\mathcal{M})$:*

$$k(X, Y) := g(\nabla_X N, Y).$$

The extrinsic curvature *or the* second fundamental form *of Σ is the pullback $\iota^* k \in \mathcal{T}^{(0,2)}(\Sigma)$.*

[Exercise: 1) Show that $\iota^* k$ is symmetric. 2) Show that we can express $(\iota^* k)(X, Y) = \frac{1}{2}\iota^*(\mathcal{L}_N g)(X, Y)$ for vector fields $X, Y \in \mathcal{T}(\Sigma)$.]

If $(\mathcal{M}, g)$ is globally hyperbolic solution to the vacuum Einstein equations with Cauchy hypersurface Σ, then the theorem below shows that $(\mathcal{M}, g)$ must be the *unique* globally hyperbolic solution to the vacuum Einstein equations (up to pullbacks with diffeomorphisms) that can be associated to the triple $(\Sigma, \iota^* g, \iota^* k)$. In fact, given a triple $(\Sigma, \iota^* g, \iota^* k)$, there always exists such a spacetime. We can therefore interpret Σ as an *initial hypersurface* ("time equal to zero") and $\iota^* g$ and $\iota^* k$ as initial data on this the hypersurface. The corresponding globally hyperbolic $(\mathcal{M}, g)$ may be thought of as the evolution of this data.

Theorem 7.1 (Choquet–Bruhat 1952, Choquet–Bruhat–Geroch 1969)**.** *Consider $(\Sigma, \overline{g}, \overline{k})$, with Σ a 3-dimensional manifold, $\overline{g}$ a Riemannian metric and $\overline{k}$ a symmetric $(0, 2)$-tensor field, such that the following* constraint equations hold:

$$R[\overline{g}] - \overline{g}^{ik}\overline{g}^{jl}\overline{k}_{ij}\overline{k}_{kl} + (\operatorname{tr}\overline{k})^2 = 0,$$
$$^{\overline{g}}\nabla^i \overline{k}_{ij} - {}^{\overline{g}}\nabla_j \operatorname{tr}\overline{k} = 0.$$

Then there exists a globally hyperbolic spacetime $(\mathcal{M}, g, [T])$, such that:

1. *$Ric[g] = 0$,*
2. *There exists a smooth embedding $\iota : \Sigma \to \mathcal{M}$, such that $\iota^* g = \overline{g}$ and $\iota^* k = \overline{k}$.*
3. *$\iota(\Sigma)$ is a Cauchy hypersurface of $(\mathcal{M}, g)$.*

$(\mathcal{M}, g, [T], \iota)$ is called a globally hyperbolic development *(GHD).*

An extension *of a GHD $(\mathcal{M}, g, [T], \iota)$ is a GHD $(\widetilde{\mathcal{M}}, \widetilde{g}, [\widetilde{T}], \widetilde{\iota})$, such that there exists a smooth isometric embedding $\psi : \mathcal{M} \to \widetilde{\mathcal{M}}$ satisfying $\psi \circ \iota = \widetilde{\iota}$ ("initial data preserving") and $\widetilde{g}(\psi_* T, \widetilde{T})|_{\psi(\mathcal{M})} < 0$ ("time-orientation preserving").*

There exists a GHD that is an extension of all GHD called the maximal globally hyperbolic development *(MGHD). The MGHD is unique up to isometry.*[48]

Analogues of the above theorem hold when considering the Einstein equations coupled with appropriate matter models, like the Maxwell equations, in which case appropriate initial data need to be prescribed for the matter.

[48] If $(\mathcal{M}, g, [T])$ and $(\mathcal{N}, \widetilde{g}, [\widetilde{T}])$ are two MGHD, then there exists a diffeomorphism $\psi : \mathcal{M} \to \mathcal{N}$, such that $\psi^* \widetilde{g} = g$ and $[\psi^* \widetilde{T}] = [T]$.

Definition 7.2. *Let $(\mathcal{M}, g)$ be an 4-dimensional spacetime with a spacelike hypersurface(-with-boundary) Σ and corresponding induced metric $\overline{g}$. The hypersurface Σ is said to have an* asymptotically flat end *if there exists an open subset $\Sigma_{\text{ext}} \subseteq \Sigma$, such that:*

1. *Σ_{ext} is diffeomorphic to $\mathbb{R}^3 \setminus \overline{B_R(0)}$, with $\overline{B_R(0)}$ the closed ball of radius R centred around $0 \in \mathbb{R}^3$.*

2. *There exists a coordinate chart (x^1, x^2, x^3) on Σ_{ext}, such that with respect to this coordinate chart:*

$$\begin{aligned}\overline{g}_{ij} &= \delta_{ij} + o_2(r^{-\alpha}),\\ k_{ij} &= o_1(r^{-\alpha-1}),\end{aligned}$$

for $\mathbb{R} \ni \alpha > \frac{1}{2}$, with δ_{ij} the Kronecker delta and $r := \sqrt{(x^1)^2 + (x^2)^2 + (x^3)^2}$. Here the "Little-o" notation $o_k(r^{-\alpha})$ groups functions $f : \Sigma_{\text{ext}} \to \mathbb{R}$ that satisfy for $I \in \mathbb{N}^3$, $|I| = k$:

$$\lim_{r\to\infty} \sup_{\partial B_r(0)} |r^{\alpha+k} \partial_{x^1}^{I_1} \partial_{x^2}^{I_2} \partial_{x^3}^{I_3} f|_{\partial B_r(0)}| = 0.$$

We say Σ is a one-ended asymptotically flat hypersurface*(-with-boundary) if $\Sigma \setminus \Sigma_{\text{ext}}$ is compact.*

7.1 Conservation of the total energy-momentum

With the notion of asymptotic flatness, we can define the following notions of total *energy*, *linear momentum* and *angular momentum* associated to a one-ended asymptotically flat hypersurface:

Theorem 7.2. *Let Σ be one-ended asymptotically flat hypersurface, with induced metric $\overline{g}$ and second fundamental form k. Let (x^1, x^2, x^3) be a global coordinate chart with respect to which asymptotic flatness is defined.*

(i) *Then the* ADM energy[49] *$E \in \mathbb{R}$ and* ADM linear momentum *$\mathbf{P} \in \mathbb{R}^3$ are defined as follows:*

$$\begin{aligned}E &:= \frac{1}{16\pi} \lim_{r\to\infty} \int_{\partial B_r(0)} \sum_{j,m=1}^{3} (\partial_j \overline{g}_{jm} - \partial_m \overline{g}_{jj}) n^m \, r^2 d\sigma,\\ P_i &:= -\frac{1}{8\pi} \lim_{r\to\infty} \int_{\partial B_r(0)} \sum_{j=1}^{3} (k_{ij} - \overline{g}_{ij} \operatorname{tr} k) n^j \, r^2 d\sigma,\end{aligned}$$

with $d\sigma = \sin\theta d\theta d\varphi$ the standard volume form on the unit round sphere in $\mathbb{R}^3$, $n = (x^1, x^2, x^3)^T$ the outward unit normal to $\partial B_r(0)$ in $\mathbb{R}^3$.

The quantities E and P_i are finite and independent of the particular choice of coordinate chart (x^1, x^2, x^3).

(ii) *If we make the following stronger assumptions on Σ:*

$$\begin{aligned}\overline{g}_{ij} &= \left(1 + \frac{2M}{r}\right) \delta_{ij} + O_2(r^{-1-\epsilon}),\\ k_{ij} &= O_1(r^{-2-\epsilon}),\end{aligned}$$

[49] The acronym "ADM" stands for Arnowitt–Deser–Misner.

for some $\epsilon > 0$,[50] *then* $\mathbf{P} = 0$, $E = M$ *and the* ADM angular momentum $\mathbf{J} \in \mathbb{R}^3$ *associated to* Σ *is determined by the following expressions:*

$$J_i := -\frac{1}{8\pi} \lim_{r\to\infty} \int_{\partial B_r(0)} \sum_{j,m,n=1}^{3} \epsilon_{ijm}(k_{mn} - \overline{g}_{mn} \operatorname{tr} k) x^j n^n \, r^2 d\sigma,$$

with ϵ_{ijk} *the Levi-Civita symbol. The quantity* $\mathbf{J}$ *is finite and independent of the particular choice of coordinate chart* (x^1, x^2, x^3).

(iii) *Assume now that there exist coordinates* (t, x^1, x^2, x^3) *such that:*

$$g_{\mu\nu} = m_{\mu\nu} + o_2(r^{-\alpha}), \tag{7.1}$$

with $\alpha > \frac{1}{2}$ *and* $m_{\mu\nu}$ *the Minkowski metric components with respect to Cartesian coordinates. Then* $(E, \mathbf{P})$ *along the level sets* $\{t = const.\}$ *are independent of* t. *Assuming that* $\operatorname{tr} k = 0$ *along* $\{t = const.\}$, *the same is true for* $\mathbf{J}$.

Proof. See for example Chapter 3 of Demetrios Christodoulou, *Mathematical Problems of General Relativity I*, European Mathematical Society, 2008. □

By considering appropriate coordinates in the Kerr black hole exterior, it can be shown that $E = M$ and $\mathbf{J} = aM(0, 0, 1)^T$, with M, a the Kerr parameters.

7.2 Black hole uniqueness theorems

In this section, we will state the *black hole uniqueness theorems*, which roughly state that asymptotically flat, stationary black hole spacetimes must be isometric to members of the Kerr family. To state these theorems more precisely, we first need a suitable notion of stationarity.

Definition 7.3. *A spacetime* $(\mathcal{M}, g)$ *with a one-ended asymptotically flat hypersurface* Σ *is* stationary *if there exists a Killing vector field* T, *which is timelike on* Σ_{ext} *and complete (i.e. the domain of its integral curves is* $\mathbb{R}$*).*

Let $\mathbb{R} \times \Sigma_{\text{ext}} \subseteq \mathcal{M}$ be the spacetime region obtained by flowing along the integral curves of T, starting from Σ_{ext}. We will use this definition (in place of future null infinity), to define the domain of outer commutations and the black hole region.[51]

Definition 7.4. *We define the* domain of outer communications $\mathcal{M}_{\text{ext}}$ *as follows:*

$$\mathcal{M}_{\text{ext}} := I^+(\mathbb{R} \times \Sigma_{\text{ext}}) \cap I^-(\mathbb{R} \times \Sigma_{\text{ext}}).$$

Then the black hole region $\mathcal{BH}$,[52] *the* future event horizon $\mathcal{H}^+$ *and the* past event horizon $\mathcal{H}^-$ *are defined as follows:*

$$\mathcal{BH} := I^+(\mathbb{R} \times \Sigma_{\text{ext}}) \setminus \mathcal{M}_{\text{ext}},$$
$$\mathcal{H}^\pm := \partial\mathcal{M}_{\text{ext}} \cap I^\pm(\mathbb{R} \times \Sigma_{\text{ext}}).$$

It can be shown that $\mathcal{H}^\pm$ are null hypersurfaces, so they are foliated by non-intersecting null geodesics, the "null generators" of $\mathcal{H}^+$.

Finally, we define the notion of *axisymmetry* of a spacetime.

[50] Here we applied the "Big-O notation" $O_k(r^{-\alpha})$, to group functions $f : \Sigma_{\text{ext}} \to \mathbb{R}$ satisfying $r^{\alpha+k} \sup_{\partial B_r(0)} |\partial_{x^1}^{I_1} \partial_{x^2}^{I_2} \partial_{x^3}^{I_3} f|_{\partial B_r(0)}| \leq C_k$ for some constant $C_k > 0$ and $I \in \mathbb{N}^3$, $|I| = k$.

[51] EXERCISE: Why must T be timelike everywhere in $\mathbb{R} \times \Sigma_{\text{ext}}$?

[52] The white hole region can be defined analogously as $I^-(\mathbb{R} \times \Sigma_{\text{ext}}) \setminus \mathcal{M}_{\text{ext}}$.

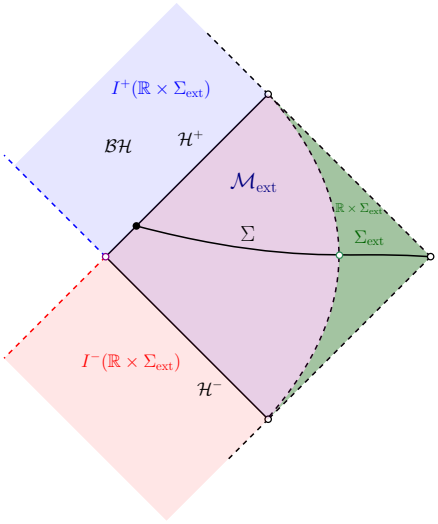


Figure 34: An picture of the objects introduced in Definition 7.4 in a Penrose diagram of subextremal Reissner–Nordström.

Definition 7.5. *A spacetime $(\mathcal{M}, g)$ is* axisymmetric *if there exists a Killing vector field Φ with periodic integral curves and an* axis of rotation*, which is a two-dimensional, totally geodesic submanifold*[53] *of $\mathcal{M}$ at which Φ vanishes.*

Now we obtain the following *rigidity theorem* for stationary, axisymmetric vacuum spacetimes:

Theorem 7.3 (Carter 1971, Chruściel–Costa 2008, Chruściel–Nguyen 2010)**.** *Let $(\mathcal{M}, g)$ be a stationary and axisymmetric spacetime solution to $\mathrm{Ric}[g] = 0$ with a one-ended asymptotically flat hypersurface Σ. Assume additionally that*

- *$\mathcal{M}_{\text{ext}}$ is globally hyperbolic,*
- *$\mathcal{H}^+$ is connected,*
- *If $\mathcal{H}^+ \neq \emptyset$, then $\partial\Sigma \subset \mathcal{H}^+$ and $\partial\Sigma$ intersects each null generator of $\mathcal{H}^+$ exactly once.*

Then $\mathcal{M}_{\text{ext}}$ is isometric to the domain of outer communications of a Kerr spacetime with $|a| \leq M$ (if $\mathcal{H}^+ \neq \emptyset$) or to the Minkowski spacetime (if $\mathcal{H}^+ = \emptyset$).

One can remove the axisymmetry assumption, but this comes at a high cost: the restriction to *analytic* spacetimes.

Theorem 7.4 (Hawking 1972, Chruściel–Costa 2008, Chruściel–Nguyen 2010)**.** *Let $(\mathcal{M}, g)$ be a stationary and* ***analytic*** *spacetime solution to $\mathrm{Ric}[g] = 0$ with a one-ended asymptotically flat hypersurface Σ. Assume additionally that*

- *$\mathcal{M}_{\text{ext}}$ is globally hyperbolic,*
- *$\mathcal{H}^+$ is connected,*
- *If $\mathcal{H}^+ \neq \emptyset$, then $\partial\Sigma \subset \mathcal{H}^+$ and $\partial\Sigma$ intersects each null generator of $\mathcal{H}^+$ exactly once,*

[53] This means: every geodesic that is initially tangent to the submanifold, stays tangent to the submanifold.

Then $\mathcal{M}_{\text{ext}}$ is isometric to the domain of outer communications of a Kerr spacetime with $|a| \leq M$ (if $\mathcal{H}^+ \neq \emptyset$) or to the Minkowski spacetime (if $\mathcal{H}^+ = \emptyset$).

It remains an open problem to verify whether the analyticity assumption can be removed in the above theorem!